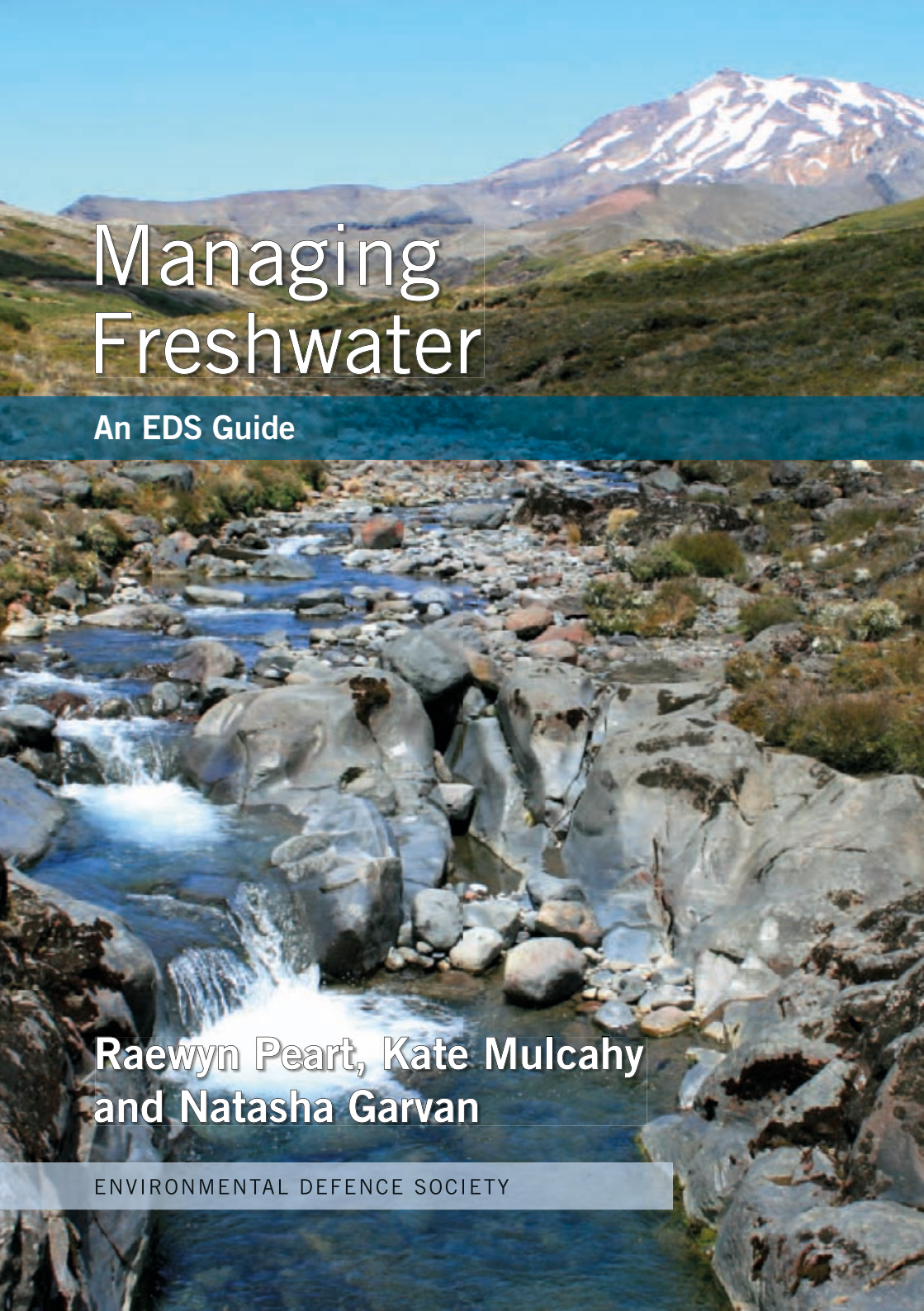




Managing Freshwater

An EDS Guide

Raewyn Peart, Kate Mulcahy
and Natasha Garvan

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New Zealand

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First published 2010
This edition printed 2012

Published by:
Environmental Defence Society Incorporated
P O Box 95 152
Swanson
Auckland 0653
New Zealand
Phone (09) 480 2565
manager@eds.org.nz
www.eds.org.nz
www.rmaguide.org.nz

ISBN 978-0-9582641-8-1

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Artwork: Neale Wills – Wilsy Design & Production

Cover image: Raewyn Peart



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Glossary of Technical Terms

Amphidromous	Travels between fresh and salt water usually in search of food and/or refuge
Anaerobic	Absence of oxygen
Anoxic	Deficiency of oxygen
Benthic	Bottom dwelling
Biofilms	Thin layers of slime
Catadromous	Spends most of its adult life in freshwater and migrates to salt water to spawn
Diadromous	Migrates between salt and freshwaters
Ecological flow	Amount of water and level of flow required in a water body to provide for the healthy ecological functioning of flora and fauna
Elvers	Young native eels
Endemic	Only found in New Zealand
Environmental flow	Ecological flows plus water flows and levels required to maintain important human values
Ephemeral stream	Only flows for short periods of time after a rain event
Eutrophication	Waters rich in mineral and organic nutrients that promote a proliferation of plant life, especially algae
Geothermal water	Water heated within the earth by natural phenomena to a temperature of 30 degrees Celsius or more; and includes all steam, water, and water vapour, and every mixture of all or any of them that has been heated by natural phenomena (section 2, RMA)
Hypertrophic	Saturated with nutrients
Intermittent stream	Only flows at certain times of the year
Invertebrate	Animal without a backbone
Kaitiakitanga	The exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources; and includes the ethic of stewardship (section 2, RMA)

Littoral zone	The shallow water and vegetated area along the shore of a lake
Macroinvertebrate	An invertebrate large enough to be seen without the use of a microscope
Nocturnal	Active during the night
River	A continually or intermittently flowing body of fresh water; and includes a stream and modified watercourse; but does not include any artificial watercourse (including an irrigation canal, water supply race, canal for the supply of water for electricity power generation or farm drainage canal) (section 2, RMA)
Transpiration	Emission of water vapor from the leaves of plants
Turbidity	The cloudiness of water caused by particles being suspended in the water
Wetland	Permanently or intermittently wet areas, shallow water, and land water margins that support a natural ecosystem of plants and animals that are adapted to wet conditions (section 2, RMA)

Preface

This Guide has been written for people who want to know more about how to promote effective management of New Zealand's freshwater systems and who would like to obtain better environmental outcomes. It is also designed to assist people engaged in activities impacting on freshwater. The Guide focuses on what 'good practice' freshwater management might look like rather than what the minimum environmental requirements are under the relevant legislation.

At the time of writing this Guide, the government-appointed Land and Water Forum has recommended a number of changes to the legal and administrative framework for managing freshwater. After any of the recommended changes have been adopted by government, this guide will be updated.

Content of guide

The first three chapters of the Guide describe freshwater bodies in New Zealand, the range of wildlife which are dependent on them and the values associated with them. Chapters 4, 5 and 6 then describe the impacts that human activities are having on freshwater bodies and ecosystems. Chapters 7 to 10 describe how freshwater is managed in New Zealand, by whom and the legal framework within which management decisions are made. Chapter 11 identifies opportunities for members of the public to participate in statutory decisions which affect how freshwater is managed. Chapter 12 describes elements of best practice freshwater management and how these might be applied in New Zealand.

Companion guides

There are five companion volumes to this Guide:

- *The New Zealanders' Guide to the Resource Management Act 1991* which provides more detailed information about participating in processes under the Resource Management Act 1991 (RMA)
- *Managing the Marine Environment: An EDS Guide* which provides detailed information on the management of the marine area, and the impact of catchment management on marine habitats and species
- *Strengthening Second Generation Regional Policy Statements: An EDS Guide* which describes how an effective and consistent approach can be applied to the development of second generation regional policy statements. It includes a chapter on addressing freshwater management issues in regional policy statements.

- *The Community Guide to Coastal Development under the Resource Management Act 1991* which provides detailed information on managing coastal development under the RMA
- *The Community Guide to Landscape Protection under the Resource Management Act 1991* which provides detailed information on addressing landscape management issues under the RMA

Web-based versions of all guides can be found at www.rmaguide.org.nz. Hard copies can be obtained from the Environment Defence Society Incorporated, PO Box 95 152 Swanson, Auckland, 0653. See www.eds.org.nz for more details.

Disclaimer

The material in this guide is provided in good faith, but with no guarantee as to it being comprehensive or accurate. The Guide is not a substitute for proper professional advice where that is needed.

Environmental Defence Society

The Environmental Defence Society is a non-profit environmental advocacy organisation consisting largely of resource management professionals. Freshwater issues are a major focus of the Society's work. If you would like to find out more about the Society, or support its environmental programmes, please visit the society's website at www.eds.org.nz.

Acknowledgements

The Environmental Defence Society would like to thank the Ministry for the Environment, Environment Canterbury, Meridian Energy, Metrowater, The Royal Forest and Bird Protection Society Incorporated, the Lion Foundation and the ASB Community Trust for supporting the production of this Guide.

The Society would also like to thank the following reviewers who provided comments on the content of a draft version of the Guide:

- Maree Baker, Partner, Anderson Lloyd
- Vicki Barker, Senior Analyst Resource Management Practice, Ministry for the Environment
- Dick Bellamy, Emeritus Professor, University of Auckland
- Hamish Cuthbert, Sustainability and Resource Policy Manager, Meridian Energy
- Neil Deans, Manager, Nelson Marlborough Region, Fish and Game New Zealand
- David Hamilton, Lakes Chair, University of Waikato
- Jason Holland, Senior Analyst, Ministry for the Environment
- Mike Joy, Director, Centre for Freshwater Ecosystem Management and Modelling, Massey University
- Cen Kaighin, Research Associate, Landcare Research New Zealand
- Garry Law, Principal, Law Associates Limited
- Graham McBride, Principal Scientist, National Institute of Water and Atmospheric Research
- Kelvin Nicolle, Senior Tutor and Waterwatch Coordinator, Lincoln University
- Siobhan Quayle, Ministry for the Environment
- Gary Taylor, Chairman, Environmental Defence Society
- Stuart Taylor, Ministry for the Environment
- Keir Volkerling, Consultant
- Bill Vant, Aquatic Ecologist, Environment Waikato
- Peter Weir, Managing Director, Earnslaw One
- Christina Wells, Ministry for the Environment
- Glen Wigley, Ministry for the Environment

EDS would also like to thank the people and agencies who provided photographs for reproduction (names shown in the text) and Joshua Gear for providing material on the Waikato-Tainui Raupatu Claims (Waikato River) Settlement Act. The final contents of the Guide remain the responsibility of the authors.



Neil Deans

one

Freshwater Bodies

Introduction

By international standards New Zealanders enjoy an abundance of clean fresh water. New Zealand's fresh water bodies include glaciers, snow, rivers, streams, lakes, wetlands and groundwater, many of which are interlinked.

Water flows in a constant cycle. Water from the ocean is evaporated by the sun's energy and transported to the atmosphere. It is then carried in air currents, which are forced into cooler altitudes, when they pass over a land mass. As the air cools, the water droplets condense and become visible as clouds, and then become so heavy that they fall to the ground as rain or snow.

Some rain is absorbed by soil, plants and animals and re-evaporated back into the atmosphere (a process called transpiration). Some flows back to the sea via streams, rivers, wetlands and estuaries (known generally as surface waters). And some water seeps down into the ground, (known as groundwaters) where it may either flow underground to the sea, flow back to the surface to join rivers or streams, or lie dormant underground.

Rivers and streams

A 'river' is defined in the Resource Management Act 1991 (RMA) as 'a continually or intermittently flowing body of fresh water; and includes a stream and modified watercourse; but does not include any artificial watercourse (including an irrigation canal, water supply race, canal for the supply of water for electricity power generation or farm drainage canal)'.

New Zealand has 70 major river catchments, 40 in the South Island and 30 in the North Island, and over 425,000 kilometres of rivers and streams.¹ It is difficult to state with great accuracy the number and extent of smaller streams in New Zealand. Although some streams are permanent fixtures, others are intermittent (only flowing at certain times of the year) or ephemeral (only flowing for short periods of time following particular rain events) and their large number makes counting them an extremely challenging task. However, attempts have been made to assemble an inventory of streams on a regional basis. For example, a 2008 study estimates that there are 16,650 kilometres of permanent streams in the Auckland region, as well as 4,480 kilometres of intermittent streams and 7,110 kilometres of ephemeral streams.²

Because of New Zealand's geography (being a narrow country with ridges of high mountains running along the centre) and its high rainfall, rivers and streams are often relatively short, fast flowing and unstable.³ The large variety of

rock types in New Zealand means that there are many different types of rivers and they vary significantly.

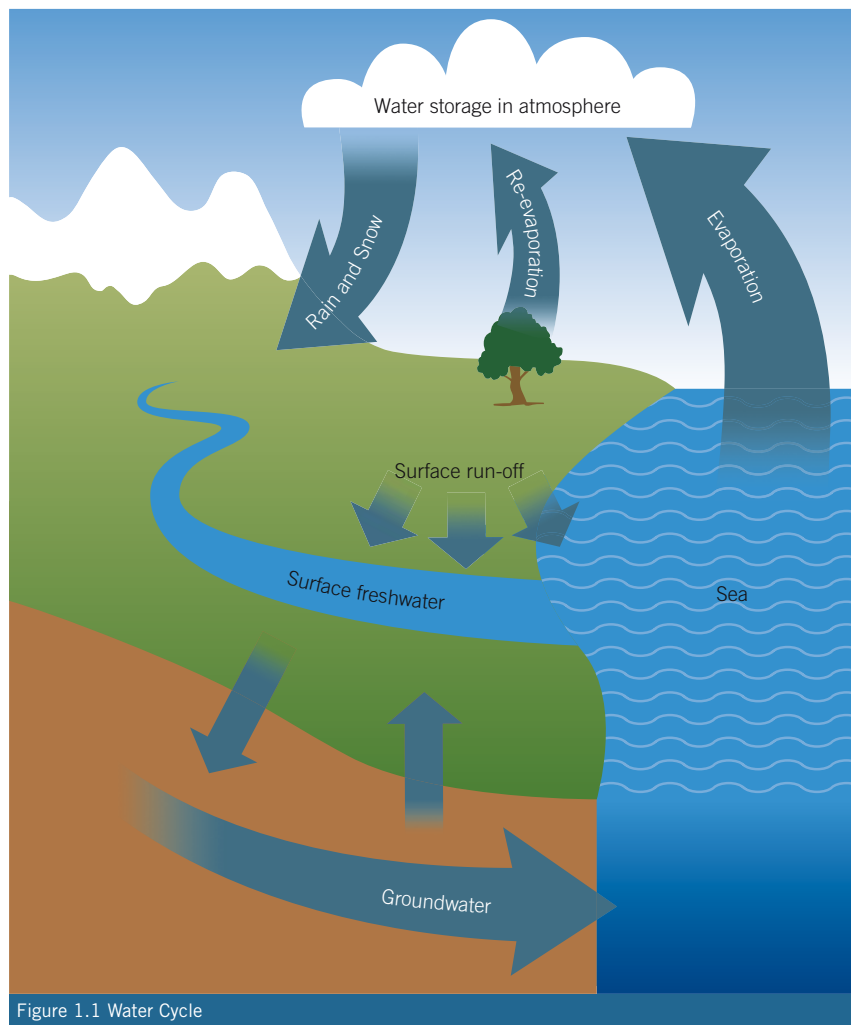


Figure 1.1 Water Cycle

North Island rivers

In Northland and on the Coromandel Peninsula rivers tend to be short, and in some cases, as sea levels rose after the last ice age, their lower reaches have become flooded and now form arms of the sea. The Hokianga Harbour is an example of this phenomenon. In the Waikato, the main rivers are alluvial and flow through flood plains that have been created by deposited sediment. New Zealand's longest river is located here, the Waikato River which flows for 425 kilometres, from the slopes of Mount Ruapehu north-west to the Tasman Sea at Port Waikato.⁴ The Tongariro power development has added a significant extra catchment to the Waikato River, incorporating the headwaters of the Whanganui and Rangitikei Rivers.⁵



Figure 1.2 Waimea River, Hokianga Harbour (Craig Potton) *In Northland, the mouths of many rivers have become arms of the sea, as is the case with the Waimea River in the Hokianga Harbour, shown here*

In the western uplands of the North Island, rivers such as the Whanganui and the Mokau incise deeply and often flow through long gorges. On the central volcanic plateau and in the Bay of Plenty, recent volcanic activity has shaped the course and character of the rivers, with the Tarawera and Kaituna Rivers draining lakes that formed in volcanic craters.⁶

South Island rivers

South Island rivers are equally diverse. There are significant differences between the rivers east and west of the main divide. West of the Southern Alps, there are rivers that have headwaters in very steep mountains. These rivers tend to flow quickly and feature rapids and scenic gorges. The Buller, Motueka and Grey rivers are good examples of this type of river. In contrast, Canterbury features slower flowing braided rivers. In Southland the rivers are alluvial. In Otago, rivers such as the Clutha and Taieri have their origins in the Southern Alps and flow through basins in schist blocks. Snowmelt is a significant factor in the seasonal variation of flow of the mountain-sourced rivers.⁷

Meandering rivers

Meandering rivers are common throughout New Zealand. They typically contain only one channel. The rivers are relatively slow flowing, and have pronounced bends, which are called meanders. The meanders are formed when moving water erodes material from the outside of a bend, and deposits it on the inside, creating a snaking pattern as the river flows downstream.



Figure 1.3 A meander on the Motueka River, Tasman District (Fish and Game New Zealand)

Braided rivers

Marlborough and Canterbury are well known for their braided rivers, which are glacial in origin. A braided river is one that, over some part of its length, flows in multiple channels across an alluvial-gravel fan. These fans can be very wide, potentially stretching over areas several kilometres wide.

Braided rivers often occur in areas where there are steep mountain ranges that have the capacity to create their own weather, like the Southern Alps. Similar conditions give rise to braided rivers in Canada, Alaska, and the Himalayas. The height of the mountains causes precipitation to fall on them at any time of the year, leading to high levels of erosion, and the creation of floodplains. The erosion is exacerbated by the steepness of the mountains and glacial activity. Eroded material runs downstream, and yields the gravel that lines the channels, shaping the river into interlinked braids of varying sizes.

Although there are many braided rivers in New Zealand (163 of New Zealand's river systems, located in 11 of New Zealand's regions, feature braided sections) the majority of significant braided rivers are found on the eastern side of the South Island, like the Waimakariri River in Canterbury.⁸



Figure 1.4: Wairau River (Fish and Game New Zealand) **Braided rivers are commonly found on the eastern side of the South Island, such as the Wairau River shown here, near Blenheim**

Braided rivers amount to much more than just the braids that can be seen from above: the gravel riverbed can be several metres deep. Surface water continually moves down through the gravel, connecting and mixing with the groundwater below, such as occurs along the lower Selwyn River in Canterbury. The level of the groundwater rises as a result of the inputs from the river and then may appear across the flood plain as small springs. The channels have a tendency to move around, because of the variability of the flow of water and supply of sediment to the river system. Together the river, floodplain and groundwater make up a complex three dimensional environment, which is constantly changing.

River classification

A *River Environment Classification* system has been developed for New Zealand rivers. It is GIS-based and classifies rivers at a range of spatial scales on the basis of six hierarchical levels. Four are related to the catchment (climate, source of freshwater flow, geology and land cover) and two are focused on specific sections of the river (position of the section of the river within the catchment network and the landform of the valley in which the river section is located). The classification system has been applied to all rivers in New Zealand and is useful in informing the selection of appropriate management tools to manage pressures on particular types of rivers.⁹

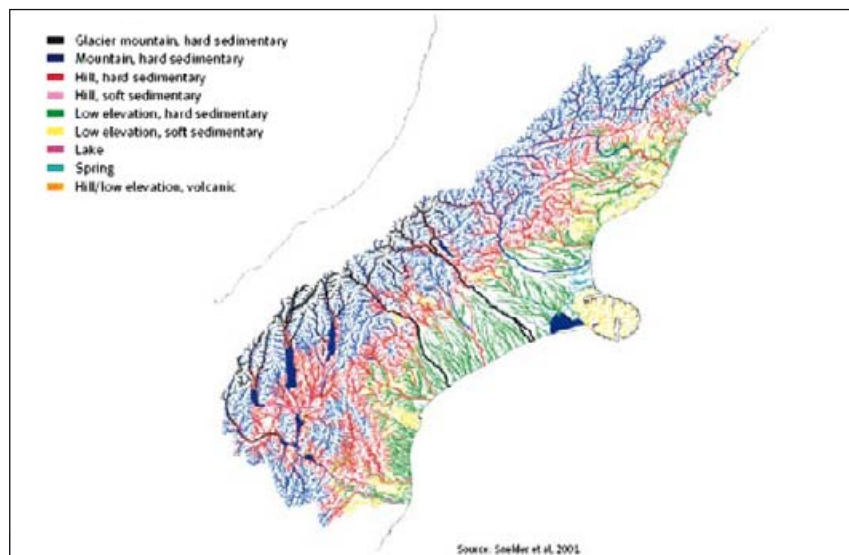


Figure 1.5 Example of the river classification system applied to Canterbury rivers

Source: Ton Snelder

Lakes

A ‘lake’ is defined in the RMA as ‘a body of fresh water which is entirely or nearly surrounded by land’. Lakes cover about 1.3 per cent of New Zealand’s land area (excluding offshore islands). New Zealand has 3,820 lakes which are larger than one hectare¹⁰ and 775 which are at least 0.5 kilometre long. About 40 per cent of New Zealand’s lakes are in catchments where the predominant land cover is pasture and less than 2 per cent of them are in towns and cities. The diverse landscapes of New Zealand have resulted in the formation of a number of different types of lake.¹¹

Sources of New Zealand Lakes	
Formed by glaciers (found in the South Island)	38%
Formed by rivers	16%
Formed by dunes	15%
Formed by humans	8%
Formed by landslides	5%
Formed by volcanoes (found in the North Island)	4%
Formed by coastal barriers	4%
Formed by other methods or their origins are unknown	10%

Figure 1.6 Sources of New Zealand lakes
(Source: Te Ara the Encyclopedia of NZ)



Figure 1.7 Lake Taupo (Raewyn Peart) **New Zealand has 3,820 lakes which are larger than one hectare. Lake Taupo is the largest lake in New Zealand, covering 616 square kilometres**

The largest lake in New Zealand is Lake Taupo, located in the central North Island, which covers 616 square kilometers. It was formed by the inundation of a massive volcanic caldera 26,500 years ago. The eight next largest are glacial lakes located in the South Island high country including Lake Te Anau which is 348 square kilometers in size.¹²

Glacial lakes

Glacial lakes such as Manapouri, Wakatipu, and Wanaka were created during the glacial periods of the last two million years. As glaciers grew, they eroded large steep-sided valleys, and carried with them piles of rock and soil that acted as natural dams once the glacier retreated. The eroded basins then filled with water. The deepest of these lakes, and the deepest in New Zealand, is Lake Hauroko in Western Southland which reaches a depth of 462 metres, well below sea level.

Volcanic lakes

New Zealand has a number of volcanic lakes, which are mainly found in the Taupo volcanic zone and the area around Auckland. Lakes Taupo and Rotorua fill calderas, which are enormous volcanic depressions caused by the land collapsing after a huge eruption of ash. Other volcanic lakes were created when explosive eruptions formed craters, or when the drainage of rivers was blocked by lava flows and ash deposits.



Figure 1.8 Lake Okareka (Raewyn Peart) *There is an extensive network of volcanic lakes in the Rotorua area including Lake Okareka shown here*

Lakes are relatively common in volcanic areas, because when the drainage pattern is disrupted by volcanic activity, water will tend to accumulate in the low areas. Many have no surface outlet, draining underground. Some have quite variable levels. Some volcanic lakes receive an input of hydrothermal or volcanic fluids and are therefore high in minerals and other chemical substances which yield unusual colours and making them potentially unsafe to swim in.

The Rotorua area has an extensive network of 20 volcanic lakes including Lakes Rotorua, Rotoiti, Rotoehu, Rotomā, Ōkātina, Ōkareka, Rotokakahi (Green Lake), Tarawera, Rotomahana, Tikitapu (Blue Lake) and Rerewhakaaitu.

Landslide lakes

Landslides can form lakes when they dam a river, causing water to build up behind the debris. For example, Lake Waikaremoana, in Te Urewera National Park (located between the Bay of Plenty and Hawke's Bay) was created by a huge landslide that occurred roughly 2,200 years ago.

Dune lakes

Dune lakes, such as the Kai Iwi lakes in Northland, are typically found on the west coast of the North Island.¹³ They are mostly shallow lakes fed predominantly by rain water, with their water level fluctuating with rainfall. Dune lakes form in stable sand dunes, where depressions in the sand are below the watertable, and rainwater accumulates in the depressions. Many of these lakes have been severely impacted by excess nutrients and invasive species. For example, Lakes Kapoi, Rotokawau and Waiporohita in Northland are hypertrophic, meaning they are saturated with nutrients and have frequent algal blooms.¹⁴

Human-made lakes

There are more than 60 human-made lakes in New Zealand, created for power generation, irrigation, water supply and recreation. Narrow valleys are frequently dammed to create reservoirs that hold water for domestic and industrial use, such as those in the Waitakere and Hunua ranges which supply water to Auckland. Very large artificial lakes are created to generate hydroelectric power, such as Lake Benmore on the Waitaki River, which has an area of 74 square kilometres. The amount of water flowing out of the lake is controlled and the power of the falling water is used to turn turbines which generate electricity. The water levels of some naturally occurring lakes are artificially controlled for irrigation and to reserve flow for electricity generation. Lake Hood in Ashburton was created to provide recreational opportunities for the community.

Lake water levels

Although lake water usually appears to be fairly static, complex changes happen constantly which are very important for the health of the freshwater body. The level of the lake reflects the relative amounts of water entering and leaving: some water is constantly lost through evaporation, seepage and possibly outflow into rivers, while it is being replenished by groundwater, rainfall, rivers or glaciers.

Lakes with no outflow into a river or stream will usually vary in level depending on the amount of inflow from sources such as groundwater or rainfall. Lakes with an outflow will maintain a relatively stable water level because the amount of water leaving the lake will simply increase when inflow increases. Rivers transport sediment into lakes making them shallower over time. Eventually lakes are totally filled in by this process. Lakes are therefore usually relatively young in geological terms, being mostly under 10,000 years old.

Wetlands

Wetlands are defined in the RMA as including ‘permanently or intermittently wet areas, shallow water, and land water margins that support a natural ecosystem of plants and animals that are adapted to wet conditions’.

Wetland composition varies widely because of regional and local differences in soils, topography, climate, hydrology, water chemistry, vegetation and other factors, including human disturbance. Wetlands have three things in common. The first of these is, unsurprisingly, water. Secondly, they all have wetland soils, which have low levels of oxygen or none at all, because they are intermittently or permanently saturated. Low levels of oxygen are also due, in part, to bacterial breakdown of high levels of organic matter. The soil is often fine textured, deep enough to store water, and has poor downward drainage. Thirdly, wetlands all support wetland plants, which have adapted to growing in wet conditions.

Wetlands of different types can be found all over the country – along hillsides where there is a change in slope or geology, on stream and river floodplains, in low-lying areas among flat landscapes, in coastal areas where sand dunes trap water runoff, and along the margins of rivers, lakes and estuaries. Their varying characteristics give rise to many combinations of features.¹⁵

Wetlands can be supplied by water from a number of different sources – including rain, groundwater, and surface runoff. Some wetlands occur at the edges of rivers, lakes or coastal areas. Many wetlands were once rivers that have partly drained away. Some wetlands, particularly in the central North Island, are geothermal in origin.

Class of wetland	Characteristics
Bog	Peatlands which only receive inputs of fresh water from rain
Fen	Mainly peatlands which receive inputs of groundwater and soil nutrients
Swamp	Wetlands which receive a rich supply of nutrients, and often also sediment, mainly from surface runoff
Marsh	Wetlands usually on mineral soils which receive inputs from groundwater or surface water and have a fluctuating water level
Seepage	Areas where surface water and/or groundwater flows over a sloped area
Shallow water	Shallow areas on the margins of lakes and rivers
Ephemeral	Areas of ponding on mineral soils which dry out completely at times
Pakahi gumland	Wetlands in poor soils which are mainly rain fed
Saltmarsh	Shallow saline areas

Figure 1.9: Classes of New Zealand wetlands

(Source: Johnson & Gerbeaux, 2004)

Wetlands are often highly productive systems because they contain plenty of nutrients as well as water. The nutrients are derived from the decay of plant material as well as from sediment washed in from flooding rivers. However, some wetlands have low nutrient levels and are relatively unproductive. For example bogs that are fed only by rainfall have simpler plant communities, and slower decay of plant material, which accumulates as peat.

Wetlands were a primary food source for traditional Māori society:

‘The importance of lowland swamps in the traditional Māori landscape was multifaceted. They watered and gave access to vast areas of country, birds were attracted to them for food and native fish that came to spawn. Dominating the swamps were rushes, reeds, flax and kahikatea or white pine. Mature, fruiting kahikatea were a seasonal mecca for birds and people. Waikaka (mudfish), a traditional delicacy for presentation at feasts, hibernated during summer drought beneath kahikatea roots. They and myriad indigenous fish species such as inanga, koaro and kokopu migrated through the estuaries and lagoons into pools enclosed by flax and raupo in the gaps in the kahikatea forests. It was these conditions that made tidal swamp-plain rivers like the Waihou, Waikato and Manawatu such superb fisheries. It was these same estuaries and lagoons that would later be taken by the Crown for drainage scheme spillways.’

Park, G, 2001, 19-20

Many of New Zealand's wetlands have been drained in order to expand pastureland or urban areas. Others have been damaged through the impacts of stock grazing and other human activities. Only around 10 per cent of New Zealand's pre-human settlement wetlands remain. This is a very small amount compared to the 40 per cent thought to remain in the more densely populated Netherlands and United Kingdom and the 90 per cent which remain in France.¹⁶



Figure 1.10 Waimango Swamp, Karikari Peninsula (Raewyn Peart) *Wetlands are highly productive ecosystems, and used to be common around New Zealand, but 90 per cent have been lost since human settlement*

New Zealand is a party to the *Ramsar Convention on Wetlands* which aims to improve the conservation of wetlands around the world. New Zealand has designated six wetlands of international importance under the convention which are:¹⁷

- Awarua Wetlands / Waituna Lagoon, Southland
- Farewell Spit, Nelson
- Whangamarino, Waikato
- Kopuatai Peat Dome, Waikato
- Firth of Thames, Waikato
- Manawatu River Estuary, Foxton

Groundwater

Groundwater is rainwater that has travelled through the soil, or from river and lake beds, to aquifers. These are underground rock formations that are sufficiently permeable so as to contain or conduct significant amounts of water. Groundwater makes up a large proportion of New Zealand's fresh water resource, totaling around 612 billion cubic metres.¹⁸

There is groundwater almost everywhere in New Zealand. In some areas, people rely on groundwater as their main water resource, such as the residents of Christchurch City and half of the rural population of the Waikato region. Groundwater is pumped to the surface for human use through wells or bores. Sometimes the water from aquifers is mineralised and requires treatment before it can be used for people, stock, irrigation or industry.

Although one might assume that groundwaters are stationary bodies of water, they are actually part of flowing water systems that incorporate surface water as well. But very little is often understood about the way that groundwater moves and how it interfaces with water on the surface. This can make effective management difficult.

Aquifer types

Groundwater aquifers can be split into two types – confined and unconfined.¹⁹ In confined aquifers, the water rises up inside the aquifer under natural pressure, because it is confined between impermeable sediments above and below the water. The pressure the water is under, can sometimes make it rise up and flow out of a bore, without being pumped. In unconfined aquifers, the water is not under such pressure, and tends to be found at shallower levels.

The most extensive aquifers are shallow, unconfined, sand and gravel sediments. In particular, they are found in areas such as the shallow, porous gravels that occupy old river plains like the Heretaunga Plains in Hawkes Bay and the Wairarapa and Southland Plains. These aquifers contain relatively young, well oxygenated and fast flowing groundwater. The mineral content in these aquifers is generally low, but the shallow water table makes them vulnerable to pollutants, such as nutrients and bacteria from land use. Other productive forms of aquifer include the fractured basalts of the Auckland region and the Coromandel's coastal sand aquifers.



Figure 1.11 Bubbling sands, part of the Te Waikoropupu Springs complex (Raewyn Peart) *Groundwater is common throughout New Zealand. In some areas it bubbles to the surface such as at Te Waikoropupu Springs near Takaka which are thought to be the largest cold water springs in the southern hemisphere* ²⁰

It can take a very long time for the water in aquifers to be exchanged. Some underground water is very old, with the oldest thought to be in the Moutere aquifer near Nelson, where the water has been underground for more than 20,000 years.²¹ In other areas the residence time can be much shorter. A study of aquifers in the Rotorua area identified a mean residence time of underground water of between 14 and 170 years.²²

This long residence time means that there is usually a significant delay between pollution of groundwater occurring, and it being detected, often only when the contaminated water flows to the surface. As a result, nutrients levels in streams fed by aquifers in agricultural areas are likely to continue to increase as the clean ‘older’ aquifer water is progressively replaced by newer, contaminated water.

Christchurch City's water supply

One of the most significant aquifers for water supply is the Christchurch-West Melton aquifer. This provides all of the city of Christchurch's water supply, and has been estimated to have an economic value of \$60 million per year. Water is pumped up from the aquifer from more than 50 sites around the city. At each site there are one or more wells drilled to depths of between 22 and 190 metres. The aquifer is recharged by rainfall and the Waimakariri River.

Source: White P A, B M H Sharp and R R Reeves, 2004; <http://www.ccc.govt.nz/Water/HowWeGetOurWater/>

Geothermal

Some aquifers are recharged with geothermal water. ‘Geothermal water’ is defined in the RMA as being ‘water heated within the earth by natural phenomena to a temperature of 30 degrees Celsius or more’. Geothermal groundwater is heated, either by the energy of nearby volcanic activity, or because it has been located very deep underground where the earth’s temperature is hotter. Sometimes this water can reach the surface through cracks in rock. Waiwera and Parakai in the Auckland region, Hot Water Beach on the Coromandel Peninsula and Hanmer Springs in the South Island are good examples of such geothermal springs.



Figure 1.12 Hot Water Beach (Raewyn Peart) *Geothermally heated water can reach the surface through cracks in the rock. At Hot Water Beach the heated water percolates through the sand and provides hot water bathing on the beach*

Geothermal groundwater has a very high mineral and salt content, as these chemicals dissolve in the hot water as it moves up from deep underground, often making it unsuitable for drinking. However, geothermal water and steam is prized as a tourist attraction and for use in geothermal power generation. The extraction of geothermal water in New Zealand is strictly regulated, as over-extraction is likely to allow cooler water to mix with the aquifer, reducing the heat and damaging the resource.

Geothermal power generation currently supplies around 10 per cent of New Zealand’s electricity, with stations mainly located within the Taupo volcanic zone. Mighty River Power is due to open a new geothermal power station at Rotokawa, north of Taupo, in 2010, which will be New Zealand’s second largest geothermal power station.²³

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Raewyn Peart

two

Freshwater Life

Introduction

There is a wide variety of biota that makes up freshwater ecosystems and that are dependent on freshwater bodies for their survival. This includes fish, birds, and invertebrates which depend in turn on plants and communities of microscopic organisms such as algae, bacteria and fungi. All these organisms combine to form complex communities and food webs, which encompass both the aquatic environment itself, and the adjacent land and vegetation.

The *primary producers* of energy in fresh water communities are bacteria and algae as well as mosses and liverworts, which all utilise sunlight. These are then consumed by *primary consumers* which typically include insects, molluscs such as snails, crustaceans such as freshwater crayfish and worms. Primary consumers also include microscopic zooplankton that graze on the phytoplankton.¹

These primary consumers are the food source for *secondary consumers* which are mainly fish but also include the larvae of insects such as the dobsonfly and stonefly and some birds. These secondary consumers can also be eaten by *tertiary consumers* such as eels, trout and birds. In addition, bacteria and fungi may also act as *decomposers*, breaking down dead leaves and wood in the water.²

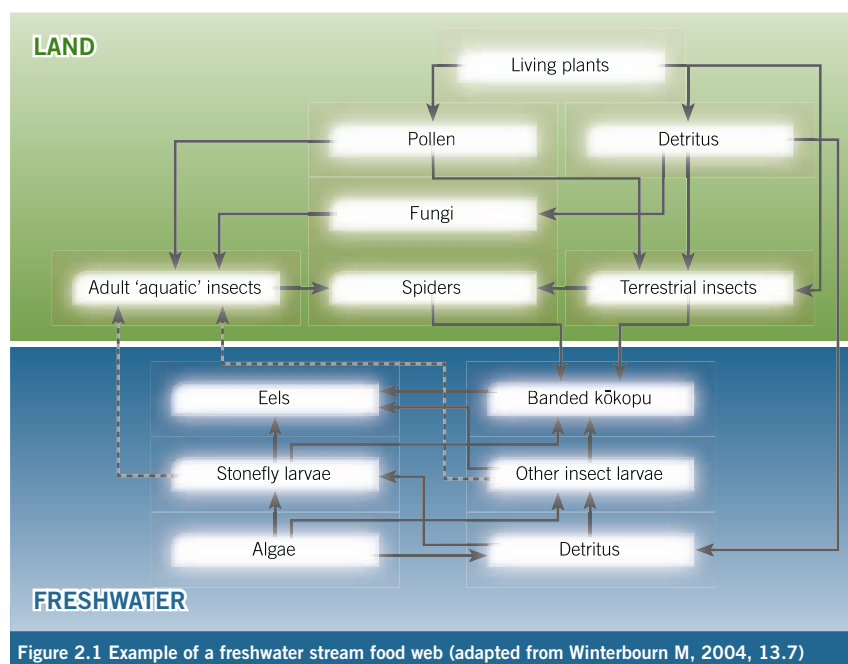


Figure 2.1 Example of a freshwater stream food web (adapted from Winterbourn M, 2004, 13.7)

There is a close ecological linkage between the aquatic environment and adjacent land. Dead leaves and branches, which fall into the water from terrestrial plants, provide an important source of energy and nutrients. Pieces of wood also provide important habitat for invertebrates and fish. Stones and boulders become coated with microscopic layers of organisms (biofilms) which provide a food source for other stream life. Some aquatic insects have a larval stage in freshwater but with the adult forms living on shore. Some fish spawn in vegetation growing along the waters' edge. This means that land use adjacent to rivers and lakes can have a major impact on freshwater life.³

The following sections describe indigenous freshwater species. Introduced species are described in Chapter 6.

Freshwater invertebrates

There is a multitude of invertebrates (animals without a backbone) which can be found in New Zealand's freshwater habitat: freshwater crayfish, shrimp, crab, worms, flatworms, leeches, snails, bivalves, molluscs, amphipods, water fleas, seed shrimps (ostracods), larvae of beetles, midges, crane fly, caddisfly, mayfly, stonefly and sandfly. Many of these species are only found in New Zealand.

Freshwater crayfish (koura)

New Zealand has two native freshwater crayfish: the northern koura (*Paranephrops planifrons*) is found throughout the lakes, streams and wetlands of the North Island and northern and west coasts of the South Island. It grows to around 7 centimetres long. The southern koura (*Paranephrops zealandicus*) is found in the east and south of the South Island and on Stewart Island. They are slightly larger than their northern cousins, and have hairy pincers, but in other respects are very similar.⁴

Koura are mainly nocturnal and shelter between rocks and stones. In lakes they burrow into the muddy bottom during the day, emerging to feed at night. They can also burrow into the bottom of temporarily dry water bodies to await the return of the water. Koura feed on aquatic insects and vegetation, and tend to remain in the same water body their whole lives.⁵

Koura are unable to survive in polluted water, and as a result, their absence or presence can be used as an indicator of the health of a waterbody. They are in gradual decline, and have been identified as threatened, mainly as a result of the degradation and loss of their habitat.⁶



Figure 2.2 Koura (Department of Conservation)

Freshwater shrimp

New Zealand's only native species of freshwater shrimp, *Paratya curvirostris*, is found in lowland streams, living amongst vegetation. It feeds on detritus such as algal films and bacteria which it scrapes off rocks and wood with its pincers. These shrimp go through a male phase of development before turning into adult females. Similar to many indigenous fish species, the freshwater shrimp is diadromous and includes a marine stage in its lifecycle. The shrimp is common around New Zealand and is fairly tolerant of the degraded condition of many lowland streams.⁷

Freshwater crab

There is also only one species of freshwater crab (*Amarinus lacustris*) in New Zealand which is found in a variety of habitats including lakes and slow-flowing streams and rivers. The crab feeds on animal and vegetable detritus found in the silt in which they live, and grow to around 10 millimetres wide. They are also found in South Eastern Australia and the Norfolk Islands. It is likely that many populations of this crab have not yet been discovered, but nevertheless they are considered to be threatened.⁸

Freshwater mussel (kākahi)

There are three species of freshwater mussel in New Zealand with *Hyridella menziesii* being the most common. Freshwater mussels can be found in lakes, rivers and streams. They are filter feeders, siphoning water and filtering out small algae and bacteria. This filtering activity can help increase water clarity. The mussels are slow growing and can reach sizes of up to 8 centimetres and ages of up to 50 years.

Freshwater mussels once formed extensive beds in many lakes and were harvested by Māori for food. More recently they have been in decline and are now considered threatened. The reasons for this decline are not fully understood. One theory focuses on the close relationship between the mussels and indigenous fish species. The life cycle of the mussels includes a larval stage, with the larvae attaching themselves to fish such as Koaro. Populations of indigenous fish species have been in sharp decline and this may explain the reduction in mussel populations.⁹ Another possible reason for the mussel's decline is the inundation of its preferred spring-fed habitats with sediment.

Other invertebrates

There are over 200 species of smaller macroinvertebrates (invertebrates which can be seen without a microscope) in New Zealand's freshwaters, with still more species yet to be discovered. Some species, such as mayflies, stoneflies and caddisflies, are only usually found in healthy streams. Other species such as worms, midges, snails and leeches are more tolerant of degraded conditions and can be found in muddy and polluted streams. This means that the presence or absence of particular species in a waterbody can be used as an indicator of water quality.¹⁰

Macroinvertebrate water quality measures

Because some species of macroinvertebrates are very sensitive to pollution, while others are more tolerant, the presence or absence of particular species can be used as an indicator of overall water quality. Two common measures of macroinvertebrates are used to indicate water quality

The *Macroinvertebrate Community Index* (MCI) looks at the entire macroinvertebrate population structure and provides a score that indicates overall water quality. Generally, an MCI score of less than 80 indicates poor water quality and a score of greater than 119 indicates excellent water quality.

The *Percentage of the total abundance comprising Ephemeroptera, Plecoptera, and Trichoptera taxa* (%EPT) measures the contribution to the total abundance of macroinvertebrates belonging to the sensitive mayfly, stonefly and caddisfly species. A low %EPT indicates degraded water quality whereas a high %EPT indicates good water quality.

Source: <http://www.mfe.govt.nz/environmental-reporting/freshwater/river/macroinvertebrates/>

Freshwater fish

There are around 38 species of native freshwater fish in New Zealand, many of which are found nowhere else in the world.¹¹ Less than five per cent of our rivers and streams have been properly sampled¹² and new species are still being discovered. Many species are very poorly understood by scientists, but it is thought that only one species has become extinct – the Southern grayling (*Prototroctes oxyrhynchus*) which was last seen in the 1920s.¹³

Native New Zealand freshwater fish are different from those in other countries in that they mostly live on the bottom (benthic), are active during darkness hours (nocturnal), and have a marine stage in their life-cycles (diadromous). These characteristics evolved in response to the peculiarities of the local environment. Before human settlement, the streams and wetlands in New Zealand were heavily shaded by indigenous evergreen forest which produced relatively little forest litter and the waterbodies were unstable due to high rainfall and short catchments.¹⁴

Because indigenous fish are shy, secretive, benthic and nocturnal, they are not easy to see. This meant that they were not very obvious to the early European settlers, who concluding that there was a lack of native sports fish, decided to introduce fish species such as trout into New Zealand waters.

Around 60 per cent of native New Zealand fish have a marine stage in their life cycles. Either the adult goes to sea to spawn or the larvae are washed out to sea.¹⁵ This characteristic has an evolutionary advantage in New Zealand's unstable physical environment. It enables species to recolonise rivers and streams from the sea, over wide geographical areas. Diadromy can have considerable disadvantages however, in human-modified systems, because structures such as dams and culverts and heavily polluted areas of rivers can form barriers preventing fish from travelling between rivers and the sea. Some diadromous species are able to vary their behavior, and remain inland, if geographical constraints prevent them from accessing marine areas.

Galaxiids

A family of fish called Galaxiids make up a large proportion of the native freshwater fish in New Zealand. There are probably at least 25 species of Galaxiids in New Zealand of which around 80 per cent are threatened. The juveniles of five species (giant kōkopu (*Galaxias argenteus*), kōaro (*Galaxias brevipinnis*), banded kōkopu (*Galaxias fasciatus*), shortjaw kōkopu (*Galaxias postvectus*) and īnanga (*Galaxias maculatus*)) make up the whitebait fishery.

The immature forms are caught when the young swim back inland after being washed out to sea as larvae.

Galaxiids take their name from the little silvery gold spots, that many of them have on their bodies, which were thought to resemble stars in a galaxy. They have no scales and a dorsal fin that is positioned quite far back on their bodies. Galaxiids tend to be good climbers, particularly when they are small, and use their fins to hold onto rocks. They can climb waterfalls using surface adhesion as well as their pectoral and pelvic fins to wriggle up wet surfaces. They mainly eat insects.¹⁶



Figure 2.3 Banded kōkopu (Department of Conservation)

The most common of the Galaxiids in the whitebait catch is the īnanga. They live in shoals in open pool areas of rivers, growing to around 10 centimetres long, and living only for about a year. Mostly during the Autumn, large shoals of adults migrate downstream to the coast, spawning when spring tides flood the grasses that line estuary shores. The adults then die.¹⁷ The eggs remain in the grasses, and hatch when inundated one month later by the next spring tide. The young then swim out to sea. They remain in the sea over the winter and return as adults to a river the following spring.¹⁸

Another native New Zealand Galaxiid, the kōaro (sometimes known as ‘mountain trout’) are found in fast flowing mountain streams. These fish prefer streams with some native bush cover, possibly because they use the damp edges

of the streams to spawn. They are long and slender, and covered in a mottled gold and brown pattern. Koaro are generally diadromous, and travel between fresh and saltwater, but have been very successful at developing landlocked populations in some lakes such as Tarawera, Taupo, Manapouri and Wanaka.¹⁹

The banded kōkopu is covered in thin pale vertical bands and lives in small river tributaries, in areas where there is vegetation cover giving complete shade. They are common in the coastal areas of the west of the country. They like small streams and are even found in the urban areas of Auckland and Wellington.²⁰

Giant kōkopu mostly live in slow moving rivers close to the coastline. They are also found in landlocked inland lakes. Although most Galaxiids are very small, these fish can grow to nearly half a metre long, although specimens of around 200-300 millimetres are far more common. Giant kōkopu are found mostly in coastal areas, and prefer slow moving or still waters such as lakes or swamps. They tend to spend much of their time hidden beneath debris or overhanging vegetation, awaiting their insect prey.²¹

Other members of the Galaxiid family include the pencil galaxiids, which are pencil shaped fish with small fins. They live in the gravels of stream beds, feeding on aquatic insects from under stones. Mudfish are also galaxiids, with five species present in New Zealand. They live among debris in wetlands, and are able to spend some time lying inactive in the mud, if the water disappears.²²

Bullies

Bullies are the second largest group of native New Zealand fish with seven species.²³ They are mostly small, reaching about seven centimetres long. Four of the species have some individuals which spend their first few months at sea and then move into freshwater. Consequently, adult bullies can be found in a wide range of New Zealand's freshwater bodies including places accessible from the sea as well as landlocked water bodies. All migratory species are becoming rare with three species (redfin, bluegill and common bully) undergoing significant declines.

Lamprey (piharau)

The lamprey (*Geotria australis*) is another of New Zealand's native freshwater fish. Although they spend most of their lives at sea, they spawn in fresh water. They look similar to eels, but have a circular sucker armed with rows of horny teeth instead of a mouth.²⁴

Lamprey has an interesting life cycle. The first life stage is as larvae when the young have no eyes, nor mouth, but a hood that can be orientated upstream to filter feed. At about one year of age the larvae metamorphoses into the

adult form. The adults feed by attaching their sucker mouth to other animals and rasping holes in their flesh. Lampreys continue to be an important food source for Māori, and a traditional fishery still occurs on a small scale in the Whanganui River near Pipiriki and in other parts of Taranaki. They are also important to Southland Māori.²⁵

Torrentfish

Torrentfish (*Cheimarrichthys fosteri*), closely related to the blue cod, are small fish that live in the fast flowing rapids of stony rivers and streams. They have dark bands along their sides, a flattened head and large pectoral fins, which they use to anchor themselves to the riverbed. Torrentfish spend part of their lives at sea, and juveniles enter river systems in spring and autumn, gradually moving upstream. Torrentfish are not good climbers, and therefore only penetrate inland rivers where the gradient is relatively low. They are quite common in most parts of New Zealand, but rare in Otago and Southland.²⁶



Figure 2.4 Torrentfish (Stephen Moore)

Smelt

There are two species of smelt native to New Zealand. Smelt have distinctly forked tails, and an extra fin – a fleshy lobe between their dorsal fin and tail. They have scales that are easily dislodged, and give off a distinct cucumber smell. They live in the upper water layers and are frequently seen in rivers and lakes feeding on drifting food organisms. Common smelt (*Retropinna retropinna*) are found throughout New Zealand, whilst Stokells smelt (*Stokellia anisodon*) are found only in parts of the east coast of the South Island, not far from the sea.²⁷

Eels (tuna)

There are two types of native eel in New Zealand: the long fin eel (*Anguilla dieffenbachia*) and short fin eel (*Anguilla australis*). The long fin eel is one of the largest freshwater eels in the world and is found only in New Zealand. It is declining in abundance and is classified as threatened. A recent arrival, the Australian spotted eel (*Anguilla reinhardtii*), is found only in northern rivers. This eel was first spotted in New Zealand in 1997. It has a mottled or blotched back and yellow pectoral fins.²⁸



Figure 2.5 Long fin eel (Amber McEwan)

Eels live mainly in rivers and lakes but can be found in all types of New Zealand's waterways, usually well inland from the coast. Longfin eels prefer clean, inland waters while shortfin eels prefer lowland lakes and swamps. Eels are extremely good climbers and can climb waterfalls and dams by leaving the water and wriggling over damp areas. They are slow growing fish, beginning life at one millimetre long but growing to up to two metres in length over 60 years.²⁹

Long-fin eels have an unusual life cycle. The adults breed only once at the end of their life. In order to do so, they swim around five thousand kilometres to spawning grounds in the tropics, thought to be in deep ocean trenches near Tonga. The adult eels die after they have spawned and never return to New Zealand. The fertilised eggs develop into larvae which drift back to New Zealand on ocean currents. The larvae develop into small 'glass' eels (elvers) which swim up rivers and slowly grow into adults.³⁰

In earlier times, fishermen used to regularly catch huge eels weighing up to 40 kilograms. These would have been more than 100 years old. But since commercial eel fishing became popular in the 1960s numbers have declined. Today, although it is still possible to spot eels, there are very few large ones remaining.

Conservation status

There has been a significant decline in freshwater biodiversity since 1970, with the biggest decline occurring during the past decade. Catchments where the land cover is predominantly pasture, urban or exotic forest have significantly less native fish than catchments with native forest cover.³¹

More than 65 per cent of freshwater fish species in New Zealand have been identified as threatened (see Figure 2.2). This list of threatened fish species is currently being updated and is likely to expand. So it is somewhat surprising that most of these species are not classified as protected wildlife and some are still legally harvested including the longfin eel, giant kōkopu and shortjaw kōkopu.

Species	Conservation status
Grayling	Extinct
Lowland longjaw galaxias	Nationally critical
Canterbury mudfish	Nationally endangered
Northland mudfish	Nationally endangered
Possible new non-diadromous galaxias "Teviot"	Nationally vulnerable
Eldon's galaxias	Nationally vulnerable
Dune lakes galaxias	Nationally vulnerable
Dwarf īnanga	Serious decline
Longfin eel	Gradual decline
Possible new non-diadromous galaxias "southern sp"	Gradual decline
Roundhead galaxias	Gradual decline
Giant kōkopu	Gradual decline
Flathead galaxias	Gradual decline
Dwarf galaxias	Gradual decline
Gollum galaxias	Gradual decline
Bignose galaxias	Gradual decline
Upland longjaw galaxias	Gradual decline
Dusky galaxias	Gradual decline
Possible new non-diadromous galaxias "Northern sp"	Gradual decline
Brown mudfish	Gradual decline
Black mudfish	Gradual decline
Shortjaw kōkopu	Sparse
Lamprey	Sparse
Possible new non-diadromous galaxias "Von sp"	Range restricted
Chatham Island mudfish	Range restricted
Tarndale bully	Range restricted
Stokells smelt	Range restricted

Figure 2.6 Threatened species of freshwater fish

Source: Hitchmough R, L Bull and P Cromarty, 2007

Freshwater Birds

Over 160 species of birds have been recorded being present on New Zealand's rivers and estuaries and many of these are unique to New Zealand. They inhabit a wide variety of different freshwater habitats ranging from deep open water, to mountain torrents, to swampy vegetation. Five endemic species spend all or crucial parts of their life cycle on rivers and are particularly adapted to them – blue duck, wrybill, black stilt, black-billed gull and black-fronted tern. Many species of freshwater bird are threatened (see Figure 2.11) and a number have become extinct during human occupation, including a pelican, a swan and a little bittern.

Blue duck (whio whio)

The whio whio (*Hymenolaimus malacorrhynchos*) forages in the fast flowing, turbulent waters of high country streams. It mainly feeds on the larvae of caddisfly. The ducks have some unusual features which make them particularly adapted to their habitat. They have a streamlined head and large webbed feet which enable them to manoeuvre in fast moving water. The ducks have a thick, fleshy 'lip' that overlaps their lower bill and allows them to scrape off insect larvae from rocks without damaging their bills.³²



Figure 2.7 Male blue duck (H R Veitch)

The duck is endemic to New Zealand and has no close relative anywhere else in the world. Only around 640 pairs remain in the North Island and just under 700 pairs remain in the South Island, a total population of between 2,000 and 3,000 individuals. The species is classified by the Department of Conservation as nationally endangered. One of the major causes of decline in the population is predation. Habitat loss from forest clearance for agriculture and poor riparian management are also problems. In addition, changes to river flows from the operation of hydro-electric power stations and irrigation have made some rivers unsuitable for blue duck.³³

Wrybill (ngutu pare)

Wrybills (*Anarhynchus frontalis*) breed on braided rivers in Canterbury and inland Otago and then fly to Northland and harbours in the Auckland region during the summer. They are found in large numbers in the Firth of Thames. The wrybill is particularly adapted to feeding on braided rivers as it has a beak which curves to the right, allowing it to more easily catch insects hidden under river stones. It was this unique feature which led the birds to be collected for use as museum specimens in the late 1800s and early 1900s.³⁴

The wrybill population has been declining over the past few decades and there are only 4,000 to 5,000 left in the wild.³⁵ This decline is probably due to the loss of habitat in braided rivers caused by hydro-electric power development, extraction of water for irrigation, and the invasion of weeds into shingle riverbeds.³⁶ The Department of Conservation has classified the species as nationally vulnerable.



Figure 2.8 Male wrybill nesting (H R Veitch)

Black stilt (kakī)

Black stilts (*Himantopus novaezelandiae*) wade out into deep slow moving water and reach down to capture mayfly and caddisfly larvae from the river bottom. The species used to be widely spread around the entire South Island and lower North Island but it has declined drastically since the 1960s. Black stilt now only breed on the braided rivers and wetlands of the Mackenzie Basin.³⁷

By 1981 the population of black stilt had declined to a low of just 23 birds and the population has been identified as nationally critical. This drastic decline was a result of predation by feral cats and mustelids and loss of habitat through hydro-electric development, the drainage of wetlands and the spread of willows and lupins onto river beds. Because the population of black stilt reached such low levels, individuals have also started breeding with the more numerous pied stilt resulting in hybrid offspring. A captive breeding programme, commenced during the 1980s, has led to an increase in the population to around 200 birds.³⁸



Figure 2.9 Black stilt on its nest (H R Veitch)

Black-fronted tern (tarapirohe)

Black-fronted terns (*Sterna albobriata*) breed in shallow depressions on braided rivers on the eastern side of the South Island and in the Nelson region. They feed on rapidly flowing rivers, touching the surface of the water with their bill, to capture emergent mayflies and stoneflies. Black-fronted terns also eat earthworms and grass grub larvae from farmland. Most birds move to the coast during autumn and winter to feed on planktonic crustaceans in the sea.³⁹

There are only around 5000 black-fronted terns surviving and the species has been classified as nationally endangered by the Department of Conservation.



Figure 2.10 Black-fronted tern (B J Harcourt)

Community type	Bird species (Threatened species in bold)
<i>Open water divers</i> Dive for fish and invertebrates in open, deep water in lakes and rivers; nest in overhanging vegetation and rock outcrops	Cormorants (black , little, little black, pied), spotted shag, NZ scaup
<i>Deep water waders</i> Have medium to long legs and can forage in water deeper than 200mm; breed in open ground	Heron (white-faced, white), royal spoonbill , stilt (black , pied), Spur-winged plover, oystercatcher (South Island pied, variable), bar-tailed godwit, golden plover
<i>Shallow water waders</i> Short legs and feed in water less than 80mm deep; breed in open ground	Wrybill , dotterel (banded , New Zealand , black-fronted), knot, turnstone, red-knecked stint
<i>Dabbling waterfowl</i> Feed by dabbling while floating in open water or grazing in wetlands and pasture; nest in dense vegetation	Black swan, paradise shelduck, grey duck , NZ shoveler, teal (grey, brown)
<i>Torrent specialists</i> Specialise in foraging in fast flowing white water for aquatic insects	Blue duck
<i>Aerial gulls and terns</i> Dive for invertebrates and small fish; nest in open shingle bars and islands in rivers	Tern (Black-fronted , white-fronted , Caspian , white-winged black), gull (black-billed , red-billed, black-backed)
<i>Swamp specialists</i> Live in dense swamp vegetation eating seeds, invertebrates and for some species vegetation, fish and amphibians	Australasian bittern , crake (marsh , spotless), pukeko, banded rail, fernbird
<i>Riparian species</i> Use both terrestrial and aquatic habitats	NZ pipit, welcome swallow, NZ kingfisher, harrier hawk

Figure 2.11 Summary of freshwater bird communities

Source: O'Donnell, 2004

Freshwater communities

There are a wide variety of different freshwater wildlife communities in New Zealand. The following sections describe some of the communities associated with braided rivers, lakes and groundwater.

Braided river wildlife

As well as making them aesthetically dramatic, the multidimensional and dynamic structure of braided rivers makes them very important habitats for a variety of wildlife. The extended floodplains which typically surround the rivers create a myriad of localised environments, ranging from large unstable channels to well-established spring complexes, quiet side braids and more turbulent main braids. Together they create a mosaic of aquatic habitats for small organisms.

The instability of the floodplain environment means that organisms found there tend to have a high turnover. But the communities survive, because whilst a particular type of habitat may be destroyed in one part of the river, it will remain intact, or be forming, in others. This ensures that the relative proportion of each habitat in any particular floodplain remains roughly constant over time and allows mobile species to move to find suitable new habitat.

Braided rivers support diverse communities of plants, invertebrates, fish and birds. For example, pied and black stilts, banded dotterels, oystercatchers, plovers and some gulls all use braided rivers at some point during their life cycles.

Many bird species migrate to the braided rivers in late winter and early spring to coincide with increased river flows generated by snow melt. The greater volume of water inundates previously dry channels and this increases the habitat for aquatic invertebrates. This, in turn, increases the availability of food for the birds which enables them to recover from their migration and to breed and raise chicks successfully.⁴⁰

Braided rivers support a wide variety of fish species, including many of New Zealand's native galaxiids, a large proportion of which are endangered. Introduced species of trout and salmon can also be found there in large numbers.

Lake biota

In lakes, the area near the shore (littoral zone) is the most productive area and is usually rich with life. These are the shallow areas where light can penetrate down to the lake bed, and therefore where plants can grow. The plants provide important habitat for algae, invertebrates and fish.⁴¹

Most lake plants grow near the water's surface, either rooted near the shoreline or floating. Some plants however can survive at considerable depths – the deepest known plants are some species of bryophyte which have been found at depths of more than 70 metres in Lake Coleridge in Canterbury, where the water is particularly clear.

Many aquatic animals live on phytoplankton.⁴² These are tiny floating organisms such as algae and cyanobacteria that grow where there is sufficient light for them to carry out photosynthesis. The phytoplankton are eaten by zooplankton (such as water fleas and copepods) and juvenile forms of other animals. Invertebrates that prefer water with little movement live in lakes, such as freshwater crayfish and insects such as caddisflies stoneflies, midges mosquitos and pondskaters.

Many lakes contain a variety of native fish such as īnanga, kōaro and kōkopu. They also contain several introduced fish including 'sports fish' such as brown and rainbow trout as well as 'pest fish' such as koi carp (see chapter 6 for a description of these introduced species).

New Zealand lakes also support water fowl, including some found only around lakes. These include grey teal, grey ducks, scaup, New Zealand dabchick (on the North Island) and crested grebe (South Island). Introduced species include Canada geese, mallard ducks and black swans. There are also Australian coots and a number of species of shag.

Wetland biota

The particular environmental features created by a wetland give rise to a large number of species which rely on these characteristics for their survival. Wetlands are a harsh and stressful environment for plants. Therefore, those that grow there have had to adapt to living in the specific conditions that the wetland offers. Kahikatea and Pukatea, which are tall trees found throughout New Zealand, possess buttress roots and therefore are stable in wet soils. Many smaller plants have spongy tissues in their leaves that act as a reservoir and passageway for oxygen to move down to the roots in wet, oxygen-deprived soils. Some plants have underground storage organs that help them survive and regrow rapidly after the sudden floods and prolonged droughts common in wetlands.

Wetlands support a high diversity of smaller plants including 47 species of rush and 72 species of native sedge. Some endangered plant species depend totally on wetlands.⁴³ In the Waikato region, for example, the remaining wetlands make up only five per cent of the region but support almost a quarter (13) of the region's threatened plant species. Threatened wetland plants include

a swamp helmet orchid, a clubmoss, a hooded orchid and a carnivorous bladderwort.⁴⁴

Today wetlands cover less than two per cent of New Zealand's land area but support 22 per cent of native bird species (more types of native bird life than any other habitat in New Zealand).⁴⁵ For example, the Australasian bittern, brown teal, fernbird, marsh crake and white heron all rely for their continued survival on New Zealand's wetlands. In addition, 30 per cent of New Zealand's native freshwater fish are found in wetlands, including īnanga, shortfinned eels, kōkopu, bullies and mudfish.⁴⁶ Wetlands are also very important for a range of invertebrate species including freshwater crayfish.⁴⁷



Figure 2.12 Fernbird (Neil Deans) *The fernbird is one of the threatened species supported by wetlands*

Groundwater biota

Although on the face of it, groundwater aquifers might appear to constitute the type of environment that should be devoid of life, they actually support important ecosystems that play a significant role in the maintenance of healthy groundwater resources. The ecosystems are based on organic carbon brought in when surface waters recharge the aquifer.⁴⁸ The carbon gets bound into

thin layers of slime (biofilms) that form on all surfaces, whatever their size. Groundwater crustaceans, commonly called well shrimps, feed on the biofilms, keeping them in check and preventing them from clogging up the fine pore spaces in the aquifer, thereby maintaining water quality and flow.

In recent years, scientists have found that groundwater crustaceans are widespread throughout the groundwaters of New Zealand and they are very diverse. Many species lack scientific names and are endemic to particular areas of New Zealand. More are still unknown to science. Crustaceans living in groundwater are generally colourless and blind, and can be up to 25 millimetres long, although some are much smaller, so that they fit into the tiny crevices within aquifers.

Scientists have found that groundwater invertebrates become more abundant in areas close to a pollution source (for example, areas where land irrigation of waste water is used, causing more organic matter to runoff into the groundwater) in response to a corresponding increase in their biofilm food.⁴⁹ This abundance of invertebrates is able to keep the biofilms in check, maintaining water flow through the aquifer and protecting water quality.

However, the system cannot cope with endless amounts of pollution. When very large amounts of organic matter enter the aquifer, it overwhelms the invertebrates, and pores in the aquifer become clogged. High bacterial populations can use up all the available oxygen, killing the invertebrates. This results in reduced flow through the aquifer and low water quality. Introduction of other toxic materials to the aquifer, principally organic ions such as zinc, arsenic, nickel, cadmium and copper, can also affect these invertebrates adversely.

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Value of Freshwater

Introduction

New Zealand's freshwater bodies including rivers, lakes, wetlands and aquifers have enormous value for a wide range of reasons including intrinsic values, Māori cultural values, recreational activities, tourism, water supply, commercial fisheries and other ecological services. These values are described in more detail in the sections below.

Intrinsic values

The intrinsic values of freshwater bodies are largely based on their ability to sustain healthy aquatic ecological systems and the rare and noteworthy species that live in them. These aquatic ecological systems are linked to bordering terrestrial vegetation and the marine area because of the movement of species between these areas during their life cycles. Maintaining sufficient instream flows and water quality to sustain the full range of aquatic life within freshwater ecosystems is therefore important. Maintaining viable linkages between aquatic and terrestrial habitats is also important for the overall sustainability of the system.

As noted in Chapter 2, freshwater ecosystems are home to many endemic New Zealand species, whose continued survival relies on sensitive and appropriate freshwater management. The intrinsic value of these habitats and species also gives rise to some of the cultural and recreational value of wetlands, whether through providing both kai and cultural meaning for Māori, or through the underpinning of eco-tourism and nature-based recreation.

Māori cultural values

Water is a fundamental taonga for Māori. The Māori creation myth explains its existence – Ranginui and Papatuanuku were the sky father and earth mother. They lay locked in a tight embrace, and their children were forced to live in the cramped dark space between them. Having discussed what it would be like to live in the light, their children decided to separate them. While Ranginui remained in the sky Papatuanuku was pushed to the ground. Ranginui's tears for his lost wife are the rain that falls to earth. In time water returns to the heavens through evaporation – this is Papatuanuku's tears for Ranginui. Waterways are considered to be the arteries of Papatuanuku.¹

In Māori culture everything has a life force sustained by water, which is the source of life. Water has an inherent mauri, or life essence, and this mauri must be healthy to enable the water to sustain healthy ecosystems and support cultural uses. Consequently, Māori culture has a great respect for water. Traditionally when water was collected, prayers were given. Water is used for ceremonial and ritual purposes and in some areas rivers were used to demarcate boundaries between adjacent hapū.²

Specific water bodies have particular significance for iwi and hapū. Some rivers are home to taniwha, which appear in the form of eels or other freshwater species, and which are the embodiment of tipuna (ancestors) of the current generation. It is traditional at gatherings for Māori to present mihimihi (introductory speeches) which include whakapapa (ancestral ties) and links with specific geographical features including awa (rivers).³

Rivers, lakes and wetlands are an important location for mahinga kai (food gathering area). Fish and shellfish were a very important part of traditional Māori diet and participation in harvesting practices enables traditions to be maintained. The mana of a marae is increased if visitors are provided with food sourced from local mahinga kai. Eels in particular are a taonga, being important for hui, tangi and gift exchanges. Māori also harvested a wide range of other species.

Māori and freshwater

Māori regulated fresh water and freshwater fisheries through kaitiaki and tohunga, whanau and hapū as vital resources for pā and kāinga. What developed was a complex system of management including the use of rāhui or temporary restriction and tapu for more permanent bans on the use of a water resource. Many iwi have proverbs about the unity of people and water. For example in the Whanganui area there is a saying 'ko au te awa, ko te awa ko au' (I am the river and the river is me).

Water formed part of an undivided entity whereby iwi did not distinguish between lakes, lagoons, rivers, swamps, their associated beds or the adjoining land. Māori land was not restricted to the waters edge but extended to the bed of the water body.

Waters were also ranked. From the sacred puna wai to the water in common use, wai māori and those of very limited use such as wai kino. Water and identity are intimately linked. For Māori certain things are believed to have or could be imbued with a mauri, or life essence and water or water bodies were believed to have a mauri of its own. Water with a healthy mauri will sustain healthy ecosystems and support cultural uses.

Mahinga kai was pertinent to the survival of iwi. They depended upon the knowledge of mahinga kai and participation in gathering from mahinga kai was important for maintaining traditional practices. It was also important that water quantity and quality was maintained as they impacted on these cultural values and traditional knowledge and practices.

Quoted (with macrons inserted) from Wai Māori, 2008, Discussion on freshwater: A wai Māori perspective, <http://waimaori.maori.nz/innern/publications/resources/Discussion%20on%20Freshwater%20-%20A%20Wai%20Maori%20Perspective.pdf>

The strong cultural and spiritual links with the country's water bodies which Māori have developed over many generations, and their embodiment as ancestors, mean that there is a close bond between iwi, hapū and their local water bodies that has been hard for European sensibilities to fully understand. Thus historian Ann Parsonson has identified that 'for Waikato Tainui, one of the greatest impacts of the confiscations in respect of the river has been the removal of their capacity to protect the river in the decades of rapid change that followed. Their authority and their tikanga were ignored, as if they had not existed for hundreds of years. As mining, farming, sewage disposal and hydro electricity development all took their toll on the health of the river, Waikato Tainui were not consulted.'⁴

As well as being of cultural significance, water is of considerable economic value to Māori. Key sectors for Māori-owned business include agriculture, forestry and tourism which all rely on the freshwater resource.

Spiritual, historic and cultural values

Numerous settlements in New Zealand are located on the banks of rivers or lakes, and as a result, many New Zealanders have grown up closely associated with freshwater bodies. Family camping, fishing and bach holidays are often associated with rivers and lakes as is also the case for the coast. These natural systems can therefore form a strong component of what attaches people to a particular locality. Freshwater makes a significant contribution to 'sense of place'. Lakes and rivers are also strongly reflected in New Zealand art and literature, indicating their important role in national cultural life.



Figure 3.1 Camping holiday (Neil Deans) *Family camping holidays by lakes and rivers are a longstanding tradition in New Zealand*

Landscape, natural character and amenity values

Water bodies such as lakes, rivers and wetlands are important elements of natural landscapes in New Zealand and contribute significantly to landscape, natural character and amenity values. The presence of water, along with naturalness (lack of human artifacts) and relief (hills, mountains and cliffs), are consistently identified in New Zealand and international research as important elements of landscapes which are highly valued by a wide range of people.⁵

There have been several studies of the landscape values of rivers in New Zealand. Each river is valued by local residents for slightly different attributes. For example, in the Clutha district it is the dramatic relief, rocky outcrops and dynamic movement of the main rivers which are highly valued. In the Lower Waitaki it is the river as a landscape feature and as a location for recreational activities which is valued. In the Lees valley it is the broad open basin landscape with views, scenery, remoteness and indigenous vegetation which is important.⁶ Often it is the broader landscape of which the river is a part, including indigenous vegetation along the banks and vistas, which are more significant to people than the individual characteristics of the river itself.⁷

Many lakes also have very high landscape and natural character values. These are associated with elements such as their natural shape, movement of water, patterns and reflections on the water, juxtaposition with surrounding land forms, presence of wildlife and historical associations with the area. Lakes modified by human activity, such as the Taupo, Coleridge, Tekapo, Pukaki and Manapouri hydro lakes, can still retain very high levels of natural character.



Figure 3.2 Clarence River (Fish and Game New Zealand) *Many rivers have very high landscape, natural character and amenity values such as the Clarence River in South Marlborough shown here*

Tourism

New Zealand's clean, green image (which is in part due to its relatively clean rivers and lakes) attracts hundreds of thousands of tourists each year, many of whom come to take part in a multitude of water-based recreation activities. The tourism industry contributed over \$21 billion to the New Zealand economy during the year ended March 2009 and provided over 16 per cent of the country's total export earnings.⁸

The annual international tourism survey indicates that international tourists are involved in many water related activities. During 2009 over 165,000 international tourists participated in canoeing, kayaking and rafting, over 100,000 in fishing, over 121,000 in swimming, over 600,000 in boating and over 600,000 visited volcanic and geothermal attractions.⁹ Although these figures include activities undertaken in the sea as well as in lakes and rivers, they nevertheless indicate that water is a significant drawcard for the tourism industry.



Figure 3.3 Emerald Lake, Tongariro National Park (Raewyn Peart) *New Zealand's freshwater bodies are a major drawcard for international tourists, including the Emerald Lake on the very popular Tongariro crossing*

Water supply

It is clear that New Zealand's freshwater resources are extremely valuable to the New Zealand economy as providing water supply for a range of uses. However, the precise value is unknown. There have been several attempts to put a value on water supply, but these need to be treated with caution, as they do not encompass all values.

A study by the Institute of Geological and Nuclear Sciences provided estimates for the value of water in various uses, although the importance and reliability of their findings was constrained by significant limitations in the data available. The study found that the combined value for domestic use of the 100 surface water catchments of largest value and the 100 groundwater aquifers of largest value was \$499 million per year. The value of the water used for stock was \$9 million per year (on the basis of data from the 20 catchments of largest value), and water used for industrial purposes was valued at \$34,215 million per year.¹⁰ In addition, a separate study by Harris and Skilton (2007) estimated that the value of water used for irrigation in highly allocated regions is \$127 million per year.¹¹



Figure 3.4 Lower Nihotupu Dam, Waitakere Ranges (Raewyn Peart) *Five dams in the Waitakere Ranges supply around 26 per cent of Auckland's bulk water supply.*

Energy generation

Hydroelectric generation is crucially important to New Zealand contributing around 55 per cent of New Zealand's total electricity supply. The most important hydro lakes are Tekapo, Pukaki, Hawea, Manapouri and Te Anau in the South Island and Lake Taupo in the North Island. Geothermal energy generation contributes another six per cent of total electricity supply.

Commercial fisheries

There is a significant commercial fishery in longfin and shortfin eels. The eels are processed in New Zealand and then exported to markets in Europe and Asia. Catches peaked in 1975 when 2,434 tonnes were harvested and processed by 35 factories. They have declined significantly since then and now less than 700 tonnes of eel are commercially harvested each year. Only two processing plants now operate in the North Island (at Te Kauwhata and Levin) and there are two licensed fish receivers in the South Island. Eel exports totalled over \$6 million during the 2006-07 year.¹²

River mouths and estuaries are also important habitat for some commercially caught species. There is a well-established set net fishery for yellow-eyed mullet and flatfish (which includes black flounder). Yellow-eyed mullet is a marine fish but sometimes moves upstream into low-lying rivers and brackish lakes.

The black flounder is the only flatfish species in the New Zealand which spends most of its life in freshwater. Little is known about the life cycle of this endemic species. Black flounder larvae are found in the sea and then migrate into river estuaries to grow. Individuals have been found up to 100 kilometres inland where the river gradient is low.¹³ Black flounder is mainly harvested from lakes Wairarapa, Onoke (Ferry) and Ellesmere.

Introduced species of catfish and koi are harvested commercially, mainly as a bycatch of the eel fishery. Some freshwater fish are harvested for the aquarium trade including bullies and torrentfish. Freshwater shrimps are also sold for use in aquariums and as food for large ornamental fish. Some freshwater invertebrates are harvested and sold to pet shops to feed aquarium fish.¹⁴

The introduced species silver and grass carp are used for the biological control of unwanted algae and aquatic plants. Grass carp farming is producing stock for weed control and also for the restaurant trade. Watercress is harvested as an important food source and some is sold in specialised food outlets. A few large facilities area also cultivate water cress for human consumption.¹⁵

There is a sizeable industry in farming freshwater fish. Several fish farms are licensed to farm species comprising whitebait. They are taken from the wild as juveniles, and in one case raised to adults and sold to pet shops or other fish farms. Goldfish farming is also a significant activity. Koura farming is now established and supplies up-market restaurants. There is a large prawn farm at Wairakei.¹⁶ Several salmon farm operations are located in the Waitaki power scheme canals.

Recreational activities

New Zealanders spend a significant amount of time enjoying the freshwater resources that the country has to offer, so that freshwater resources may offer values to New Zealanders of health and wellbeing that are not easily quantifiable. In a 2004 survey, 79 per cent of New Zealanders identified themselves as recreational users of freshwater,¹⁷ and a wide variety of recreational activities involving freshwater are pursued across the country, including jet boating, fishing, swimming and bird watching.

In 2004 the Ministry for the Environment identified 105 water bodies of potential national importance for recreational activities, including 54 rivers, 30 lakes and 21 wetlands. The study found that, in particular, these lakes and rivers were important for fishing, whitebaiting, walking and tramping, sightseeing, picnicking, canoeing, wakeboarding, swimming, jet skiing, diving, yachting, bird watching, waka ama paddling, rowing, windsurfing, and jet boating. Wetlands were popular with the public for walkers, bird watchers and environmental education.¹⁸ In 2007/2008 the most popular recreational activities in New Zealand were fishing and swimming. A national survey found that in any one month, 197,671 people swam 'outdoors, in or by a beach, river, lake or the sea.'¹⁹

Fishing

Over 12 months (2007-2008) 5.7 per cent of New Zealanders participated in freshwater fishing, making it 21st equal in rankings of most popular sports during 2007/2008.²⁰ New Zealanders fish both for sport and for food. Sport fishing is mainly focussed on trout and salmon, which were introduced to New Zealand by early European settlers. Specifically, brown trout (*Salmo trutta*) from England and rainbow trout (*Oncorhynchus mykiss*) and salmon from North America were released. At first the trout thrived, but once they had eaten the small native species their numbers began to decline. Restocking and the

introduction of smelt to the fisheries have restored stocks, and tourists come from all over the world to fish for them.

Rivers in the Nelson region such as the Wangapeka, the Buller and the Grey, amongst many others, are all popular sites for tourists and locals hoping to catch a trophy trout. The increasing availability of private helicopter flights means that more and more rivers are accessible.²¹ Salmon fishing in New Zealand is based mainly on the east coast rivers of the South Island, such as the Rakaia, Rangitata and Waitaki as well as the Paringa, Taramaka and Hokitika in the west. The value of recreational fishing on a popular river can exceed \$2 million a year.²²

Salmon in New Zealand are mainly Chinook (also known as Quinnat or King) salmon (*Onchorhynchus tshawytscha*). They are fished as they travel upstream to their spawning grounds from the sea, with the season running from December to March, peaking in February. Atlantic Salmon (*Salmo salar*) have only become established in the Te Anau catchment.



Figure 3.5 Freshwater angler (Neil Deans) *Recreational fishing in rivers and lakes is a very popular pastime in New Zealand*

Boating

Boating of all types is popular on New Zealand's waterways, from the tourist attraction of the steamship on Lake Wakatipu to sailing on Lake Taupo. New Zealand has a number of rowing facilities at places such as Lake Karapiro and

Lake Ruataniwha. Canoeing and kayaking are also popular. Whitewater New Zealand (formerly the New Zealand Recreational Canoeing Association) has 16 member clubs across New Zealand corresponding to over 800 active members, and there are many more who participate in canoe racing, dragon boating, whitewater rafting and waka ama paddling.



Figure 3.6 Whitewater kayaker at the Buller Festival, Murchison (Fish and Game New Zealand)
Whitewater kayaking is a very popular sport with Whitewater New Zealand having over 800 active members



Figure 3.7 Recreational boats, Lake Taupo (Raewyn Peart) Boating is a favoured sport on many freshwater bodies

Jet boating is an activity which is particularly popular in New Zealand, at least partly because it is a Kiwi invention. The modern jet boat was invented in the 1950s by Bill Hamilton, who wanted to build a boat that could negotiate Canterbury's shallow braided streams. Jet boating is now practised throughout New Zealand. Jet boats' manoeuvrability and ability to go at high speeds over very shallow water allows people to access remote rivers that would otherwise be inaccessible. It has also become a popular tourist attraction. For example in Queenstown there are a number of operators offering a range of jet boating activities on rivers such as the Kawarau and Shotover, from short adventure rides to longer scenic trips. Jet boating clubs exist all over the country and it is estimated that there are around 6000 jet boaters in New Zealand.

Ecosystem services

As well as the more familiar provisioning and cultural services set out above, freshwater systems provide a wide range of very valuable environmental regulating services including the assimilation and metabolism of waste and the regulation of natural hazards.

Economists at Massey University have attempted to value the ecosystem services provided by New Zealand's biodiversity. They derived a value of \$44 billion for terrestrial biodiversity for the 1994 year. When comparing the value per hectare for different types of ecosystems, estuaries came out on top, closely followed by wetlands and then rivers and lakes. This indicates that New Zealand's freshwater systems are extremely important in providing the country's ecosystem services.²³

Wetlands are particularly significant in this regard. The ability of wetlands to absorb and contain large amounts of water can be extremely important for flood control, as they can absorb heavy rainfall, releasing it only gradually over time so that flooding is reduced.²⁴ They also stabilise downstream water flow and groundwater levels in periods of low rainfall, by holding water, and ensuring that a steady flow is released.

Wetlands act as filters, cleaning water before it enters the downstream environment. As water enters a wetland, its rate of flow decreases, allowing particles to settle out. Plant surfaces provide for filtration, add oxygen to the water, and absorb solids. Nutrients can be reduced. Accordingly, when wetlands are drained or filled in, or where stopbanks are built alongside adjacent rivers, a catchment will lose some of its natural ability to cope with unusual patterns of precipitation and pollution.

Terrestrial ecosystem type	Value \$NZ/hectare (direct and indirect)
Estuarine	39,980
Wetlands	34,163
Rivers	14,911
Lakes	14,904
Horticulture and cropping	5,558
Mangroves (only partial value)	3,526
Agriculture	1,029
Forest	930
Intermediate agriculture-forest	693
Forest-scrub	610
Native scrub	547
Intermediate agriculture-scrub	411

Figure 3.8 Value per hectare of New Zealand ecosystems (1994)

Source: Patterson M and A Cole, 1999, 51

Wetlands that are peat forming can also act as significant sinks for atmospheric carbon, with some bogs sequestering significant volumes of carbon in wet peats. It has been estimated that wetlands may store as much as 40 per cent of global terrestrial carbon.²⁵ These carbon stores can be vulnerable to release, particularly where wetland management results in highly fluctuating water levels, or a permanent reduction in water levels. Conversely, well managed and stable peat forming wetlands can remain a store of carbon for extremely long periods of time.

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Aalbert Rebergen

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Freshwater Pollution

Introduction

Water quality is highly variable throughout New Zealand. Although many alpine rivers and lakes have very high water quality, pollution problems in lowland waterways are real, and in many cases are becoming more serious. Over the past decade, river water quality has significantly deteriorated in the lowland areas of Northland, Auckland, Waikato, the east coast of the North Island, Taranaki, Manawatu-Wanganui, Canterbury and Southland regions.¹ It is also thought that 30 per cent of New Zealand's lakes, primarily those in North Island lowland areas, have very poor to extremely poor water quality.² Key pollutants include nutrients, bacteria, sediment and organic matter.

Nutrients

Nutrient enrichment is becoming a serious problem for New Zealand waterways, particularly due to growing urban areas and the economy's reliance on agriculture, which generates large amounts of nutrient pollution. The development of better systems for waste treatment, such as the application of effluent to land instead of directly into waterways, has helped reduce the impact of point source pollution on waterways. However, levels of phosphorus and nitrogen that run off land from diffuse urban and agricultural sources have generally been on the increase.

An excess of nutrients in water can cause rapid growth of plants and algae. In a healthy ecosystem, the natural nutrient levels regulate the amount of plant growth that can occur. When excess nutrients enter the water, this can upset the balance. This may cause exotic weeds to take over, or it may lead to eutrophication. This process occurs when plants and algae grow rapidly, taking over the water body. When they die, bacteria decompose the plant material, using up the available oxygen in the water during the process. This causes the deaths of fish, invertebrates and plant life which are reliant on water-borne oxygen. In the most severe cases, a waterway may become anoxic – seriously lacking in oxygen. This has happened seasonally to parts of some lakes in the Rotorua area.³

Another effect of eutrophication can be blooms of some species of algae which produce toxins. This can be fatal to aquatic creatures and render water unfit for consumption or contact recreation. Avian botulism is similar, derived from bacteria that need anoxic conditions. It is a sporadic killer of birds in lakes during New Zealand summers.

Lakes

Although eutrophication can occur in all types of water bodies, lakes are particularly vulnerable. This is because, unlike rivers where water is constantly replaced, lake water is held in a basin for many years and is only slowly renewed. This allows nutrient pollution to remain in one area for long periods of time.

Lakes at high altitude and in unmodified catchments generally have high water quality. But areas where the catchment is predominantly in pasture are associated with the worst lake water quality. The water quality of many lakes is thought to have deteriorated over the past 20 years, particularly in shallow, lowland, North Island lakes. For example, the eight lakes monitored by Environment Waikato are all highly to extremely nutrient enriched. However, the quality of some lakes has improved where catchment restoration work has been undertaken.⁴

Lakes with some of the lowest water quality include Lakes Hakanoa and Mangakawhere in the Waikato, and Ellesmere and Forsyth in Canterbury. These tend to be small, shallow lakes, which due to their smaller volume, respond more rapidly to changes in the catchment.⁵



4.1 Lake Forsyth, Banks Peninsula (Raewyn Peart) *Lakes are particularly vulnerable to eutrophication, especially shallow lowland lakes such as Lake Forsyth shown here*

Rivers

Many rivers are also affected by excess nutrients, particularly in lowland areas. In an analysis of river water quality monitoring data for the period 1998 to 2007, over 50 per cent of sites in the Auckland, Waikato, Canterbury and Southland regions were found to have median total nitrogen levels that exceeded the New Zealand guideline value. In addition, over 50 per cent of the sites in Northland, Auckland, Waikato and Southland had median total phosphorus levels that exceeded the guideline value.⁶ Rivers in the Waikato

region showed the greatest increase in nutrient loadings, with 45 per cent of monitored sites showing increased total nitrogen concentrations and 21 per cent of sites showing increased total phosphorus.⁷ The most nutrient affected rivers are scattered around the country, and include the Maitai (Southland) Waingongoro (Taranaki), Waihou (Waikato) and the lower Manawatu.



4.2 Waihou River, Thames (Raewyn Peart) *Many lowland rivers are recipients of excess nutrients, with the Waihou River being one of the most affected*

Groundwater

Groundwater is also susceptible to pollution by nutrients. Indeed, the 2007 state of the environment report found that 39 per cent of the groundwaters in New Zealand that were monitored had higher than normal nitrate levels. This is of particular concern when the aquifer is used as a source of drinking water. For example, nitrogen found in water in the form of nitrate, can cause blood disease in babies.⁸

Once groundwater is contaminated it is very difficult to clean up. Contamination may occur over a long period of time, and therefore may not be noticed until it is too late. In addition, nitrates that leach into groundwater may eventually find their way into surface water downstream. This can have important implications for freshwater management, as pollution may not appear in the surface water until a long time after the activity that originally caused the pollution has occurred.

Consequently, a rigorous groundwater monitoring programme is important. There are almost 1,100 sites for monitoring groundwater quality in New Zealand focused on aquifers which are an important source of supply or which are considered to be at particular risk of pollution. However, not all catchments are covered. Testing groundwater for nutrient pollution can be a good indicator of the general health of the groundwater, suggesting whether other pollutants such as faecal pathogens and pesticides are likely to be present.⁹



4.3 Unnamed stream near Omarama in the Mackenzie Country immediately upstream (top photograph) and downstream (bottom photograph) of dairy farms (Fish and Game New Zealand)
Agricultural activities can result in the pollution of freshwater bodies

Drinking water

A drinking water standard has been established as a guide for water quality in relation to human health.¹⁰ However, this should not be mistaken for an environmental standard which would more properly establish allowable nitrogen levels in receiving waters and then work back to acceptable concentrations in groundwater bodies which flow into them.

The 2005 New Zealand drinking water standard of 11.3 milligrams of nitrate per litre is exceeded at nearly five per cent of groundwater test sites, which are commonly shallow, unconfined aquifers. It is unknown how many of these aquifers are used for drinking water.¹¹ Nitrate concentrations are increasing in some parts of the country, such as Canterbury (probably due to intensification of farming) and decreasing in others. Aquifers containing nitrate concentrations that breach health standards are found in most regions, but they are most common in the Waikato and Manawātū, probably because of the prevalence of agricultural activity and market gardening over shallow water tables in these regions.¹²

Pathogens

Animal and human faecal matter contain large numbers of bacteria which grow in the gut. Although many are harmless, some of these can cause serious health problems in humans, if they reach water used for consumption or recreation. This can occur as a result of poor farming practices or inefficient waste management systems. For example, *Campylobacter* is the most commonly reported source of food and waterborne illness in New Zealand. Typhoid is easily transmitted via water contaminated by faeces. *Giardia* and *Cryptosporidium* are parasites which are found in the gut of humans and animals and are also spread via faeces. They both survive for long periods in freshwater bodies and are widespread around New Zealand.

The most common, and easily measured, bacteria found in faecal matter are faecal coliforms, of which *E. coli* is one species. The commonly used Colilert test measures thermo-tolerant coliforms which primarily originate in the intestines of warm-blooded animals. The results are often presented as *E. coli* counts.

Viruses can also be found in human and animal faecal matter. These include the enteric viruses which cause illnesses such as polio, meningitis and hepatitis. Aquatic viruses can cause gastrointestinal and respiratory illnesses.



4.4 Swimming water quality sign, Banks Peninsula (Raewyn Peart) *Many councils regularly test freshwater swimming spots for *E. coli* and most have low levels of contamination*

Over the summer of 2009-2010, 57 per cent of the freshwater swimming spots tested by regional councils were deemed safe to swim, as low levels of bacteria were found in at least 95 per cent of the samples taken there. However, 11 per cent of the swimming spots tested frequently had high levels of *E. coli* bacteria, meaning they were generally unsuitable for swimming. The data available suggests relatively stable water quality levels over the past six years.¹³ There has been deterioration in some sites, such as Coe's Ford in the lower Selwyn River near Christchurch, which is no longer suitable for swimming.

Faecal bacteria are also a problem in groundwater. Twenty-three per cent of around 700 monitored sites do not comply with the New Zealand drinking water standard. These aquifers are mostly found in Auckland, Otago and Taranaki. Bacteria counts are highest in shallow, unconfined aquifers.¹⁴

Organic matter

Agricultural runoff, urban wastewater and industrial waste can transfer organic material to water bodies within a catchment. This can include faeces from humans and animals; proteins, vegetables and sugars from food preparation; and cleaning soaps. When organic matter enters a healthy aquatic environment where there is plenty of oxygen dissolved in the water, naturally occurring aerobic (oxygen using) bacteria will eat the organic material, using up oxygen in the water to do so. Consequently their numbers grow beyond what you would normally see in a healthy aquatic ecosystem.

Freshwater bodies undergo a natural process of re-aeration, where oxygen from the atmosphere diffuses into the water, and this process guards against excessive oxygen depletion. But in systems which are overloaded, oxygen depletion can still occur despite this natural replenishment system. Oxygen depletion can have very serious effects on the health of the ecosystem, as fish and plants need oxygen levels to be maintained in order to survive. Some animals need relatively high levels of dissolved oxygen and seasonal depletion can be enough to remove entire populations. If all the oxygen is used up, anaerobic bacteria (non-oxygen-using bacteria) will take over, decomposing the waste material and producing gases such as methane, hydrogen sulphide and carbon dioxide.¹⁵

The potential harm caused by organic waste can be measured by its potential to remove oxygen from the water. There is a standard measure known as the biochemical oxygen demand (BOD) which is the amount of dissolved oxygen needed by aerobic bacteria to consume the waste (over a 5-day period where the water temperature is 20 degrees centigrade, measured in terms of the number of grams of oxygen consumed per cubic metre). Thus, if waste deposited in the catchment has a BOD loading which is too high, too much oxygen will be removed from the water by the aerobic bacteria thereby jeopardising plant and animal life.

Over the period from 1989 to 2002 total BOD has decreased steadily in rivers across the country. About half of the 77 monitored sites have shown significant decreases in organic waste and none have become more polluted by organic waste.

Mataura River

The Mataura River in Southland is an example of one river that has improved as a result of a reduction in point source discharges. In 1975 15.5 tonnes of organic waste from a meatworks were discharged into the river every day. Improvements to the factory's effluent treatment system meant that by 2000 the factory was discharging just three tonnes per day. Accordingly, the river has shown improvements in health, such as less surface scum and foam. Unfortunately, the river still has elevated nutrient and bacteria levels as a result of non-point discharges.

Source: Ministry for the Environment, 2007, 293

The Ministry for the Environment suggests that ‘the improved water quality is probably the result of better management of point sources of pollution, such as dairyshed and factory wastewater discharges.’¹⁶ Investment in upgrading treatment facilities has therefore had a tangible impact on the health of New Zealand’s freshwater bodies.

Sedimentation

An increase in the rate that sediment is transported off land within a catchment can bring about marked changes in freshwater ecosystems. The rate of sedimentation of water bodies can be affected by natural factors, such as the type of rock found on the river bed and in the surrounding area. Some types of rock (such as sandstone, mudstone and gravel) are easily eroded, resulting in high suspended sediment loads. However, excess sediment loads are commonly the result of human activity. Mining was once a serious source of sediment, indeed rivers were managed as places for disposing of waste, but better management has reduced this problem.

The type and extent of vegetation cover over the soil has a major impact on sedimentation rates. Sedimentation rates are typically lowest where land is under mature indigenous forest cover and increase significantly when the forest is removed and replaced with pasture. Sedimentation rates often increase again during the transformation of rural catchments into urban areas. The removal of vegetation cover and undertaking of earthworks during the construction stage of urban development can release very large amounts of sediment. However, these rates typically reduce when urban areas become established and soil is covered with paved areas, established gardens and lawns.

An excess of sediment can affect water depths, water coverage, the type of sediment found on the floor (for example the size of the particles) and the clarity of the water. When an excess of sediment enters the catchment it can be suspended in the water, reducing light by affecting water clarity and turbidity (the amount of suspended material in the water). Such environmental changes can bring about changes in vegetation, be detrimental to fish habitats, degrade spawning areas and deplete invertebrate populations.

Many fish native to New Zealand are adapted to spend much of their lives nestled in the rocky beds of rivers and other water bodies. Increased sedimentation fills in the gaps between the rocks, reducing their habitat. Many native species are also strongly affected by reductions in water clarity. Organisms living on the river or lake bed can become smothered by the sediment as it settles out (see Figure 4.1).

The visual clarity of New Zealand's rivers varies widely but has been gradually improving.¹⁷ Rivers with very high levels of sediment, where visibility is sometimes as low as 10–40 centimetres, include the lower Manawātū, the Waitara in Taranaki and the Waipaoa in Gisborne district. On the other hand, rivers with the best visual clarity (such as the upper Motueka, Clutha and Monowai in the South Island high country) have visibility of up to 10 metres.¹⁸

The Australia and New Zealand Environment Conservation Council guidelines (2000) state that clarity of less than 70 centimetres, averaged for upland and lowland rivers, will not meet the requirements for ecosystem protection. The recommended minimum for human recreation is 1.6 metres, so clearly at times some New Zealand rivers have been below the recommended guideline.

Another impact of the increased sedimentation of rivers is the increased flood risk as river beds rise. For example, the Manawātū River flooded in 2004 displacing 2,300 people, destroying 4 bridges and seriously damaging 21 others.¹⁹ At the peak of the 2004 flood, the section of the river passing through Palmerston North was shifting 28 tonnes of sediment every second. Analysis of the sediment indicated that about a quarter of it was valuable topsoil, thought to have washed off from steep pastoral land.²⁰

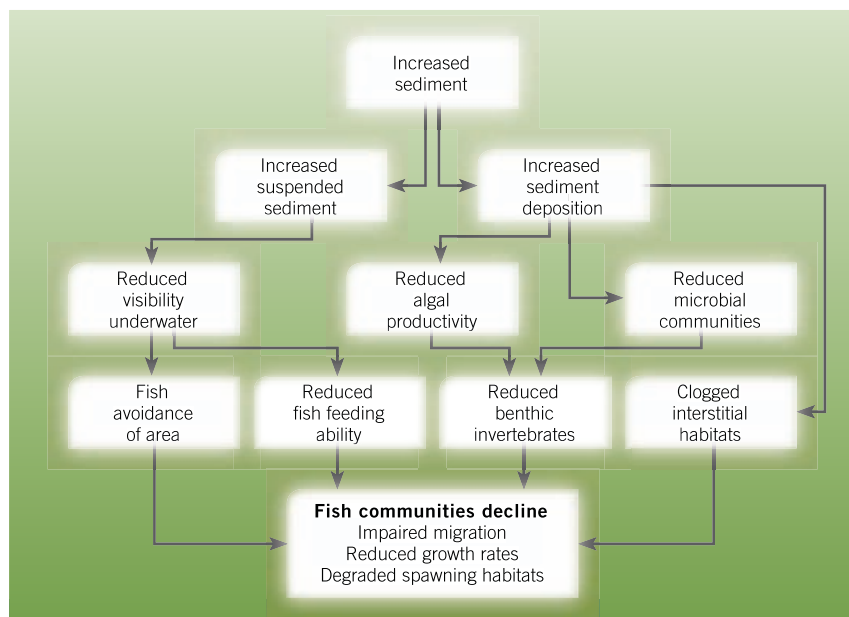


Figure 4.5 Impacts of sediment on stream ecosystems

(Source: adapted from Harding J and I Boothroyd, 2004, 36.5)

Other pollutants

A wide variety of other pollutants can find their way into freshwater ecosystems, produced by industrial processes, mining and other human activities.

Arsenic is used in a variety of industries such as timber treatment, herbicides, insecticides, mining, smelting, and pulp and paper production. Cyanide is used in some mining processes, which may also discharge a cocktail of other heavy metals. Mine drainage can lead to pollutants being discharged where mines have exposed mineralised deposits or mine waste has been left in contact with water. Such drainage can continue for many years from sites of historic mines. Run-off from roads carries pollutants such as zinc, copper, lead and hydrocarbons. Pesticides, including herbicides, fungicides and insecticides can pollute groundwater, and render it potentially harmful to humans and stock. Of the 163 groundwater bodies sampled in 2006, 19 per cent had detectable traces of pesticides.²¹ Pharmaceuticals, detergents and other household chemicals can enter freshwater bodies through effluent from wastewater treatment systems.

Saltwater intrusion

Saltwater intrusion is another issue that can be a problem for freshwater resources. It is not common in New Zealand but it is a serious resource management issue for councils in some areas of Northland, Waikato, Manawatū, Tasman, Lower Hutt and Canterbury. Saltwater intrusion into low lying coastal aquifers can occur naturally, but it can also be caused by the extraction of too much water from the aquifer.

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Fish and Game New Zealand

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Freshwater Abstraction

Introduction

On an international scale, New Zealand has an abundance of fresh water, and in 2008 was ranked 10th out of 173 countries for water availability on a per capita basis.¹ However, demand for water per capita is two or three times higher than in other OECD countries and is growing. The volume of water allocated for activities such as irrigation, domestic use and manufacturing increased by around 50 per cent between 1999 and 2006.

Although New Zealand has a relatively large amount of water, it is not evenly distributed, and resource limits are being reached in some parts of the country. This can result in poor water quality, water shortages and water-constrained economic opportunities. It can also result in the loss of in-stream values. This represents an important management challenge for councils and central government.

In 2006 almost 20,000 consumptive resource consents to take water were in place in New Zealand. Irrigation accounted for 78 per cent of these, 11 per cent were for manufacturing processes, nine per cent for public water supply and two per cent for stock watering. However, it should be borne in mind that the vast majority of water takes for stock consumption are not the subject of resource consents, as they fall under the permitted use category in the relevant regional plan. Sixty-six per cent of the resource consents related to groundwater takes, although the volume of water taken from surface water sources was higher. Added together, the total allocation of water in 2006 was 676 cubic metres every second, which is the equivalent of twice the average flow rate of the Waikato River.²

In some areas, freshwater is fully or over-allocated. A study undertaken in 2000 identified 134 fully or over-allocated resources of which 112 were rivers or streams.³ However, consent holders often do not take the total amount of water they are allocated or extract the water at the same time. The proportions of allocated water actually taken are highly variable. Demand for water changes according to various factors, for example the time of year, crop types, growth stage of the crop and weather conditions.

Groundwater

Water that is taken from groundwater reserves can be replaced by rainwater, but if too much is taken, the level of water in the aquifer can permanently decrease. This is known as lowering the water table. Although wells inevitably lower the

water table to some extent, and this is necessary if water is to run into the well, they often do so only on a seasonal basis.

Where the water table is permanently lowered, this may have effects on surrounding streams and wetlands, as many may rely on groundwater flow particularly in the dry season when there is not enough surface water available to support them. Land may subside and in coastal areas, salt water may infiltrate fresh water as the water table drops. In addition, users with shallower wells may lose their supply.

It is often hard to establish whether the water table has in fact been permanently lowered. Although the water level may be lower during a period of successive dry years, it may recover in subsequent wetter years.

Rivers

Abstraction of large amounts of water from a water body can have significant effects on its ecological health. When large amounts of water are removed from a river, not only does the level of flow decrease, but the natural variability of the flow is affected. This is because abstraction generally occurs at times when the flow is higher. These changes can destroy the river's natural character, as the



Figure 5.1 Opouri River during low flow (Fish and Game New Zealand) *Extraction of water from rivers and streams can result in a lack of water during low flow periods. Here, the irrigators are still in operation despite the river drying up.*

varying flow is part of what has caused the river to develop the characteristics that it has. Unusually high flood flows that occur during periods of heavy rain can be important for keeping the river downstream clean, reducing nuisance plant growth, clearing away weeds and helping maintain its braided nature.⁴

Reducing flow variability and flow levels can impact on fish, invertebrates and bird life. It can also affect the recreational values of a river. For example, kayakers and boaters require a minimum water level in order to undertake their activity.

Some rivers are particularly susceptible to damage from abstraction. Decreasing the flow in a braided river may eliminate some side braids altogether, as the reduced flow cannot sustain more than the main channel. This results in a decline in available habitat for braided river species – for example the dwarf galaxias, a rare native fish, tends to live in these side braids. Many invertebrates live there and they can be an ideal foraging habitat for birds such as the rare black fronted tern.⁵

Dams are often used to store water for abstraction. Their construction in a previously natural environment can have serious adverse environmental effects. Dams do not just prevent water from moving down a river, but also gravel and silt that would normally be carried along by the water. Sediment that would naturally be transferred downstream can thus build up behind the dam. This can exacerbate downstream and coastal erosion as gravel and silt that would normally replace eroded areas is not available.

Water exiting the dam may flow at a greater rate than would have been the case under natural conditions. This can cause scouring and erosion downstream as the water flows by with more force. Erosion of the river bed and riparian areas results in the loss of habitat for many species, for example native fish that normally inhabit the crevasses between pebbles on the river bed. In addition, dam structures can prevent fish passage along the water body. This problem can be mitigated if effective fish passes are used to allow the fish through the structure.

Geothermal

Abstraction of geothermal fluid for energy generation can result in a reduction in the temperature and pressure in geothermal fields. This in turn can lead to a loss of geothermal features such as geysers as well as result in ground subsidence.

Agriculture

Irrigation is increasingly used as a means to expand and intensify agricultural activity in New Zealand.⁶ In 2006, the total consented irrigated land was just over 970,000 hectares, with 80 per cent of this being in Canterbury and Otago.⁷ The majority of the water for which the consents were allocated is taken in Canterbury and Otago, which account for 55 per cent and 18 per cent respectively.



5.2 Rural irrigation (Fish and Game New Zealand) Irrigation is increasingly being used as a means to expand and intensify agricultural activity

The growth of irrigation in certain areas of the country is due to a desire to sustain and intensify farming enterprises in areas which are relatively water-poor. Dairy farming also tends to use a lot of water compared to other types of farming – in a Taupo study, dairy farming was found to use on average 83,000 litres per day whereas drystock farms used 14,000 litres per day. This meant that dairy farms were using six times more water per hectare than drystock farms.⁸ Despite this, farmers in water poor areas such as Canterbury and Otago, remain keen to intensify their farms because of the increased financial returns.

The east coast of the South Island receives relatively little rainfall, because the location of the Southern Alps intercepts the prevailing weather, causing precipitation to fall before it reaches the region. Between 2001 and 2008 there was a decline in refills of deep inland aquifers in Canterbury, although higher than average rainfall in 2009 broke this decline.⁹ Under low flow conditions,

peak demand which is mainly for irrigation, cannot either be met by current abstraction methods or sustain the current methods of use. Methods to store surface water from lakes and rivers need to be found if the resource managers are to keep up with demand. However, suitable places to store water are few and far between, as they generally require the flooding of land. This is often land which has other important uses and environmental values.

Central Plains Water

The tensions between the desire to provide for growth in the region and the desire to protect its dwindling natural water resources are illustrated by the ten year debate over a project to irrigate part of the Canterbury plains. In 2001 Central Plains Water (CPW) lodged an application to establish an irrigation scheme on the Canterbury plains between the Waimakariri and Rakaia rivers. The plan was to build a 55-metre high dam across the mouth of the Waianiwhi Valley in the Malvern Hills, flooding the valley. The dam would store 280 million cubic metres of water, which would be extracted from the Rakaia and Waimakariri rivers at a rate of 40 cubic metres per second, and distributed from the lake via a 10 kilometre long tunnel and extensive network of distribution canals.

The project was intended to double the land suitable for dairying from 22,000 hectares to 46,000 hectares, irrigating 60,000 hectares of land. In November 2005, CPW was granted requiring authority status by the Minister for the Environment, enabling the Trust to apply to local councils for a notice of requirement to designate the land required for the scheme.

There was significant opposition from environmental and recreational groups, and local people, some of whom faced the prospect of their homes being compulsorily acquired and submerged beneath the reservoir. Concern was expressed about the unsustainability of taking large amounts of water from the braided rivers, which are extremely sensitive and unique natural environments.

Many people argued that flooding an area of fertile farmland in order to irrigate the plains which are naturally dry, poor farmland was not a sustainable approach to environmental management. In addition, concern was expressed about the environmental effects of creating additional dairy land in an area short of water – more dairy farms could mean more nutrients being leached into the depleted groundwater, significantly worsening water quality.

Ultimately the hearing committee rejected the proposals to create the dam and the tunnel, but accepted a scaled down version of the project, in which water will be extracted from the Rakaia and Waimakariri rivers at a lesser rate, and conveyed to the plains via canals. The approvals are currently subject to appeal.

Subsequent to this decision, the Environment Canterbury councilors were removed and replaced with commissioners who were given stronger legislative powers to manage water issues in Canterbury.

Source: Environment Canterbury 'Central Plains Water Scheme notified consent'

<http://www.ecan.govt.nz/GET-INVOLVED/CONSENT-PROJECTS/CENTRAL-PLAINS-WATER/Pages/Default.aspx>



Figure 5.3 Waimakariri River (Damon McKinley) The Waimakariri is one of the rivers affected by the Central Plains Water scheme

Industry

Large volumes of water are required for a range of industrial processes in New Zealand, at various stages of the industrial process. Water is used variously for cooling, heating, cleaning machinery and manufactured products, and as part of the manufacturing process itself. For example, fresh produce must be washed in potable water that has been treated to comply with domestic and international food standards.

Water is used in many industrial processes as a cooling agent. Cooling systems are used to remove heat from various sources such as machinery or heated processed material. Water is circulated to absorb the heat. The water can then be either returned to the water body from where it was extracted, or cooled using air contact and evaporation in a cooling tower.

Cooling towers vary in size from small rooftop units to very large structures. They serve to dissipate heat in the atmosphere so that wind and air distribute it over a wide area. Evaporative cooling towers result in permanent removal of water from the river but avoid the need to return hot water which can adversely affect river life. 'Once-through' cooling is used in two power stations on the Waikato River (Wairakei and Huntly), and despite the large river flow, results in appreciable increases in river water temperature.

Water is also used in air pollution control systems, which clean hot waste gases before they are discharged into the atmosphere. Sprays of water collect the particles as the gases pass through various chambers. The resulting mix of water and solids (mainly iron oxides, coal dust and ash) flows to a treatment facility.

Glenbrook Steel Mill

The Glenbrook steel mill has around one million cubic metres of water in circulation. Around 20,000 cubic metres is added from the Waikato River each day as some of the water – about one per cent – is discharged into the Manukau harbour. The amount taken every day is enough to supply about a third of Hamilton's population with water.

When water is abstracted from a natural water body there is a danger that fish and other marine life will be sucked up with the water. This was once a particular problem at the Glenbrook Steel Mill, but measures have been put in place to minimise this risk. During the 1980s New Zealand Steel developed a new system to prevent uptake of fish – the speed of water flow around the area of the uptake was increased, using high pressure jets. Flow is increased to such an extent that the fish are unable to swim against it and tend to avoid the area for the more tranquil areas away from the uptake. Fish travelling downstream are swept past. The intake is covered with a grid of slats that prevent fish from swimming through and also create turbulence between the slats that acts as a further deterrent. In addition, siting the intake away from the riverbank helps to prevent interference with whitebait, which swim close to the edge of the river.

The water abstracted from the river is used to transport ironsand from the Waikato North Head mine and for the steelmaking process at the mill. Around 7,000 cubic metres of water a day are used to transport the ironsand. It is mixed with water to form a slurry that can be pumped the 18 kilometres between the mine and the mill by pipe. Once the slurry reaches the mill, the ironsand is separated from the water, and the water is returned to the Waiuku estuary via a drain. As a result of the turbulent transfer process, the water picks up clay minerals, which must be removed before the water can be returned to the natural environment. A chemical agent is used to settle the clay particles out of the water.

At the mill, water is used for cleaning waste gases, producing steam in the boilers, cooling, spraying and rinsing product, transporting solid wastes, and general plant cleaning. Water is also used to cool the hot steel after it has been cast into slabs and as it is rolled into sheets. This water can become contaminated with iron oxide scale dust and other solids. Water is also mixed with chemicals to create various types of solution which are used in the process of treating the steel.

Source: *New Zealand Steel, Water – an essential resource*, <http://www.nzsteel.co.nz/responsibilities/-environment>

Domestic water use

Water for domestic use comes from a variety of sources. Some areas, such as Christchurch and the Heretaunga Plains rely on groundwater as their main source, whereas in other areas water is taken from reservoirs and rivers. The Auckland region is supplied 370,000 cubic metres of water a day from a range of sources – 60 per cent of the water comes from five dams in the Hunua ranges, 25 per cent from five dams in the Waitakere ranges, 10 per cent from the Waikato River, and 5 per cent from groundwater in Onehunga.¹⁰ In isolated areas such as Waiheke Island, where centralised water provision services are limited, many properties are reliant on on-site rainwater collection.

It is estimated that nationally domestic water use is currently at 46 per cent of allocation. This means that New Zealand has the capacity to support population growth in most areas at current allocation levels, although the capacity for growth varies. For example, in 2004 the Waikato region was using 23 per cent of the 150,000 m³ per day allocated for domestic use, whereas Palmerston North City was using 81 per cent of its allocated 63,850 m³ per day.¹¹

The installation of water meters, and related volumetric charging, can help to improve efficiency of use on individual sites. For example, between 2000 and 2003 in Tauranga, water meters were installed on properties to manage demand and improve efficiency. The programme has apparently been successful, as whereas water use was 400-500 litres per person per day during the 1995/96 and 1998/99 years, it had dropped to 325 litres per person per day during the 2002/03 year.¹²

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Raewyn Peart

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Activities impacting on freshwater

Introduction

A wide range of human activities impact on freshwater bodies including agriculture, forestry, horticulture, urban development, mining and the introduction of invasive species. Climate change is likely to place further pressure on lakes, rivers and streams.

Just over 50 per cent of the total length of rivers and streams in New Zealand are within catchments which have predominantly natural land cover such as indigenous bush or tussock. Slightly less (43 per cent) are in catchments modified by agriculture, five per cent are in catchments modified by plantation forestry and only one per cent are in urbanised catchments.¹

Water quality tends to be poorest in rivers and streams located in urban and farmed catchments, because these land uses produce significant quantities of pollutants which are transported by rain into groundwater, rivers and lakes. The most polluted water bodies in farmed catchments have seen the most deterioration in water quality over the last decade.

Natural factors

Natural factors can cause changes to the quality of freshwater, and can also exacerbate the damage caused by human-made pollution. Conversely, human activities tend to exacerbate naturally existing conditions. Algae grows best in a warm climate, so water bodies in warmer regions are more likely to experience algal blooms than those in colder parts of the country. In addition, other factors can encourage the growth of algae. Wind can create waves and currents, particularly on shallow lakes, and lift sediment from the bed into the water. This causes reductions in the clarity of the water and increases the amount of nutrients available to support algal growth.

The amount of rainfall that a water body receives controls the amount of water that flows in rivers and aquifers, and affects water quality by carrying pollutants from land to the water. At times of heavy rainfall, particularly after a dry period, much higher levels of pollutants may enter water bodies.

The geology of the catchment will have an impact on the level of sediment carried by the water: mountainous regions where the catchments are steep-sided, or where the sediment is soft, are more likely to experience natural erosion causing rivers draining from these catchments to carry high levels of sediment. Water moving slowly through volcanic rocks or geothermal areas is

much more likely to contain high concentrations of minerals such as sulphates, and metals such as arsenic, than water in other areas.²

Although these factors are important to consider when evaluating water quality, the main concern for managers of water resources are the impacts generated by New Zealand's human population, including pollution from urban, industrial and agricultural sources and other effects of human activity in the area surrounding the waterway such as heating due to removal of riparian shading.

Water pollution can be divided into two types – *point source discharges* and *non-point source discharges*. Pollution from point sources refers to contaminants that enter the catchment from one isolated source such as a sewage treatment plant or a dairy milking shed. Non-point source pollution refers to contaminants that enter the catchment from multiple diffuse sources, such as runoff from pasture or urban areas.

Agriculture

Agricultural pasture makes up at least 40 per cent of the land in New Zealand, and agricultural activity affects nearly all catchments.³ Since European settlement agriculture has changed the landscape of New Zealand dramatically, altering the land cover within catchments, and in some cases becoming a serious source of both point and non-point source pollution.



Figure 6.1 Rural area near Takaka (Raewyn Peart) *Agricultural activity affects nearly all catchments in New Zealand and in some areas is a significant source of waterway pollution*

Key causes of impacts on water bodies are summarised in Figure 6.2 and described in more detail in the sections below.

Impact on Freshwater body	Cause of impact
Native riparian vegetation cover and diversity reduced	Deforestation Livestock grazing, browsing and trampling of remnant vegetation Exposure to wind and sun causing drying Wetland drainage
Soil condition degraded	Increased bare ground Compaction and reduced water infiltration
Channel stability reduced	Trampling by livestock Vegetation removal
Channel width reduced	Pasture grasses armour against erosion and trap sediments Soil creep from hillslopes into channels Channel width increases at livestock crossings
Riverbed sediment texture decreased	Siltation of the streambed by fine material
Suspended sediment load and turbidity increased	Trampling and grazing leading to bank erosion and sediment suspension Hillslope instability Reduced entrapment in riparian vegetation Stock crossings
Nutrients increased	Stock defecation in stream channel Runoff from dung on hillslopes Leaching of urine Reduced entrapment in riparian soils and vegetation
Agrichemicals entering water bodies	Poorly managed application of agrichemicals Poorly managed sheep dip operations
Faecal microbes increased	Defecation in stream channel Stock crossings Runoff from farm tracks and raceways Reduced entrapment in riparian vegetation Farm dairy effluent

Figure 6.2 Potential impacts of agriculture on freshwater bodies

(Source: Adapted from Parkyn S and B Wilcock, 2004, 34.3)

Agricultural point source pollution

Point source pollution has generally declined over the last 20 to 30 years as treatment system technology has been upgraded, and alternative methods of disposal have been developed, including applying effluent to land rather than discharging it into waterways. Progressively stricter controls on discharge practices were introduced under a 1971 amendment to the Water and Soil Conservation Act 1967 and subsequently under the RMA. However, at specific sites, such pollution can still be a problem.

Meatworks, dairy factories, tanneries, fertiliser plants, stockyards and many other types of agricultural enterprise all pose a risk of discharging pollutants into the catchment. In the early years of European settlement there were many such enterprises scattered across the country, all discharging industrial or agricultural waste on a small scale. However in the last 30 years agricultural and industrial activity has changed, so that there are fewer, larger operations. For example, in 1970 there were more than 200 dairy factories whereas there are now only about 30.⁴ These bigger businesses tend to be better able to deal with waste effectively because of their relative financial strength and access to technology. Furthermore, the last 30 years have seen the implementation of stricter legal requirements and a shift in attitudes towards greater recognition of the importance of avoiding pollution, together with a greater interest in finding better technology to achieve this.

Whereas farmers might once have discharged effluent from dairy sheds directly into waterways, they now generally use pond treatment systems or land irrigation systems to dispose of their effluent. Two-pond treatment systems pass the effluent through two ponds, each containing different naturally occurring types of bacteria, before discharging it into the catchment. This system can remove up to 95 per cent of the organic matter from the effluent. However it does not affect the nutrient content, and may not reduce the amount of organic matter actually entering the water to safe levels, because of the sheer volume of waste. To be effective, the ponds need to be well designed and maintained and adequate for the herd size, and this is not always the case. New technology is being developed which may enable energy from wastewater solids, and algae grown on wastewater nutrients, to be harvested as biogas.⁵

Land irrigation disposes of waste water and nutrients to fields or forest rather than into the water. Soils have differing potentials to deal with nutrients, both seasonally and through their nature. Land disposal is effective in protecting the catchment from pollution so long as the land is not overloaded. At this point nutrients will wash off into streams or leach into the groundwater, posing the same risk to the ecosystem as they would have if they had been discharged directly into waterways.⁶

Agricultural non-point source pollution

Whilst pollution from point sources has been decreasing over the last decades, other types of agricultural pollution have been increasing at a significant rate. Thus the National River Water Quality Network indicated that in respect of the period from 1989 to 2007, there was a general trend towards improvement in visual clarity of water, but levels of nutrients such as phosphorus and nitrogen were clearly increasing. This probably reflects a diminution in the number of point pollution sources discharging organic matter into the catchment, whilst expansion and intensification of pastoral land use has caused increases in nutrient pollution from pastoral runoff.⁷

'There is no doubt that our declining river quality over the last 20 years is associated with intensification of pastoral farming and the conversion of drystock farmland to dairy farming, particularly in Waikato, Southland, and Canterbury. For example, between 1992 and 2002, the number of cows in the Waikato increased by 37 per cent; during the same period nitrogen levels in the region's streams increased by 40 percent and phosphorous levels went up by 25 percent.'

Source: Proffitt F, 2010, 15

Agricultural runoff has been a problem for New Zealand's waterways for many years. As native vegetation is turned into pasture, wetlands drained and stream bank vegetation removed, the land becomes less able to absorb water. As more water runs from the land to the waterways it picks up sediment, organic matter (cattle effluent) and nutrients that are deposited on pasture, transferring them to the waterways.

In recent years the condition of many lowland streams in farming areas has deteriorated. This has been coupled with the widespread conversion of sheep farms and forestry to dairying (and less commonly, deer farming), and a trend towards intensification of dairy farming, taking advantage of improvements in fertilisers and stock management, so that more cows can be kept on a given piece of land.⁸ Monitoring undertaken in the Waikato region has shown that nitrogen and phosphorus concentrations in the rivers are increasing at an average of two per cent per year.⁹

The worsening pollution problems have brought about a growing awareness of the effects that agricultural activity can have on waterways. In recent years farmers have made efforts to resolve these problems, but improvements thus far have not managed to reverse the trend towards nutrient enrichment of lowland waterways.

Nutrient pollution from runoff is increased where drains like mole and tile systems are used to dry out pasture-land. These work in clay soils and are common in Otago and Southland. They use perforated plastic tubing sunk at a depth of about 50 centimetres to carry water from pasture into larger drains.

Tunnels are also made at a slightly shallower depth, at about five metre intervals, by dragging a torpedo shaped device through the soil. This creates a network of small drains across the paddock that lead water into the plastic tubing. The system is effective at removing water from pasture, but it also increases the amount of pollution that is transferred to the waterways.

Sedimentation

The removal of hill and riparian forests for agricultural pasture increases the amount of rainfall running off the land into streams and rivers. The tree canopy in a mid-latitude forest typically intercepts and then evaporates away up to one third of the rain that falls on that canopy, effectively reducing the rain that reaches the ground. Canopy interception losses are a function of effective leaf area and canopy roughness, and in west coast locations with frequent rain, can be significantly greater than water losses to transpiration (water drawn up through the tree roots via photosynthesis). There is little difference in canopy interception losses between native and exotic forests, tall manuka and kanuka or dense infestations of gorse or broom, provided that a closed canopy exists. But such interception losses are far greater than that occurring under tussock, short pasture grasses or crops.

In contrast to the highly compacted soils often found under pastoral or arable agricultural systems, forest soils have much higher infiltration capacities and seldom generate overland flow or surface run-off. When forest is replaced by short-rooted pasture grass, increased amounts of rainfall are allowed to run off the land. In the absence of tree roots to hold the soil together, greater amounts of sediment made up of inorganic matter (rock and soil) and organic matter (mostly animal faeces), are washed from the land into groundwater, streams, rivers and wetlands.

Although most forests in New Zealand were removed between 800 and 80 years ago (in two waves with initial Māori impacts followed by extensive impacts of European settlement), the effects of these changes in the landscape continue to be felt, as the hills and riparian areas continue to be used for livestock production.

Pastoral catchments thus have a sediment load which can on average be two to five times greater than that of a forested catchment,¹⁰ except for the East Cape where major gully erosion that followed wholesale deforestation and establishment of exotic grasses has led to sediment yields of over a thousand times that of the original native forest cover. This excess sediment makes waterways muddy and murky. This is not only aesthetically displeasing but also makes the water less suitable for many native aquatic species and sport fish such as trout which need clear water for spawning and feeding.



Figure 6.3 Eroding pastureland, Northland (Raewyn Peart) *The removal of forest and replacement with pasture has resulted in an increase in sediment entering waterways*

The presence of large numbers of stock which churn up fields can exacerbate the problem. When farmers graze stock intensively on smaller fields during the winter, to minimise damage to valuable pasture, this can result in pugged (hoof damaged) bare soil with large amounts of effluent on the surface. Rainfall can wash this damaged topsoil and effluent into waterways.¹¹ The effects of this are much worse when riparian areas are stripped of growth and used as pasture in addition to the main pasture area. In recent years, some councils have recognised this and instigated programmes to provide incentives for farmers to replant riparian areas, to ensure that a buffer zone is maintained between pasture used by stock and the waterway.¹²

Nutrient pollution

Nutrient pollution, predominantly in the form of nitrogen and phosphorus, is one of the most significant byproducts of agricultural activity. Nitrogen and phosphorus are essential parts of amino acids and nucleic acids, and consequently they are essential building blocks for all life on Earth. Unfortunately when they appear in waterways in excess they can be very harmful, because they disrupt natural processes of growth and decay.

The nutrients can come from a number of sources. New Zealand's livestock excrete nearly 40 times more organic waste than New Zealand's human population. This is a large amount of organic effluent that must be disposed of. The breakdown of organic nitrate and phosphate in faeces, and the oxidation of ammonium in urine, both release large amounts of nutrients into the water.

Nitrogen and phosphorus pollution is also derived from fertilisers which are applied to agricultural pasture to promote grass growth. The amount of nitrogen fertiliser used in New Zealand has increased ten-fold over the past 15 years.¹³ Furthermore, soil in agricultural pasture often has unusually high natural levels of nitrogen due to the presence of clover, which is sown by farmers as a method of making yet more nitrogen available for plants.

'Using results from nutrient budget modelling, the Ministry of Agriculture and Forestry (MAF) reported in 2008 that the average nitrogen lost from soil on dairy farms was 39 kilograms per hectare per year, compared to 12 kilograms for deer farms and 8 kilograms for sheep and beef farms'

Proffitt F, 2010, 15

Recent work has also identified gorse as a potential contributor of nitrogen to waterways. A study which quantified nitrate leaching from stands of gorse growing in the Lake Rotorua catchment concluded that 43 tonnes of nitrogen derived from gorse were being leached into the groundwater each year.¹⁴

Removal of riparian shading

Most aquatic life in New Zealand's waterways developed in heavily shaded forest conditions. When vegetation is removed from the banks of waterways, the amount of available shade is decreased so the temperature of the water increases, as does the growth of algae and aquatic weeds. Oxygen levels in the water also reduce as the temperature rises. This makes conditions for native aquatic life less than optimal. Riparian areas are also an important habitat for many types of native fauna, whose habitat may be destroyed by the removal of the vegetation.

The removal of trees from the riverbank can destabilise the area. Then allowing stock to graze the banks will exacerbate this problem. Removal of riparian planting also removes a valuable filter. Plants, leaf litter and soil can filter out sediment nutrients, faecal material and other pollutants that run off the land towards the waterway. In addition, undisturbed waterways and wetlands provide extremely valuable flood prevention control. Interfering with them reduces their ability to absorb and slow peak flows, and to release water slowly to maintain summer low flows.¹⁵



Figure 6.4 East Cape river (Raewyn Peart) *Removing riparian vegetation from rivers decreases shade, increases the water temperature, and removes important habitat for indigenous species*

Stock grazing

Stock grazing in the areas around water bodies and in wetlands can contribute significantly to pollution. This is through direct nutrient enrichment and bacterial contamination of the water from faeces and urine damage vegetation. Stock can also decrease soil stability. Stock with access to waterways can contribute large amounts of faecal matter directly to the catchment. A study has shown that when cows cross a stream on their way to and from milking they are 50 times more likely to defecate in the water than on adjacent areas.¹⁶



Figure 6.5 Beef cattle in a river (Aalbert Rebergen) *Stock grazing in streams and wetlands can contaminate the water with faeces and urine as well damage vegetation and destabilise banks*

Wetlands are particularly sensitive to the presence of grazing stock. Because New Zealand has no native terrestrial mammals other than bats, native wetland vegetation has not developed the hardiness to cope with the presence of large grazing animals. The native herbivores that graze on wetland vegetation, in particular avian species such as native geese and the New Zealand giant coot, may impact on the vegetation above ground but do not have the weight to cause the level of damage to the earth below that stock can cause through trampling. Livestock can open up sites for weed invasion and carry weed seeds in their hooves and dung.¹⁷

Livestock grazing may cause damage to wetland birds by trampling on their nests. They can alter the type of wetland habitat available, such as creating more open water by destroying vegetation. In addition, many native fish use wetland areas for spawning, and this may be disrupted by the presence of livestock in the area. For example, a 1991 study of whitebait spawning grounds in the Bay of Plenty found that survival rates of īnanga spawn were significantly lower where cattle were present in the area. This was probably due to trampling and consumption of wetland vegetation by the cattle.¹⁸

It should be noted that the effects of grazing stock in a wetland can be complex, as it will depend on the type of animal grazing and the particular composition of vegetation present. In fact, stock have at times been used in wetlands to control particular types of plants. For example, cattle grazing can be helpful to wetland health by preventing the growth of woody vegetation, and controlling the growth of exotic species which can take over the ecosystem. Likewise, grazing sheep can help to maintain short turf that increases native species diversity. For example, on Great Barrier Island grazing outside the nesting season has been used to control dense areas of kikuyu, and to open up more land for endangered brown teal habitat.¹⁹

Grazing can also increase or decrease species' richness, depending on the particular vegetation present. If plants that are palatable to livestock dominate the area, grazing is likely to reduce their number and so allow other species room to grow. However, despite the potential benefits, the fact remains that the presence of stock in wetlands can have significant adverse effects as a result of trampling and pollution from effluent. Methods such as the targeted use of herbicides, mowing or fire can have the same ability to improve conditions for non-dominant species without the damage that livestock can cause.

Stream modification

Streams can be often modified in order to increase the workable area of farms. Such modifications can include culverting or piping the stream as well as diversions and realignments of the stream bed. In Taranaki, a recent studied

estimated that over 700 kilometres of streams had been modified within the region.²⁰ These works can have significant impacts on freshwater bodies and result in habitat degradation and loss, reduction in water quality, changes in hydrological regime and loss of biodiversity.

Urban development

Urban development can have significant impacts on waterways through the direct discharge of wastewater and through urban runoff and land use changes as shown in Figure 6.6. These changes are described more fully in the sections below.

Impact of urbanisation	Effects on freshwater bodies
Contaminated runoff	Reduced water and sediment quality Increased sedimentation Excessive macrophyte or algal growth
Changed flow regime	Increased flood frequency and intensity Reduced baseline flows Changes to stream width and bank erosion
Increased water temperatures	Excessive plant growth Reduced oxygen Thermal stress for invertebrates and fish
Altered riparian vegetation	Reduction in instream energy sources Loss of habitat and biodiversity Loss of aesthetic value
Physical changes to rivers and streams, channelisation and piping	Bottlenecks for invertebrates Barriers to fish passage Loss of habitat in concrete-lined streams

Figure 6.6 Potential impacts of urbanisation on water bodies
(Source: Adapted from Suren A and S Elliot, 2004, 35.9)

Wastewater treatment plants

Approximately 1.5 billion cubic metres of domestic wastewater is discharged into the environment in New Zealand every day.²¹ Human effluent contains a number of substances which can be extremely damaging to freshwater ecosystems and to human health if released into the catchment untreated.

The most significant pollutants are rotting organic matter, disease-causing microorganisms, excess nutrients, ammonia and suspended solids. Domestic wastewater can also contain a range of household chemicals and pharmaceuticals, some which can bio-accumulate in the environment.

Most of the household sewage in New Zealand is piped through treatment plants to remove pollutants before it is discharged into the catchment; all towns of 5,000 or more have a reticulated sewage system which pumps everybody's waste water into a common treatment plant and discharge point. Wastewater treatment plants discharge treated effluent into rivers, streams, groundwater, estuaries, wetlands, the sea or onto land.²²

The plants vary in age and technological capability. Treatment systems use a number of stages in the treatment process, depending on how advanced they are. 'Primary' treatment means that suspended solids are filtered out of the effluent before it is discharged into the catchment, and 'secondary' treatment refers to the use of microorganisms to remove organic matter and nutrients, and decomposition in wetlands or oxidation ponds. 'Tertiary' treatment further reduces nutrients and organic matter using various types of chemical and biological treatment. Ultra-violet treatment and various other disinfection technologies are used by the most advanced plants in New Zealand to inactivate the pathogens that remain in the water.

Most treatment plants in New Zealand use at least primary and secondary treatment and a large number also use tertiary treatment. A growing number are also using ultra-violet or equivalent systems. In Whangarei district, for example, the wastewater undergoes ultra-violet treatment and is then directed through a large constructed wetland to 'polish' the water. As well as resulting in a cleaner discharge, this system provides additional habitat for wetland species. The effectiveness of wastewater treatment systems around the country varies but has improved considerably over the past two decades.

Some wastewater treatment plants are ineffective because they are too small to cope with the volume of waste they receive. Some are put under pressure by new urban developments, or by the increased amounts of effluent generated by an influx of visitors during holiday season.

Treatment of wastewater generates between 700,000 and one million tonnes of sewage sludge each year.²³ A major issue is how to dispose of this large quantity of material. Because industrial waste is often treated in the same system which treats human waste, sewage sludge (also called 'biosolids') can be contaminated with heavy metals, organochlorines and other potentially hazardous chemicals. Many treatment plants reduce the volume of sludge by digestion of the sludge at body temperatures producing useable biogas, which may be sufficient to drive the energy to run the treatment plant, as is the case

in Christchurch City. The process reduces the organic content but does not remove many contaminants or pathogens.



Figure 6.7 Mangere Wastewater Treatment Plant (Raewyn Peart) *Most household sewage in urban areas is piped to treatment plants to remove pollutants before being discharged. The wastewater treatment plant at Mangere has recently been upgraded to increase the quality of effluent.*

In the past, biosolids had commonly been dumped at sea or buried. Most contaminated biosolids are now disposed of in landfills although there have been efforts to reuse the material. For eight years Wellington City Council partnered with Living Earth Limited to produce compost from a mix of sewerage sludge and garden waste. However, in 2008 the council abandoned the scheme, citing odour problems as the reason, and now disposes of the biosolids into landfill.²⁴ New Plymouth City Council uses thermal drying of sludge to produce a stable pasteurised fertiliser which is sold.²⁵

Stormwater drainage systems can also be a source of pollution to urban catchments. Some areas use combined sewer and stormwater drains. This is common in old networks that have not yet been replaced with more modern systems. The pipes in a combined sewage and stormwater system carry both sewage and stormwater, and are designed to overflow into the catchment during incidents of moderate to heavy rain, causing sewage contamination of waterways.²⁶

Even where separate sewage and stormwater systems are used, sewage can sometimes overflow into the stormwater network, usually as a result of failures in the pipes, or faulty connections particularly in times of heavy rainfall. The converse situation can occur where stormwater enters sewage pipes. Sewage systems are designed to cope with a certain peak flow, but if this is exceeded through water inflows, overflows occur at designated relief points discharging sewage into stormwater drains, land, and the sea. For example, in the Auckland stormwater network there are around 400 designated sewer overflow points.²⁷

In 2008 there were 2,479 recorded sewerage overflows in the Auckland region during wet weather.²⁸

Septic tanks

Smaller settlements in New Zealand often do not have community wastewater treatment plants but rely on individual household septic tanks. There are around 270,000 septic tanks throughout the country.²⁹ A basic septic tank system comprises a tank, where solids are settled out, and an effluent disposal area contained within the property where the effluent soaks into the ground. This typically consists of either underground soakage trenches or subsurface drip irrigation systems. The soil and the bacteria in the soil clean the effluent.

Septic tanks are generally effective in treating sewage but can fail where they are poorly maintained, where the soil is unsuitable, or where the surrounding land area is not big enough. As a result, untreated effluent can contaminate groundwater and streams. This is a particular problem in modern rural developments, where large homes (rather than traditional baches) with modern conveniences requiring a large amount of water, are built on small sections very close together. A 2007 study identified 250 communities in New Zealand, containing a total of 42,000 houses, which had a significant number of failing on-site wastewater treatment systems.³⁰

Designs of septic tanks are being improved, and these more modern and effective systems are increasingly being utilised.

Urban runoff

Urban development can place extreme pressure on nearby waterways. Urbanisation increases the amount of impervious surface area on the surrounding land and so prevents rainwater from soaking into the soil and reaching the groundwater. This results in less water replenishing the groundwater resource whilst runoff into streams occurs more quickly and in greater volume. Furthermore, vegetation which normally decreases runoff as a result of transpiration and evaporation is often absent in urban areas.

A study of four catchments in the foothills of the Waitakere Ranges concluded that 'extensive areas of impervious surface and stormwater drainage networks, which discharge directly into stream channels, led to substantial degradation of aquatic ecosystems'.³¹



6.8 Driveway, Gisborne (Raewyn Peart) *Large areas of paving in urban areas prevents rainwater from soaking into the soil and increases runoff into waterbodies*

Runoff from urban areas can affect stream ecosystems, as activities in urban areas generate large amounts of pollutants which are carried along by the water. Urban runoff commonly contains high levels of sediment; natural sediment traps such as vegetation and wetlands are few and far between in urban areas, and large amounts of sediment can be generated by construction works. A study undertaken for the Auckland region identified earthworks as having by far the greatest potential to generate sediment on a per square kilometre basis (see Figure 6.8).

Land-use	Predicted average annual soil loss (over 20 years) (Tonnes/km ² /year)
Developed Urban – Residential	24
Pasture	46
Market Gardening	52
Developed Urban – Industrial	100
Earthworks	16,800

Figure 6.9 Annual soil loss in the Auckland Region (Auckland Regional Council 2001, 11)

Runoff can also contain bacteria from animal faecal matter, herbicides, pesticides, detergents and other household chemicals. Runoff from roads carries pollutants such as hydrocarbons and metals (especially zinc, copper and lead) which come from exhaust emissions, brake linings and tyre wear, as well as from the road itself. Unpainted galvanized roofs and guttering can also result in zinc contamination of stormwater.

These contaminants are a particular problem in the Auckland region where the middle parts of the Waitemata harbor and upper Tamaki estuary are contaminated with heavy metals as a result of stormwater flows and have poor ecological health.³²



Figure 6.10 Earthworks, Snells Beach (Raewyn Peart) *Through removing the vegetative cover from soil, major earthworks can release large amounts of sediment into waterways, if not well managed*

Loss of riparian shading

The loss of riparian shading caused by the removal of vegetation for urban development means that the temperatures of streams are also altered. The reduced base flow of the stream (caused by impervious surfaces lowering groundwater levels) causes the water to be heated even more quickly. Raised temperatures can cause damage to aquatic life which is accustomed to particular environmental conditions.

Waterway development

In urban catchments engineering solutions such as channel straightening, lining the channel with concrete, weirs, culverting and removal of instream debris are often used to contain increased flows. While these works may be effective at

managing the removal of water from urban areas at a local scale, they are often detrimental to ecosystem health as habitats are removed. Weirs and culverts may prevent the movement of fish and discharge of heated water can prevent the dispersal of aquatic organisms.



Figure 6.11 Canalisation of the Leith River, Dunedin (Raewyn Peart) *Straightening river channels and lining river beds with concrete removes natural habitats which are essential for many freshwater species*

Wetland drainage

Wetlands are often purposely drained to maximise the land available for urban development and to improve access to rivers, lakes and coastal margins for recreational activities. Drainage at the edge of peat bogs, or upstream of their water source can cause a drop in the water level of the wetland, which can stop the formation of peat and therefore change the type of plants that grow there. Drainage can also result in the loss of habitat for species such as eels.

Many low-lying Auckland suburbs were originally wetland areas. On the Auckland isthmus wetlands were present at the mouth of streams and in many depressions created by volcanic activity. Most of these have disappeared as a result of urban development and now less than 0.5 per cent of the land cover in the region is wetland.³³

Horticulture

Horticultural and arable land uses make up less than two per cent of the country's land area, a tiny portion compared with agricultural land uses. However, horticulture is a more intensive user of pesticides, using over 13 kilograms of active ingredients per hectare compared to less than three kilograms per hectare for pastoral farming.³⁴

Most of the land used for horticulture is flat or slightly undulating, which means that it does not produce large amounts of runoff. However, there is still the potential for substances from fertilisers, herbicides and pesticides to leach from the soil into freshwater. This can have marked effects on freshwater quality in some areas.³⁵ On modest slopes soil loss is often an issue in heavy rainfall events.

Forestry

Plantation forestry yields similar levels of nutrients as native forest, so it is only in association with felling that forestry activity has an impact on water quality. Although the effects of forestry on a catchment are generally short-term and localised, they can be significant. Road, track and landing construction can increase the amount of sediment, while the decay of pine needle litter and understorey vegetation after harvest, can generate small nutrient flushes (in particular nitrogen, together with smaller amounts of potassium, phosphorus, magnesium and others).



Figure 6.12 Forest harvesting, Taranaki (Raewyn Peart) Forestry harvesting needs to be carefully managed to avoid releasing sediment into rivers and streams

Trees and associated riparian vegetation provide a barrier to soil and other contaminants entering waterways. Once trees are harvested, soil erosion can occur, resulting in the sedimentation of freshwater bodies and a reduction in water clarity. Tree roots progressively decay after harvest with the beneficial effect of root reinforcement of slopes being largely lost within about seven years of harvest, leaving deforested slopes vulnerable to soil erosion and in some cases large scale mass movement including gully erosion such as occurred with the Tarndale and Mangatu 'slips' on the East Cape. The extent that these adverse effects will occur depends on the characteristics of each site

When harvesting is carried out close to streams, there is little to stop rainwater washing the soil into the waterways, unless slash is left to form a barrier. The extent of these effects depends on the topography and geology of the particular area, the type of soil and the amount of rain normally experienced. The worst effects are generally only felt for the first year after felling, but some impacts may be detectable for up to three to five years.

The negative effects of forest harvesting can be mitigated by the establishment or retention of riparian planting. This can protect the waterways from the majority of sediment carried by any surface runoff from roads or bladed tracks created by harvesting machinery, or from other disturbed areas. Retention of riparian vegetation at harvest maintains shade which lowers water temperatures and helps to stabilise stream banks.

The overall effects of forestry on the quality of New Zealand's freshwater resources can be positive when land is converted from pasture to plantation forest. Pasture causes much more sedimentation and nutrient runoff, so afforestation of an area previously used as pasture can return sediment loads to normal levels within six to eight years of planting.³⁶

On the other hand conversion from pasture to forestry can reduce the total water yield (run-off) in the catchment. A study undertaken at the Glendhu Forest in the upper Waipori Basin found that over a 10-year period there was a 30 per cent reduction in water yield after tall tussock grassland was converted to plantation forestry.³⁷

Conversion in the reverse direction, from forestry to pastoral use, increases the supply of nutrients in runoff and to groundwater. Such conversion is now common in central and eastern North Island areas once thought to be unsuitable for pastoral use because of cobalt deficiency (bush sickness). But this is now easily remedied.

Dams and diversions

There are numerous dams and diversions in New Zealand, used for a variety of purposes. Dams on small farm waterways are used for irrigation or to store water for consumption by livestock. Large scale dams for irrigation are used mainly in Canterbury and Otago.

Dams are also used to store water for human supply, and to generate hydroelectric power. Fifty-five per cent of New Zealand's electricity is generated by hydro electric power stations.³⁸ They work by generating power through the gravitational force of falling water. River water is dammed and allowed to escape through a particular channel. The falling water turns the turbines which power a generator. The amount of power generated is therefore dependant on the height of the dam and volumes of water that pass through the power station.

Some rivers have been extensively dammed to generate hydroelectric power. For example, there are eight dams on the Waikato River and two dams on the Clutha River. There are also eight power stations on the Waitaki River, linked to three artificial lakes, three natural lakes whose level is manipulated, and a network of artificial canals.³⁹

The damming of a river or stream changes its natural flow, which can have a variety of impacts on river life. The most immediate is on flows downstream which are reduced or the pattern of high and low flows altered. Algae and other nuisance plants may become a problem as the high flows that would normally flush the river system are reduced. Release of water from dams can also change the water temperature of the river downstream as the water in the dam has different thermal properties and is likely to be warmer or cooler than the river.

The sediment flow downstream of the dam will often reduce as a result of sediment being trapped in the lake. This can improve water clarity downstream, but also often results in changes to the stream bed. Organic matter can be trapped behind the dam wall, reducing the amount of this food source available to downstream life. Dams can also act as barriers to fish, preventing them from accessing upstream habitats.⁴⁰

The levels of storage hydro lakes tend to fluctuate to a greater extent than natural lake levels. High lake levels can result in shoreline erosion, the swamping of vegetation around the shoreline, and a reduction in light levels at the bottom of the lake due to the increased depth. Conversely low lake levels can also result in erosion and affect organisms which live in the littoral zone around the shoreline.⁴¹

The creation of a lake as a result of damming can provide new habitat for a range of species including indigenous invertebrates, fish and birds. However,

dams are ideal environments for exotic species such as koi carp and mosquito fish, which compete with native fish for food and also feed on them. They also favour the popular recreational fishing species of trout and salmon; the hydro lake at Benmore is among the most popular fishing sites in New Zealand.

As well as dams, there are many other structures on rivers which can affect their natural flow including structures to divert water, stopbanks, weirs and floodgates to prevent flooding and culverts and fords to create permanent river crossings. These can have the effect of changing a river or stream's natural flow, preventing the movement of fish and shrimps, and destroying the natural riverbed habitat.

Most indigenous fish are diadromous, and therefore move between freshwater environments and the sea. Dams and culverts which obstruct fish passage can therefore have a major impact on these species. In the West Coast and Southland, access to about 22 per cent of longfin eel habitat has been modified by the construction of hydro and other dams, including Lakes Te Anau and Manapouri.⁴²



Figure 6.13 Eel damaged in the turbine of the Branch River hydro scheme (Fish and Game New Zealand)
Dams and other structures can obstruct the passage of indigenous species along waterways and between the sea and freshwater systems

Mining

Mining of various types of materials is an important industry in New Zealand, contributing to many sectors of the economy: construction (aggregates and limestone for cement), energy (coal), transport (road aggregates), primary industry (coal and ironsand), agriculture (fertiliser) and manufacturing (industrial minerals).⁴³



Figure 6.14 Martha Mine, Waihi (Raewyn Peart) Mining is an important industry in New Zealand but can also have detrimental impacts on freshwater bodies

Coal and mineral mining

One of the most economically significant mining products is coal. Coal supplies around 13 per cent of New Zealand's primary energy supply. The biggest users are the Huntly power station and Glenbrook steel mill. Some New Zealand coal is valued overseas for its low ash and sulphur content, and it is exported around the world, although the majority goes to India and Japan. In total, 4.83 million tonnes of coal was mined in New Zealand in 2007 from four underground and 21 open cast mines.⁴⁴ Over 60 per cent of national production was from two

large opencast operations in the Waikato (Rotowaro) and in the South Island (Stockton).

Gold mining is also an important industry in New Zealand. It has taken place since the first gold rush of the 1860s. The industry declined in the 1920s, and then underwent a revival in the 1970s as new technology allowed gold to be extracted from more areas using opencast mines, which were dug in Otago and in the Hauraki district during the 1980s and 1990s. Gold worth \$626 million was mined in 2008, mainly from two large mines in Waihi and Otago as well as from several medium-sized alluvial operations.⁴⁵

The process of separating commercial minerals from the ground results in waste product known as mine tailings. Tailings are fine grained solids which consist of a slurry of ground up rock, water and chemical residues. Because of the volumes of waste involved, mining companies tend to want to dispose of the waste close to the site. The tailings may be deposited into open pits, underground, or constrained behind a dam. Tailings dams are located at four current or former mining sites in New Zealand. Two of these are in the Hauraki district, one in the Matamata Piako district and one in the Waitaki district. Mining activities can also result in the diversion of streams around the pit as occurs at the Rotowaro open cast coal mine near Huntly.

Runoff from mining sites and from tailings dams that are left behind can cause damage if not properly controlled, even after the mine has ceased operation. The runoff is often highly acidic and contains a cocktail of dissolved heavy metals and other substances. This is known as 'acid mine drainage'. The runoff leaches into streams and has serious effects on freshwater ecosystems, by increasing the acidity of the water and polluting them with toxic substances. Rivers are normally neutral or slightly acidic, but never naturally as acidic as can be caused by acid mine drainage.⁴⁶ If acid drainage reduces the pH of the waterbody to less than 4, all fish and most invertebrates are likely to die.⁴⁷ Acid runoff can be generated rapidly in rain events and then dry up after the storm.

Acid mine drainage has been a significant problem in the coal mining areas on the West Coast of the South Island. It is estimated that around 125 kilometres of streams are adversely affected by acid mine drainage in this area.⁴⁸ It is well recognised internationally that acid rock drainage processes from inadequate mine waste management can take prolonged periods to develop and have been documented to persist for in excess of 300 years.⁴⁹



Figure 6.15 Tui mine tailings (Ministry for the Environment)
*Leachate from the abandoned Tui mine is polluting the
 Tunakohoia and Tui streams with heavy metals*

Water discharged from coal mines can also contain high levels of dissolved iron, aluminium and other metals which are potentially toxic to freshwater life. Gold mining can cause toxic substances and heavy metals such as cyanide, arsenic, cadmium, lead, antimony and zinc to contaminate waterways.⁵⁰ Even where the water runoff from the site is not very acidic, the presence of dissolved heavy metals will affect fish and insects.

New Zealand streams and rivers almost never have naturally high heavy metal concentrations, so native species are not well adapted to their presence. Studies have observed higher rates of biological infections in benthic invertebrates in streams with higher levels of dissolved heavy metals. This may be because the invertebrates were under more stress.⁵¹ Testing during the early 1990s found elevated copper concentrations in the muscle tissue of rainbow trout in the Waitekauri River downstream of the Golden Cross gold and silver mine on the Coromandel Peninsula.⁵²

Tui Mine

One of New Zealand's most contaminated sites is the Tui mine, on the western flank of Mount Te Aroha in the Waikato region. The mine was opened in 1967 to extract metals including copper, zinc and lead and later gold and silver. Subsequently, unacceptable levels of mercury were found in the mine and in 1973 the company buying the ore pulled out. Two years later the mining company went into liquidation and the mine was abandoned.

One of the main issues associated with Tui Mine is that the polluter, Norpac Mining Limited, no longer exists and the present landowners did not cause the contamination. Therefore Tui Mine is known as an 'orphan' site. This means that there is no clear legal responsibility for the remediation of the site. Large piles of ore and tailings were left at the site behind a tailings dam. Over the years, the tailings dam fell into disrepair and became unstable. In 1980 the Hauraki Catchment Board built a gravel embankment to stop the tailings slipping onto property further down the mountain but other than this little effort was made to address the environmental risks of the site.

There are three separate components of the Tui Mine which are collectively identified as the 'site'. They are:

- The Tunakohoa Stream, located on Crown land managed by the Department of Conservation, which is affected by heavy metals discharging from the mine tunnels and leaching from the tailings impoundment. These heavy metals are contaminating the Tunakohoa Stream which flows through the centre of Te Aroha
- The Tui Stream catchment which is adjacent to, but separate from, the Tunakohoa catchment. The Tui Stream is also affected by heavy metals leaching from the tailings impoundment
- A geotechnically unstable tailings impoundment in the Tui catchment on land owned by the district council. Technical reports have found that this structure is at risk of collapse in a moderate seismic event or an extreme weather event. Such events could result in 90,000m³ of mine waste liquefying and flowing down the Tui stream on the edge of Te Aroha.

Groundwater and surface water originating from the site has low pH (is acidic) and contains elevated concentrations of trace metals, particularly copper, lead and zinc. At elevated concentrations these metals pose a risk to the aquatic receiving environment, as well as a potential public health risk through contact recreation (i.e. through children swallowing water while swimming).

In September 2008, Cabinet agreed that the Tui Mine Remediation project should be completed in two phases:

Phase One is underway and expected to be completed by March 2012. Phase One involves preparatory work (detailed design, site establishment and upgrading of access to the site) and the remediation works within the underground components of the mine. Underground remedial works will include the installation of a concrete bulkhead at the lowest discharge point from the mine, as well as introduction of lime within the mine workings to reduce the potential for acid rock drainage from the mine.

Phase Two involves remediating the tailings impoundment area, and the waste rock piles dumped in the Tunakohoa Stream during the mining operations. The tailings impoundment will be stabilised using a combination of in-situ stabilisation and blending of tailings, waste rock and lime/cement to provide a geotechnically stable landform with significantly less acid rock drainage potential. Re-contouring and stabilisation of the waste rock stock piles will also significantly reduce the contaminant loads that are currently discharging into the receiving environment.

Source: Ministry for the Environment

Gravel mining

Together with cement, gravel is needed to make concrete, which is New Zealand's most widely used construction material. Gravel is mined from river beds in many parts of the country. Rivers close to urban centres are favoured, to keep transport costs to a minimum. Most gravel extraction takes place from alluvial rivers draining catchments where greywacke forms the dominant rock type, which is supplied to the river bed by erosion upstream.⁵³

Over-extraction of river gravel can lower the level of the river bed, change the profile of the channel and alter the composition of the riverbed sediment. Lowering of the riverbed can affect bridge stability as occurred in the Oreti River in Southland and the Wairoa River near Nelson.⁵⁴ Extraction can also have downstream effects as it changes the amount of gravel supplied to the coast, potentially causing river mouth lagoons to become unstable and coastal erosion to be accelerated.

Gravel extraction can damage invertebrate communities by altering the composition of the riverbed, both at the extraction site and further downstream, decreasing habitat diversity. When gravel is removed the composition of the riverbed changes, as finer silt and sediment and exposed bedrock is left when the gravel is removed. This creates a far more uniform surface, and makes the river flow in a more consistent way than it did when gravel of varying sizes lined parts of the river bed. This means that there are no longer eddies and depositional zones in the river which are conducive to the development of diverse invertebrate colonies. A study of the Kakanui River near Oamaru found a decline in the total abundance of invertebrates, in the number of different species present and in the percentage of sensitive species such as mayflies, stoneflies and caddisflies, downstream of the gravel extraction site.⁵⁵

Invasive fish and plant species

New Zealand's freshwater bodies have been invaded by a large number of aquatic plants and pest fish. These have often been more successful at competing for resources, such as food and light, than indigenous species and significantly alter aquatic ecosystems.

Exotic weeds

Exotic competitive weed species can displace native species, upsetting the ecological balance and leading to a decline in the abundance of native vegetation. Fifty-eight species of exotic plants have been found in New Zealand's freshwater bodies.⁵⁶

Invasive freshwater plant species have been a particular problem for lakes in New Zealand, invading over 60 per cent.⁵⁷ Problematic species include oxygen weeds (such as *Egeria densa*, *Elodea Canadensis* and *Largarosiphon major*) and hornwort (*Ceratophyllum demersum*) which can form dense stands extending from the lakebed to the surface. They displace indigenous species, reduce light and oxygen levels, and fundamentally change the chemistry and substrate of the lake. *Egeria* has been associated with vegetation collapses in many shallow New Zealand lakes including those in the Waikato.⁵⁸ Water net (*Hydrodictyon reticulatum*) is another exotic algae that was problematic in the 1990s, but has since declined in abundance. Exotic weeds can substantially reduce the recreational attractiveness of lakes and clog intakes of pumps and power stations. They cause economic losses of millions of dollars annually.⁵⁹

Another well known pest species is didymo, (*Didymosphenia geminata*) a freshwater diatom (alga) which was first reported in New Zealand in 2004.⁶⁰ Didymo, also known as rock snot, is a microscopic pest that is spread in water. It can form massive blooms on the bottom of streams, rivers and lakes, attaching itself to the bed of the water body on stalks. Didymo has a negative effect on the aesthetic values of water bodies for recreational users. It can also cause significant problems for irrigation by clogging up pump screens, sprinkler nozzles and filters, and can have the potential to reduce the area available to native fish for spawning, foraging and refuge.

Didymo may cover the spawning areas of trout, reducing oxygen available to the larvae and fry, and also potentially preventing them from escaping to the surface. Didymo is also a threat to native fish, such as the threatened lowland longjawed galaxiid. This fish burrows into gravel to prevent stranding during periods of low flow but thick layers of didymo may prevent it from doing this.

When didymo smothers the pebbles on the riverbed it may also prevent other native fish from spawning.



Figure 6.16 Didymo (Fish and Game New Zealand) *Didymo* was first discovered in New Zealand in 2004 and can form massive blooms on the bottom of streams, rivers and lakes

Exotic fish

Liberation of exotic fish can cause serious problems for a lake ecosystem. Many at their juvenile stages feed on zooplankton, which can result in blooms of planktonic algae, when they are released from the grazing control exerted by the zooplankton. Exotic fish may contribute directly to a diminution in vegetation cover and may prevent vegetation from re-establishing itself. They can also reduce water quality and compete with native species for food.⁶¹

Koi carp were brought to New Zealand in the 1970s. They are now very common in the North Island. For example, koi carp now make up 67 per cent of the total fish biomass in the lower Waikato River downstream of the Karapiro Dam.⁶² They have been illegally spread through parts of the North Island, mostly the Auckland and Waikato regions, by recreational fishers who prize them for their size and fighting ability. Koi are very damaging to the environment, as they churn up the beds of rivers, lakes and wetlands when they search for food, and they have proved very difficult to eradicate.

Goldfish were imported, probably as an ornamental species, not long after European settlement. By the 1870s they had established in the wild, in places such as Lakes Taupo and Rotorua, and today they are widespread in ponds and small lakes throughout the northern North Island.

Catfish were introduced in the 1870s and are now widespread in the Waikato River system. Similar to koi carp, they feed by ingesting and expelling bottom sediments to extract invertebrates. This can result in degraded water quality through suspending sediment in the water column.⁶³

Tench were introduced into New Zealand during the nineteenth century. More recently, anglers have spread tench around the country and they are widespread in small lakes. Grass carp and silver carp were introduced for the biological control of excessive plant growth in ponds and drainage ditches. Grass carp eat large leafy aquatic plants and this can reduce the need to spray water weeds or to use draglines to clean drains. Silver carp eat phytoplankton and were therefore introduced to try to reduce it. Grass carp and silver carp are reared in hatcheries and it is unclear to what extent they may become established in the wild.

Marron are large freshwater crayfish that are native to Western Australia. They were introduced to New Zealand in 1986 for aquaculture purposes, as they are considered to be a delicacy, but some were allowed to escape. They colonise sandy areas in dams or rivers especially where organic matter accumulates. In the wild, marron pose a significant threat to New Zealand's freshwater environment and its inhabitants, especially to insects and native crayfish (koura). In 1990 marron farming was banned, as it was considered that the risk to New Zealand's environment was too great. However, it is believed

that there are still some populations of marron remaining. For example, some were discovered at a disused service station in West Auckland, and in a pond at South Kaipara Head in 2005.⁶⁴

Fishing

Fishing activity reduces the population size of the targeted species. This can be of direct concern when the survival of the species is threatened. Reducing the population size of the targeted species can also impact on freshwater habitats by removing key species from the food web. For example, the number of goldfish in Lake Omapere, Northland has reportedly increased since the eel population was reduced. Other species, such as catfish and trout may also move into the niche left by the depleted predator producing negative impacts on the aquatic environment.⁶⁵

Climate change

Climate change is likely to have significant effects on New Zealand's water resources, as changes in weather patterns affect demand for water and the rate at which water resources are replenished. It is predicted that likely climate change impacts include higher temperatures, especially in the North Island, which will result in greater demand for water for human and stock consumption. Long hot summers will reduce soil moisture, deplete groundwater supplies, and lower river levels. Less water in rivers will encourage raised water temperatures, which could in turn aggravate water quality problems, as well as altering the habitat conditions for native wildlife and potentially encouraging the growth of populations of exotic species. With higher temperatures, snowlines and glaciers are expected to retreat, affecting the flow of South Island rivers.

Changes in rainfall patterns are likely to mean that there is higher rainfall in the west and less rainfall in the east. More frequent intense winter rainfall in western areas is expected, increasing the likelihood that rivers will flood and that flash flooding will occur when stormwater drains become overwhelmed. This could result in pollution events as sewage systems becoming overwhelmed by flood water. In addition, droughts are likely to occur, especially in the east. This could cause stock losses and place pressure on groundwater resources.⁶⁶

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Raewyn Peart

seven

Freshwater management framework

Ownership of water

Under common law, naturally flowing freshwater is not owned by anyone, but is treated as a public good. This is still the legal position today.

There is an unresolved issue as to whether Māori have customary title to freshwater. The Court of Appeal, in the *Ngati Apa* case¹, found that the introduction of common law to New Zealand from England did not extinguish Māori customary title. This means that whatever customary title Māori held to freshwater, prior to the assertion of British Sovereignty in 1840, will continue to exist unless it has been lawfully extinguished. Some nineteenth century deeds of sale of Māori land did explicitly transfer the rights to use water associated with the land but others did not. The issue of whether there are currently any Māori customary rights to water has not yet been considered by the Courts, so the legal position is unclear.²

Rights to manage water

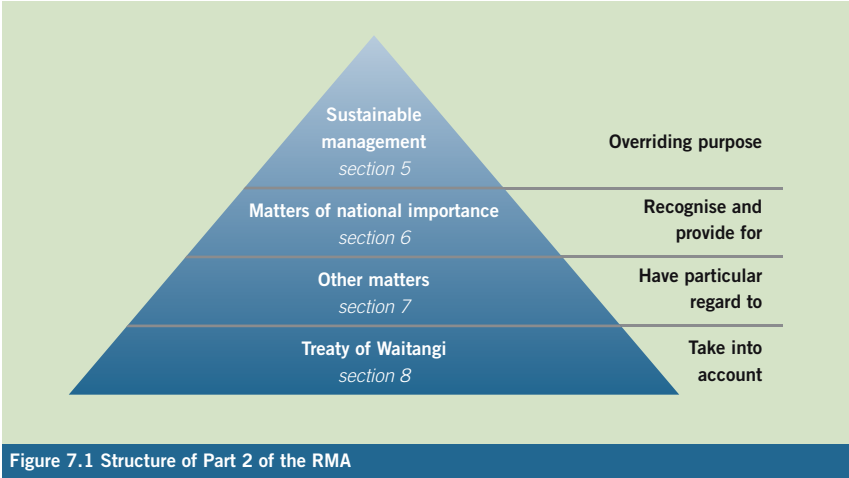
Under section 21 of the Water and Soil Conservation Act 1967 all rights to use, dam, divert, or take water or discharge into water were vested in the Crown. This legal situation was continued under the RMA (section 354). So strictly speaking, although the Crown does not own freshwater, it exercises the overall control on how freshwater in New Zealand can be used.

The Crown has delegated the control of use and pollution of freshwater to regional councils (through section 30(1)(e) and (f) of the RMA). Councils exercise this control through preparing regional policy statements and plans and determining resource consent applications in the first instance. The Crown has reserved the right to provide national direction on the content of policies and plans, and matters to be considered in relation to resource consent applications, through the development of national policy statements and national environmental standards.

Under Article Two of the Treaty of Waitangi, Māori were guaranteed the 'full exclusive and undisturbed possession' of their taonga. Māori claim that lakes and rivers are their taonga and are therefore protected under the Treaty. Treaty settlements are giving iwi rights to participate more fully in the management of freshwater.

Legislative framework

Water management in New Zealand is largely undertaken within the framework of the RMA. Part 2 of the RMA provides a purpose and set of principles to guide the management of freshwater, as well as other resources. Each section within this part of the legislation has a different level of importance when it comes to decision-making. They are described in the following sections.



Sustainable management

The purpose of the RMA is to promote ‘sustainable management’ which is defined in section 5. The definition encompasses management of the ‘use, development and protection of natural and physical resources’ so that people and communities are ‘enabled’ to provide for their own wellbeing while also safeguarding the environment.

Freshwater management needs to sustain the potential of freshwater to meet the reasonable foreseeable needs of future generations, safeguard the life-supporting capacity of freshwater (including its ability to sustain plants, birds and aquatic species), and avoid, remedy or mitigate any adverse effects of human activities on freshwater bodies. All decision-making under the RMA needs to achieve this overriding purpose.

Sustainable Management

Section 5(2)

... managing the use, development and protection of natural and physical resources in a way, or at a rate, which enables people and communities to provide for their social, economic, and cultural wellbeing and for their health and safety while –

- (a) Sustaining the potential of natural and physical resources (excluding minerals) to meet the reasonable foreseeable needs of future generations; and
- (b) Safeguarding the life-supporting capacity of air, water, soil, and ecosystems; and
- (c) Avoiding, remedying or mitigating any adverse effects of activities on the environment.

Figure 7.2 Definition of sustainable management under the RMA

Matters of national importance

Section 6 of the RMA identifies matters of national importance, which decision-makers are required to recognise and provide for. Many of these matters are of particular relevance to freshwater management. They are described in more detail below.

Matters of national importance under the RMA

Section 6

- The preservation of the natural character of the coastal environment (including the coastal marine area), wetlands, and lakes and rivers and their margins, and the protection of them from inappropriate subdivision, use and development (section 6(a))
- The protection of outstanding natural features and landscapes from inappropriate subdivision, use and development (section 6(b))
- The protection of areas of significant indigenous vegetation and significant habitats of indigenous fauna (section 6(c))
- The maintenance and enhancement of public access to and along the coastal marine area, lakes, and rivers (section 6(d))
- The relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tapu, and other taonga (section 6(e))
- The protection of historic heritage from inappropriate subdivision, use, and development (section 6(f))
- The protection of recognised customary activities (section 6(g))

Figure 7.3 Matters of national importance under the RMA

The preservation of the natural character of ... wetlands, and lakes and rivers and their margins, and the protection of them from inappropriate subdivision, use and development (section 6(a))

The term 'natural character' is not defined in the RMA. It is a difficult concept to define because it incorporates a complex mix of physical, ecological, perceptual and cultural elements. The term encompasses the natural world (and natural elements within it), how people perceive it, and what it means to them.

Notwithstanding the difficulties in defining natural character, generally accepted approaches to applying the concept on the ground, have been developed through case law and professional practice. Natural character comprises three main components: natural processes, natural elements and natural patterns.³

Natural processes include the action of rain and wind, the movement of water, the life cycles of plants and animals and the functioning of broader ecosystems. Natural processes produce *natural elements* such as river channels, banks, gorges, floodplains, vegetation and the presence of wildlife. The distribution of these elements over an area form broader *natural patterns*.

Natural character will exist to some degree in all fresh water bodies. But not all natural character is protected under the RMA. The degree of protection afforded is related to the degree of naturalness of the water body – areas where there has been little human modification will receive higher protection than areas which are highly modified.



Figure 7.4 Lake McCrae (Fish and Game) *The preservation of the natural character of wetlands, lakes and rivers and their margins is a matter of national importance under the RMA*

When assessing the natural character of a freshwater body, Paul Mosley⁴ suggests that the following should be taken into account:

- Location, setting and context within the broader environment
- Topography, landforms and land-forming processes, including specific landform features such as a gorge
- Vegetative ground cover, biota, ecosystems, and ecological processes
- Combination of natural elements, natural patterns, and natural processes
- Resilience to disturbance, potential naturalness, and likely future recovery
- Presence or absence of built structures and artificial patterns, and their relative dominance with respect to natural elements

The protection of outstanding natural features and landscapes from inappropriate subdivision, use and development (section 6(b))

Some water bodies will comprise an outstanding natural feature on their own. Many will be part of larger outstanding natural landscapes. Such landscapes will often include hillsides, water bodies, vegetation and wildlife. Natural features and landscapes are considered ‘outstanding’ if they are conspicuous, eminent or remarkable within the context of the district or region.⁵ They are likely to comprise significant undeveloped freshwater bodies such as lakes and wild and scenic rivers. Even where water bodies have been partially modified by development they can still comprise outstanding natural features or landscapes. An example is Lake Taupo, where the lake level is managed as a hydro storage system, but which is still iconic and retains outstanding landscape qualities.



Figure 7.5 Lake Wakatipu (Raewyn Peart) *Some water bodies are outstanding natural features on their own and many are part of larger outstanding natural landscapes. They are to be protected as a matter of national importance.*

The protection of areas of significant indigenous vegetation and significant habitats of indigenous fauna (section 6(c))

Whether any particular freshwater body provides significant habitats of indigenous fauna is something that is determined within the context of the region or district. Factors that are taken into account include representativeness, rarity, intactness, size and ecological viability. Criteria which can be applied to identifying significance are shown in Figure 7.6.

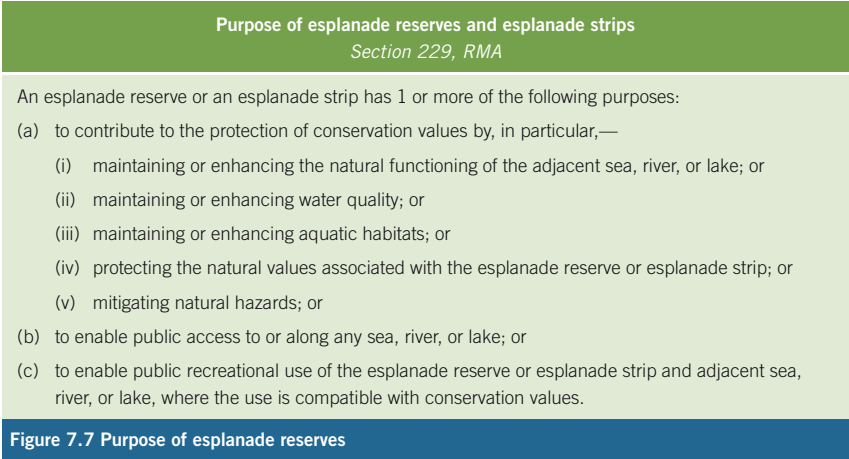
Criteria for identifying significant habitats of indigenous aquatic fauna within a region
• Is it a habitat for indigenous aquatic species or associations of such species which are classed as threatened, at risk or data deficient or which are endemic to the region? • Is the habitat type under-represented in an ecological district, ecological region or nationally? • Is it a habitat that prior to human settlement was nationally uncommon such as geothermal or kaarst ecosystems? • Is it a wetland habitat? • Is it an area of naturally occurring habitat that is large relative to other examples of similar habitat type and which contains all or almost all indigenous aquatic species typical of that habitat type? • It is an aquatic habitat that is critical to the self sustainability of an indigenous species within a catchment of the region and which contains healthy, representative populations of that species? • Is it an area of habitat that is a healthy and representative example of its type because its structure, composition and ecological processes are largely intact; and it is protected from the adverse effects of plant and animal pests and of adjacent landuse and can maintain its ecological sustainability over time? • It is a habitat that forms part of an ecological sequence that it either not common in the region or an ecological district or is an exceptional, representative example of its type? Source: Adapted from Environment Waikato, Appendix 1, Regional Policy Statement Working Draft 2010

Figure 7.6 Criteria for identifying significant habitats of indigenous aquatic fauna within a region

The maintenance and enhancement of public access to and along ... lakes, and rivers (section 6(d))

A key mechanism to provide public access to water bodies is through the creation of esplanade reserves along the edges of rivers and lakes. The default position under the RMA (section 230) is that an esplanade reserve 20 metres in width must be set aside when land is subdivided into a lot less than 4 hectares in size along the bank of any river (whose bed has an average width of three metres or more) or margin of any lake (where the lake bed is 8 hectares or more in size). This requirement can, however, be reduced by a rule in a district plan or a resource consent.

Esplanade reserves have a range of purposes in addition to enabling public access. They are a key tool for protecting the conservation values of rivers and lakes (See Figure 7.7):



An esplanade reserve vests in the territorial authority and is managed by the council as a local purpose reserve for esplanade purposes under the Reserve Act 1977. The boundary of the reserve is fixed, and therefore if erosion of the river bank or lakeshore occurs, the effective width of the reserve will decrease.

The RMA also provides for an esplanade strip to be created instead of a reserve (section 232). If an esplanade strip is created, private ownership of the land is retained and the land owner manages the land saving the council the expense of additional land management. The strip creates a public right across the land which is registered on the land title. The boundary of the strip is not fixed and it moves when the location of the river bank or lakeshore moves. The property owner can apply to the council to vary or cancel the strip so it may not be as permanent as an esplanade reserve.



Figure 7.8 Blue Lake, Rotorua (Raewyn Peart) The maintenance and enhancement of public access to and along lakes and rivers is a matter of national importance

The relationship of Māori and their culture and traditions with their ancestral ... water, sites, wāhi tapu, and other taonga (section 6(e))

Water is a fundamental taonga for Māori. In Māori culture everything has a life force sustained by water, which is the source of life. Water has an inherent mauri, or life essence, and this mauri must be healthy to enable the water to sustain healthy ecosystems and support cultural uses. Māori custom has strict rules surrounding the use of water, to ensure that the mauri of the water is retained.⁶

Traditionally water intended for different uses was never mixed – water that was to be used for the preparation of food was kept separate from all other water. According to Māori teaching, if water is altered (or contaminated) then it becomes weak and its powers are lessened. If water is contaminated, everything is affected and becomes diseased and life is confused. Thus, mixing waters used for gathering food with waters contaminated with human waste, for example through the discharge of sewage into a river, is deeply objectionable on spiritual grounds.⁷

Māori have a long standing tradition of harvesting aquatic resources for food (mahinga kai) including fish, birds and invertebrates. Degradation of the health of freshwater bodies has reduced the abundance of these resources. In

addition, tangata whenua may have lost physical or legal access to rivers and lakes to harvest food with land use and ownership changes.

Some deeds of settlement identify taonga freshwater species which are particularly important to specific iwi. For example the Ngai Tahu Claims Settlement Act 1998 lists as taonga species kōkopu/hawai (giant bully), kowaro (Canterbury mudfish), paraki/ngaiore (common smelt), piripiripohatu (torrentfish) and taiwharu (giant kōkopu) (schedule 98).

There are many places associated with freshwater bodies which are of particular significance to Māori, and the ongoing relationship of Māori with these sites is a matter of national importance. Lakes and rivers feature in creation stories and historical accounts. Many have traditionally been used as access routes or transport courses. Many lakes, rivers and wetlands are strongly associated with settlements, historic sites and tribal identity more generally.⁸

The protection of historic heritage from inappropriate subdivision, use, and development (section 6(f))

Historic sites associated with freshwater bodies include early settlements, logging remnants such as kauri dams, infrastructure associated with mining including water races and mine tailings, other early infrastructure including old bridges, culverts, ramps, seawalls, and water-powered electricity generation. The RMA indicates that these sites should be protected as a matter of national importance. This can be achieved through identifying the sites in regional and district plans (a process often termed ‘scheduling’) and including rules in the plans preventing incompatible activities.

Other matters

The RMA also identifies a set of ‘other matters’ in section 7 to which decision-makers are required to ‘have particular regard’. This is a less rigorous test than for matters of national importance. Many section 7 matters have particular relevance to the management of freshwater including the efficient use of water, the benefits to be derived from the use and development of renewable energy (including hydro and geothermal energy), the effects of climate change on water quantity, the intrinsic value of freshwater ecosystems, the maintenance and enhancement of the amenity and quality of freshwater bodies and the protection of the habitat of trout and salmon which includes many rivers and lakes.

Other matters under the RMA

Section 7

In achieving the purpose of this Act, all persons exercising functions and powers under it, in relation managing the use, development and protection of natural and physical resources, shall have particular regard to –

- (a) kaitiakitanga
- (aa) the ethic of stewardship
- (b) the efficient use and development of natural and physical resources
- (ba) the efficiency of the end use of energy
- (c) the maintenance and enhancement of amenity values
- (d) intrinsic values of ecosystems
- (f) maintenance and enhancement of the quality of the environment
- (g) any finite characteristics of natural and physical resources
- (h) the protection of the habitat of trout and salmon
- (i) the effects of climate change
- (j) the benefits to be derived from the use and development of renewable energy

Figure 7.9 Other matters under the RMA

One of the ‘other matters’ in section 7 refers to kaitiakitanga. This is defined in section 2 as meaning ‘the exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources; and includes the ethic of stewardship’. The concept of kaitiakitanga is often not well understood. It includes several key concepts, with kaitiakitanga being:⁹

- Integrated with the spiritual, cultural and social life of tangata whenua
- Holistic across land and water
- Inclusive of people within the concept of the environment
- Locally defined and exercised
- Not focused on ownership, but on authority and responsibility
- Concerned with both sustainability of the environment and utilisation of its benefits

Treaty of Waitangi

Section 8 requires all persons exercising functions and powers under the RMA to ‘take into account the principles of the Treaty of Waitangi’. This includes duties to act reasonably and in good faith and to actively protect Māori interests. Consultation is also a principle of the Treaty, but section 36A of the RMA explicitly states that neither an applicant nor a local authority have a duty to

consult any person, which includes Māori, about a resource consent application unless this is required under other legislation.

Policy and planning framework

Policies and plans are developed within the broad framework provided by Part 2 of the RMA in order to provide more detailed guidance on decision-making in individual cases. These documents are prepared at the national, regional and district levels.

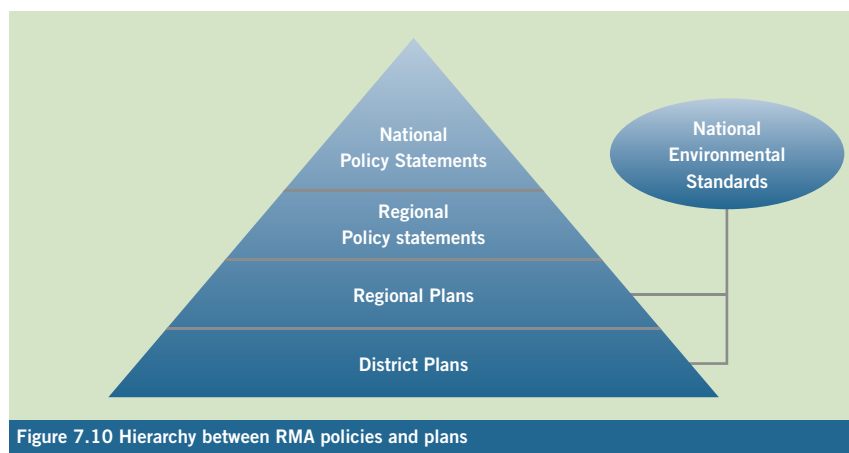


Figure 7.10 Hierarchy between RMA policies and plans

National instruments

The RMA makes provision for central government to develop national instruments which can guide the management of freshwater by councils. There are two types of such instruments, national policy statements and national environmental standards.

National policy statements enable the Minister for the Environment to establish objectives and policies (but not rules) on resource management. There is no requirement for the Minister to prepare a national policy statement to guide freshwater management. However, if he or she chooses to do so, the Ministry for the Environment has the role of preparing a draft national policy statement. This then goes through a public submission process which usually involves a public hearing of submissions by an independent board of inquiry.

In the case of freshwater within the coastal environment, the Department of Conservation prepares a draft New Zealand Coastal Policy Statement (NZCPS) which then goes through a similar public submission process and is approved by the Minister of Conservation. The preparation of a NZCPS is mandatory under the RMA. The coastal environment includes estuaries, dune lakes and other rivers, lakes and parts of the catchment which are influenced by the sea.

A national policy statement only takes effect after the Minister has recommended its adoption and it is approved by the Governor General in Council (effectively cabinet). This is the normal process for adopting regulations.

A NZCPS was adopted in 1994 and has since been replaced by the NZCPS 2010. The NPS for Renewable Electricity Generation 2011 also has implications for freshwater management. Policy E2 requires regional policy statements, regional and district plans to include objectives, policies, and methods (including rules within plans) to provide for the development, operation, maintenance and upgrading of new and existing hydro-electricity generation activities to the extent applicable to the region or district. The NPS for Freshwater Management was issued on 12 May 2011. The relevant provisions are discussed in detail below.

All regional policy statements and regional and district plans must 'give effect' to national policy statements and this may require that the council planning documents be amended. In addition, councils must have regard to any relevant national policy statement when considering an application for a resource consent. A local authority must also take any other action specified in the national policy statement. This can include making changes to a policy statement or plan without going through a public submission process (section 55(2A)).

National environmental standards provide the opportunity for the Minister for the Environment to promote the adoption of consistent standards relating to water management and other matters at the regional and district levels. Standards can be qualitative or quantitative and can include a wide range of subject matter including stipulating minimum environmental standards, methods of classifying water, and methods of ascertaining the state of freshwater bodies.

Similar to the situation with national policy statements, draft environmental standards are prepared by the Ministry for the Environment, are open for public submission (but usually only written with no hearing) and then are adopted by Order in Council on the recommendation of the Minister for the Environment.

A council must amend its plan if a rule duplicates or conflicts with a provision in a national environmental standard. Such a plan change will have immediate effect and will not undergo a public notification and hearing process. Consent authorities must have regard to any relevant national environmental standards

when considering an application for a resource consent. The conditions of existing water, discharge or coastal permits can also be reviewed where this is required to meet the standards in a national environmental standard.

NATIONAL ENVIRONMENTAL STANDARD FOR SOURCES OF HUMAN DRINKING WATER

This standard came into effect on 20 June 2008. This is designed to reduce the risk of drinking water sources, such as rivers and groundwater, becoming contaminated by requiring regional councils to consider the effects of upstream activities on drinking water sources in decisions on resource consents and plans.

Regional councils are required to decline discharge or water permits that are likely to result in community drinking water becoming unsafe for human consumption following existing treatment and must also be satisfied that permitted activities in regional plans do not result in such effects. Councils must also include conditions on relevant resource consents requiring drinking water suppliers to be notified if significant unintended events occur that may adversely affect sources of human drinking water.

A national environmental standard for sources of human drinking water came into effect on 20 June 2008. The Ministry for the Environment is currently developing a national environmental standard for ecological flows and water levels and on-site wastewater systems.

The RMA also enables the Minister for the Environment to develop *regulations* under section 360(1)(d). Unlike national environmental standards, regulations can apply directly to existing resource consents without the need for review. The Resource Management (Measurement and Reporting of Water Takes) Regulations 2010 have been made to improve the management of water resources by improving the way that water takes are measured (see Chapter 9).

National Policy Statement for Freshwater Management 2011

Overarching objectives

The NPS sets out several objectives for the management of freshwater. The overarching objective in A1 and B1 is to safeguard the life-supporting capacity, ecosystem processes and indigenous species including their associated ecosystems of fresh water, in sustainably managing the use and development of land; the taking, using, damming or diverting of freshwater; and of discharges of contaminants. Objective C1 is to improve integrated management of fresh water and the use and development of land in whole catchments. This is necessary if the other objectives are to be achieved.

There is also a key objective in respect of water quality, objective A2 that the overall quality of fresh water within a region is maintained or improved while:

- (a) Protecting the quality of outstanding freshwater bodies
- (b) Protecting the significant values of wetlands
- (c) Improving the quality of freshwater in water bodies that have been degraded by human activities to the point of being over-allocated

'Outstanding freshwater bodies' are defined in the interpretation section as those water bodies with outstanding values, including ecological, landscape, recreational and spiritual values. In order to sufficiently protect such water bodies and wetlands they must be identified in the regional planning instruments, ideally by spatially mapping them. There must also be appropriate policies and methods to ensure the quality of outstanding freshwater bodies and the significant values of wetlands are protected. The objective to protect significant values of wetlands is also included in the water quantity section in objective B4.

To achieve the objective of improving the quality of degraded freshwater, regional planning instruments must first set freshwater objectives and limits. The relevant policies related to setting freshwater objectives and limits are discussed in further detail below. Once freshwater objectives and limits have been set it will then be possible to assess those water bodies that have been degraded by human activities to the point of being over-allocated. 'Over-allocation' is defined in the interpretation section as the situation where the resource has been allocated to users beyond a limit or is being used to a point where a freshwater objective is no longer being met. Local authorities must then ensure there are appropriate controls on land use, discharges and water takes to ensure that the quality of freshwater in such water bodies is improved.

Setting and complying with freshwater objectives, limits and targets

In the Preamble it is noted that a key purpose of the NPS is to set enforceable quality and quantity limits. The NPS includes policies A1 and B1 that direct local authorities to make or change regional plans to ensure they establish freshwater objectives and set quality and quantity limits. 'Freshwater objective' describes the intended environmental outcome(s). The interpretation section states that 'limit' is the maximum amount of resource use available, which allows a freshwater objective to be met.

The objectives and limits are to give effect to the objectives in the NPS. Critically this includes Objective A1 and B1 to safeguard the life-supporting capacity, ecosystem processes and indigenous species including their associated ecosystems of fresh water. Regional Councils are also to have regard to the reasonably foreseeable impacts of climate change and the connection between water bodies when establishing objectives and setting limits.

‘Environmental flows and/or levels’ are a type of limit which describes the amount of water in a body of fresh water (except ponds and naturally ephemeral water bodies) which is required to meet freshwater objectives. As noted above, these are the intended environmental outcomes. The interpretation section provides that environmental flows for rivers and streams must include an allocation limit and a minimum flow (or other flow/s). Environmental levels for other bodies of fresh water must include an allocation limit and a minimum water level (or other level/s).

Regional councils are required to make or change regional plans to establish methods (including rules) to avoid over-allocation. This is likely to mean that rules are required to control discharges and the use of water to ensure freshwater quality and quantity limits are met. Regional councils must also change plans to provide for the efficient allocation of freshwater to activities, within the limits set.

Policy A2 provides that where water bodies do not meet the freshwater objectives, every regional council is to specify targets in the form of limits and implement regulatory and/or non-regulatory methods to assist in the improvement of water bodies to meet the targets within a defined timeframe.

Regional councils are to impose conditions on discharge permits to ensure the limits and targets can be met. Provision is also made for the creation of rules requiring the adoption of the best practicable option to prevent or minimise any actual or likely adverse effect on the environment of any discharge of contaminants where it will or may enter water.

Integrated management

As noted above, there is an objective relating to the improvement of integrated management of fresh water and the use and development of land. Policy C1 provides that regional councils are to manage catchments in an integrated and sustainable way, so as to avoid, remedy or mitigate adverse effects, including cumulative effects.

In order to implement this policy it is imperative that regional councils have knowledge of the activities that impact on the life-supporting capacity of fresh water, ecosystem processes and indigenous species including their associated ecosystems; baseline data of existing discharges and water use in a catchment; project future discharges and uses; understand any trends in water quality and quantity; and use this information to set catchment limits for discharges and water use. Ongoing monitoring will be required to ensure the policy is implemented.

Policy C2 requires regional councils to make or change regional policy statements to provide for the integrated management of the effects of the use and

development of land on fresh water, including encouraging the co-ordination and sequencing of regional and/or urban growth, land use and development and the provision of infrastructure.

Addressing over-allocation for water quantity

Objective B2 is to avoid any further over-allocation of fresh water and phase out existing over-allocation. In the interpretation section ‘over-allocation’ is defined as the situation where the resource has been allocated to users beyond a limit or is being used to a point where a freshwater objective is no longer being met.

Policy B5 requires every regional council to ensure that no decision will likely result in future over-allocation. If water bodies are over-allocated policy B6 requires every regional council to set a defined timeframe and methods in regional plans by which over-allocation must be phased out, including by reviewing water permits and consents to help ensure the total amount of water allocated in the water body is reduced to the level set to give effect to Policy B1.

Efficiency requirements

Objective B3 is to improve and maximise the efficient allocation and efficient use of water. ‘Efficient allocation’ is defined in the interpretation section as including economic, technical and dynamic efficiency.

Fresh water is to be efficiently allocated to activities within the limits set to give effect to Policy B1. Given the multiple definitions of efficiency this is likely to mean that regional councils are required to consider what the best economic use of the resource is and ensure that standards for the use of water are established to provide for technical efficiency. Examples of measures to provide for technical efficiency are set out in Chapter 12.

Regional Councils are to make or change regional plans to ensure the plans state criteria by which applications for approval of transfers of water take permits are to be decided, including to improve and maximise the efficient allocation of water. Policy B4 requires every regional council to identify methods in regional plans to encourage the efficient use of water. Such measures could include the creation of incentives for water users to use the minimum required. Specific examples are set out in Chapter 12.

Tangata whenua roles and interests

Objective D1 is to provide for the involvement of iwi and hapu, and to ensure that tangata whenua values and interests are identified and reflected in the management of fresh water including associated ecosystems, and decision-

making regarding freshwater planning, including on how all other objectives of this national policy statement are given effect to.

Specific guidance for local authorities is set out in Policy D1 which states that reasonable steps shall be taken to:

- (a) Involve iwi and hapu in the management of fresh water and freshwater ecosystems in the region
- (b) Work with iwi and hapu to identify tangata whenua values and interests in water and freshwater ecosystems in the region and
- (c) Reflect tangata whenua values and interests in the management of, and decision-making regarding, fresh water and freshwater ecosystems in the region.

Transitional provisions

The NPS includes transitional provisions that require regional councils to amend regional plans and include specific policies without using the process in Schedule 1 until any others changes to give effect to specific policies of the NPS have become operative.

Progressive implementation programme

Policy E1 of the NPS provides that every regional council is to implement the NPS as promptly as is reasonable in the circumstances. Where a regional council is satisfied that it is impracticable for it to complete implementation of a policy fully by 31 December 2014, the council may implement it by a programme of defined time-limited stages by which it is to be fully implemented by 31 December 2030. Any programme of time-limited stages has to be public notified within 18 months from 12 May 2011 and the Council is required to publicly report on progress.

As noted in the Preamble of the NPS for Freshwater Management 2011, the management of coastal water and fresh water requires an integrated and consistent approach. The NZCPS 2010 addresses issues with water quality in the coastal environment.

In the preamble of the NPS it is noted that the Minister for the Environment intends to seek an independent review of the implementation and effectiveness of the NPS in achieving all its objectives and policies, no later than five years after it comes into force. Given the time it takes to prepare plan changes and complete the Schedule 1 process, it is unlikely that the objectives and policies will have been achieved after five years. This is particularly so for the objectives and policies related to water quality as there is a substantial lag time between discharging contaminants onto land and observing the effects in a water-body.

Regional policies statements and plans

Regional policy statements, prepared by regional councils, are designed to provide an overview of resource management issues throughout the region and to identify policies and methods to achieve the integrated management of natural and physical resources (section 59). This enables a strategic and integrated approach to be applied to water management, which can be considered on a region-wide basis.

Regional policy statements do not include rules and therefore do not have direct impact on activities on the ground. However, regional plans and district plans which do have rules, are required to 'give effect to' regional policy statements. This means that regional policy statements primarily have effect through directing or influencing the content of regional and district plans.

Regional plans are the key planning document for determining what activities are able to be undertaken to use, divert or impound freshwater or to discharge contaminants into freshwater. The rules in the plan determine what activities can take place without consent (permitted activities), and where a resource consent is required, what matters will be considered when a specific consent is applied for.

Section 65(1) states a regional council 'may' prepare a regional plan for the whole or part of its district for any of its functions including freshwater management. Councils were not required to prepare regional plans to provide guidance on water management although most have done so. However, the NPS now requires every regional council to make or change regional plans to ensure the plans establish freshwater objectives, set freshwater quality limits and set environmental flows and/or levels for all bodies of freshwater in its region. Where a plan has not been prepared, all activities which use or pollute water (apart from the small number of non-commercial exceptions) will require a resource consent. When an application for such consent is processed, it will be assessed against the criteria in the RMA.

The preparation of regional plans, and undertaking the public submission and appeal process set out in Schedule 1 of the RMA, is often a resource intensive and lengthy process. Some councils have therefore explored more efficient ways of undertaking regional planning such as the preparation of joint regional policy statements and plans. This approach has been adopted by the Horizons Regional Council which has prepared the 'One Plan' combining the existing regional policy statement and six regional plans which cover air, land and water, the coast, beds of rivers and lakes, Manawatu catchment water quality, Oroua catchment water allocation and river flows.

District plans

District plans are focused on managing the effects of land use, including the effects on freshwater bodies. Land use can impact on freshwater bodies through such activities as the clearance of vegetation, earthworks which change land contours, extension of the area covered by impermeable surfaces, and enabling sediment and pollutants to reach water bodies. District plans also control subdivision, the provision of esplanade reserves along the margins of lakes and rivers, and activities on the water surface of rivers and lakes such as boating. Objective C1 of the NPS for Freshwater Management 2011 is to improve integrated management of fresh water and the use and development of land in whole catchments. This is likely to have implications for district councils in managing the effects of land use.

Water conservation orders

In 1981, provisions were first incorporated into the Water and Soil Conservation Act 1967 to provide for the protection of wild and scenic rivers. This was in response to the increasing use of rivers for hydro-electric power generation and irrigation and the need to ‘close off, wholly or partially, the developmental option in relation to the remaining wild and scenic rivers of New Zealand before it was too late.’¹⁰ These provisions were carried over into Part 9 of the RMA.

Under Part 9, a water conservation order is a tool that can be used to recognise and protect the outstanding amenity or intrinsic values that a water body provides, in either a natural or a modified state. An order can be created to protect:

- The habitat of terrestrial or aquatic species
- Fisheries
- Wild, scenic or other natural characteristics
- Scientific and ecological values
- Recreational, historical, spiritual or cultural values
- Characteristics of outstanding significance in accordance with tikanga Māori

Whether or not a value is “outstanding” is assessed on a national basis, with the exception of characteristics in accordance with Tikanga Māori.

Water conservation orders can apply to rivers, lakes, streams, ponds, wetlands, geothermal water or aquifers. They work by identifying the

outstanding characteristics or values of a water body and then put in place restrictions or prohibitions to protect those characteristics. They affect how the regional council can manage the resource. In particular they can restrict management measures that affect:

- Quantity, quality, rate of flow or level of the water body
- Maximum, minimum or range of levels or flows, or the rate of change of levels of flows
- Maximum allocation for extraction or maximum contaminant loading
- Ranges of temperature and pressure in the water body

Anyone can apply for a water conservation order. Once such an order is in place, resource consents granted after its creation must not be contrary to any restriction, prohibition or other provision of the order. In addition, regional policy statements, regional plans and district plans may not be inconsistent with the order. After a water conservation order has been in place for two years, any person can apply to revoke or amend the order so it does not necessarily provide permanent protection.

As at November 2009 there were 16 water conservation orders in force in New Zealand, in relation to lakes and rivers located all over the country.¹¹

River	Date
Motu River	1984
Rakaia River	1988
Lake Wairarapa	1989
Manganuioteao River	1989
Lake Ellesmere	1990
Ahuriri River	1990
Grey River	1991
Rangitikei River	1993
Kawarau River	1997
Mataura River	1997
Buller River	2001 and 2008 (amendment)
Motueka River	2004
Mohaka River	2004
Rangitata River	2006
Oreti River	2008

Figure 7.11 Water conservation orders in place

Source: <http://www.mfe.govt.nz/issues/water/freshwater/water-conservation/>



Figure 7.12 Rakaia River (Fish and Game New Zealand) *Water conservation orders, such as the order applying to the Rakaia River, protect the outstanding amenity or intrinsic values that a water body provides, in either a natural or a modified state*

Resource consents

Most significant uses of water, and activities which result in the pollution of water, will require a resource consent before they can be legally carried out (see chapters 9 and 10). Consents relating to the taking or impounding of freshwater are called *water permits* and consents relating to discharges into freshwater bodies are called *discharge permits*.

When considering applications for water and discharge permits, consent authorities need to consider a range of matters including the actual and potential effects on the environment of allowing the activity and the provisions of relevant national environmental standards, national and regional policy statements and regional and district plans.

Water and discharge permits can be granted for a maximum period of 35 years, but may be granted for shorter periods. The duration will usually be specified in the consent, otherwise it is five years (section 123).

When holders of resource consents wish to renew the consent (by applying for a new consent for the same activity), they obtain priority over other

applicants for the same resource (section 124B). In addition, when considering the application, the consent authority must consider (section 124(B)):

- The efficiency of the use of the resource by the applicant
- The use of good industry practice by the applicant
- Any enforcement orders or convictions lodged against the applicant, including how serious they were and how recently they occurred
- The value of investment by the applicant (section 104(2A))

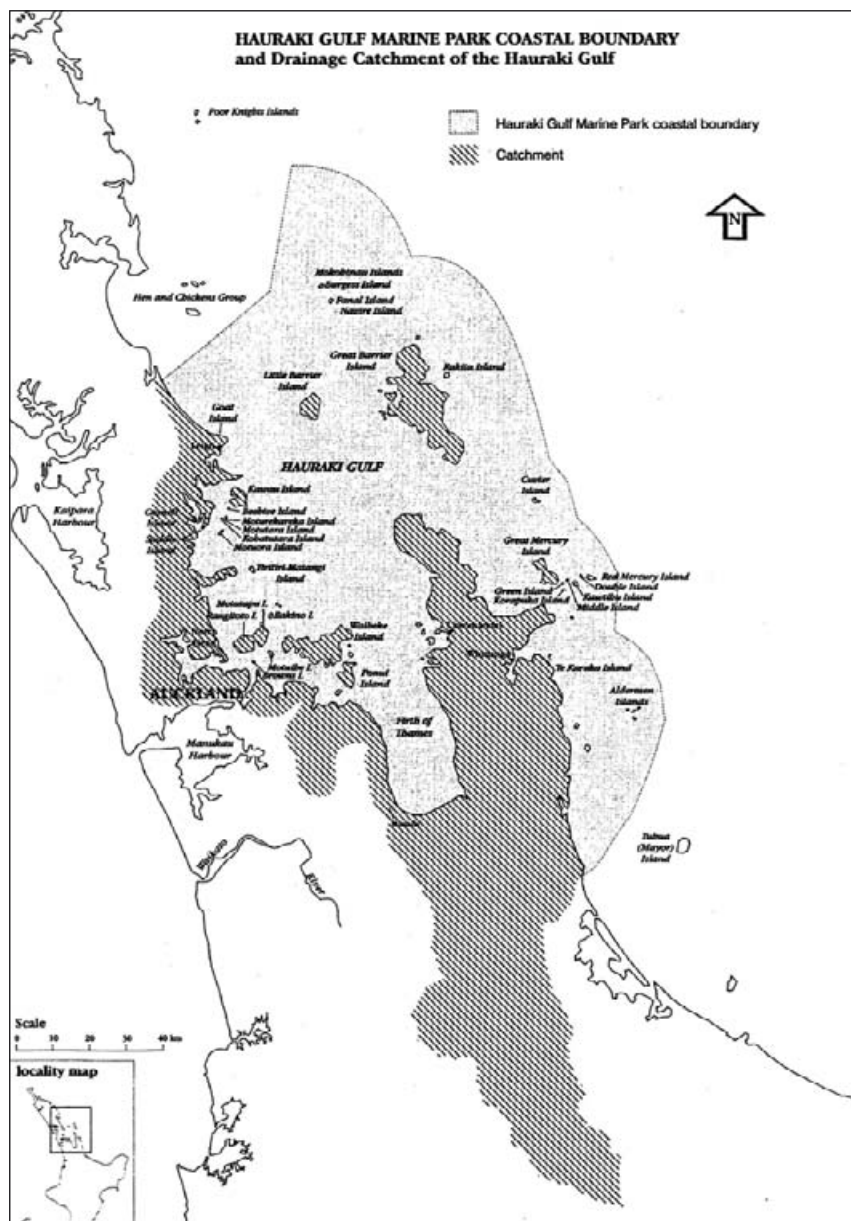
Regional legislation

The RMA provides the general framework for freshwater management in New Zealand, but in some areas regional legislation also applies.

Hauraki Gulf Marine Park Act 2000

This legislation provides special recognition for the Hauraki Gulf, including the catchments which drain into the coastal marine area, and this has implications for freshwater management. In the preamble the Act states the: 'Hauraki Gulf has a quality and diversity of biology and landscape that makes it outstanding within New Zealand. The islands of the Gulf are valued as the habitats of plants and animals, once common, now rare, and are often the only places in the world where these species exist naturally.' The area covered by the legislation is shown in Figure 7.10.

Section 13 of the HGMPA provides that all persons exercising powers or carrying out functions for the Hauraki Gulf under any Act specified in Schedule 1, which includes the RMA, must have 'particular regard to' the provisions of sections 7 and 8 of the HGMPA. This is reaffirmed in section 9(4) of the HGMPA which provides that a consent authority must have regard to sections 7 and 8 when considering an application for a resource consent for the Hauraki Gulf, its islands and catchments.



Section 7 provides that the interrelationship between the Hauraki Gulf, its islands, and catchments and the ability of that interrelationship to sustain the life-supporting capacity of the environment of the Hauraki Gulf and its islands are matters of national significance. The management of freshwater within the Gulf's catchments has a significant impact on the health of the coastal marine area and therefore is of importance under the legislation.

Section 8 of the HGMPA provides that to recognise the national significance of the Hauraki Gulf it should be managed for several specific objectives including the protection and, where appropriate, the enhancement of the life-supporting capacity of the environment and the natural, historic and physical resources of the Hauraki Gulf, its islands, and catchments [including freshwater]. This includes those resources with which tangata whenua have an historic, traditional, cultural, and spiritual relationship and those resources which contribute to the recreation and enjoyment of the Hauraki Gulf for the people and communities of the Hauraki Gulf and New Zealand.

Sections 7 and 8 must be treated as a New Zealand coastal policy statement. In addition, regional councils and territorial authorities must 'give effect' to sections 7 and 8 as though they were a national policy. Sections 9(2) and (3) of the HGMPA provide that territorial authorities must ensure that any part of a regional policy statement or a regional or district plan that applies to the Hauraki Gulf, its islands, and catchments, does not conflict with sections 7 and 8.

The HGMPA also establishes the *Hauraki Gulf Forum* to manage the Hauraki Gulf, its islands and catchments. This Forum includes representatives from various Ministries and the relevant local authorities as well as iwi appointees. The *Hauraki Gulf Marine Park* is also established under the HGMPA. It consists of conservation areas and various types of reserves administered by the Crown.

Waitakere Ranges Heritage Area Act 2008

This legislation seeks in its *purpose* to recognise the national, regional and local significance of the Waitakere Ranges Heritage Area and to promote the protection and enhancement of its heritage features for present and future generations (section 3(1)). The approximate boundaries of the heritage area are shown in Figure 7.11.

The heritage features of the area identified in the legislation include many of direct relevance to freshwater (section 7):

- Aquatic ecosystems of prominent indigenous character, including those within large continuous areas of primary and regenerating lowland and coastal rainforest, wetlands and dune systems with intact ecological sequences

- The naturally functioning streams that rise in the eastern foothills and contribute positively to downstream urban character, stormwater management, and flood protection
- The historical, traditional, and cultural relationships of people, communities and tangata whenua with the area and their exercise of kaitiakitanga and stewardship
- The evidence of past human activities in the area including those in relation to water impoundment and supply
- The public water catchment and supply system, the operation, maintenance and development of which serves the people of Auckland

The *objectives* of the heritage area of particular importance to freshwater management include to protect, restore and enhance the area and its heritage features, to maintain aquatic ecosystems in the area, and to protect those features that relate to its water catchment and supply functions (section 8).

Councils must give effect to the purpose of the legislation and its objectives when preparing or reviewing a regional policy statement, regional plan or district plan (sections 10 and 11). In addition, when considering a resource consent application, councils must have particular regard to the purpose and objectives of the legislation (section 13). Councils may also reject requests for private plan changes if they are inconsistent with the purpose or objectives of the legislation (section 12).

There is also provision for the preparation of *local area plans* by territorial authorities (district and city councils). These plans are intended to provide more detail on implementing the objectives of the legislation in local areas, but have no legal impact on decisions made under the RMA.

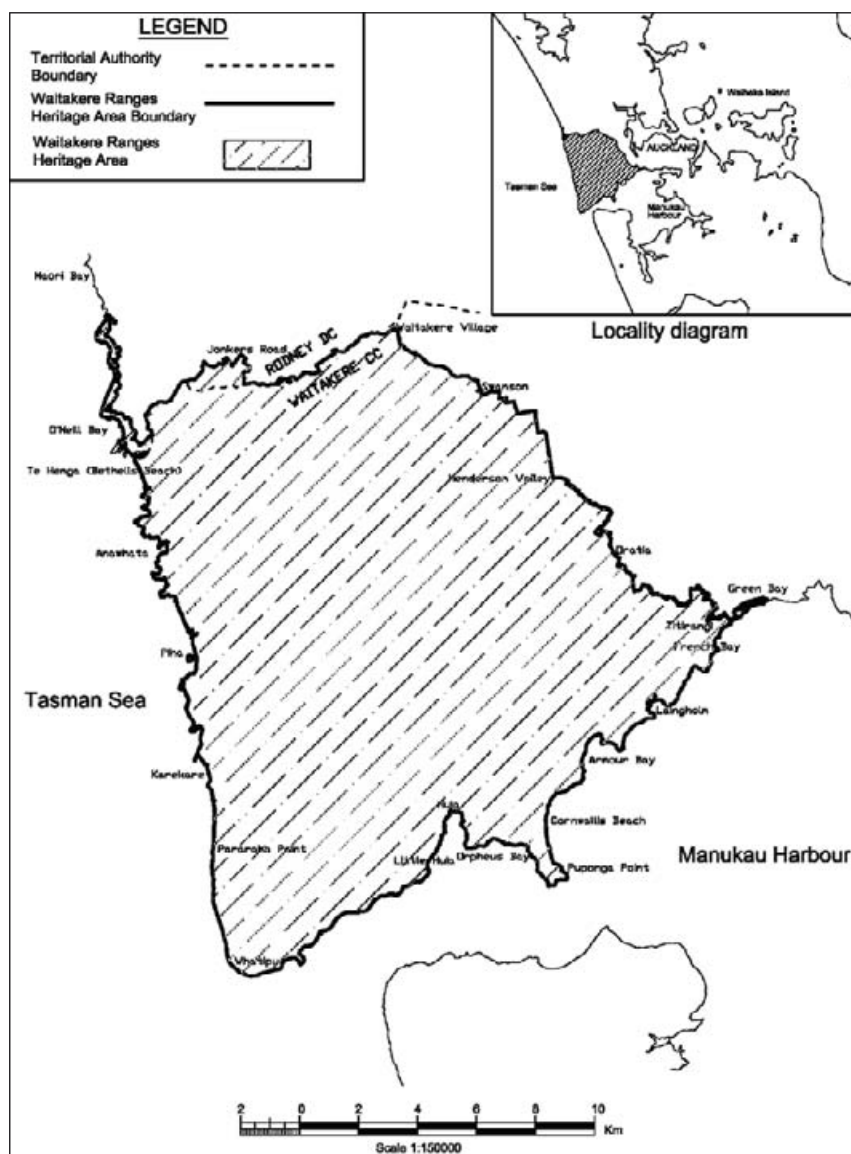


Figure 7.14 Area affected by the Waitakere Ranges Heritage Area Act (Schedule 1, Waitakere Ranges Heritage Area Act 2008)

Waikato-Tainui Raupatu Claims (Waikato River) Settlement Act 2010

The Waikato-Tainui Raupatu Claims (Waikato River) Settlement Act addresses a number of matters related to the Waikato River including resource management issues. The legislation establishes the *Waikato River Authority* which will set the primary direction for the Waikato River through a 'Vision and Strategy' to protect the health and wellbeing of the river for future generations. The Authority will be a co-governance entity overseeing river management. It will consist of 10 members appointed from the Waikato River Clean-Up Trust, iwi, and the Minister for the Environment in consultation with relevant Ministers. The Authority will:

- Advise on giving effect to the vision and strategy (see below)
- Act as a trustee of the Waikato River Clean-up Trust
- Carry out monitoring and reporting
- Periodically review the Vision and Strategy
- Request call ins under the RMA
- Appoint iwi commissioners to river-related resource consent hearings.

The *Vision and Strategy* is reproduced in the Second Schedule of the Act and is incorporated directly into the Waikato Regional Policy Statement. Environment Waikato is required to ensure that the policy statement does not remain inconsistent with the vision and strategy for any longer than is necessary to amend it. If there are any inconsistencies between the two documents, the vision and strategy will prevail.

Section 18 of the Settlement Act requires that within three months of the settlement date, the Authority must begin a review for the purpose of considering whether targets and methods should be developed for inclusion in the vision and strategy and whether the document needs to be amended.

Following this review, local authorities that come within the catchment of the Waikato River must review their district plans to initiate changes to ensure consistency between district plans and the vision and strategy. The vision and strategy will prevail over any national policy statements that affect the Waikato River, as well as the New Zealand Coastal Policy Statement. In addition, if the vision and strategy is more stringent than national environmental standards or water conservation orders it will also prevail.

The Settlement Act also establishes the *Waikato River Clean Up Trust* which has the objective to restore and protect the health and wellbeing of the Waikato River for future generations. The Trust will administer a contestable fund for projects and initiatives that will contribute to the restoration and protection of the health and wellbeing of the Waikato River. The Crown has stated that

it is contributing \$210 million to this fund which will be paid over 30 years. Provision has also been made for others to be able to contribute to the fund. The Authority is the Trust's sole Trustee.

The Trust will also have the discretion to prepare a *Waikato-Tainui Environmental Plan*. If the Trust decides to prepare an Environmental Plan, it will be prepared in consultation with Waikato-Tainui marae, and will be served on the Director General of Conservation, the chief executive of the Ministry of Fisheries, relevant local authorities and any other relevant agency. The environmental plan must be recognised in the same manner required under the RMA as any other planning document recognised by an iwi authority when council is reviewing or changing a planning document under the RMA. The environmental plan must also be had regard to when a consent authority considers an application for resource consent under section 104 of the RMA.

Joint management agreements must be entered into between each local authority and the Trust, and the agreements must provide for the local authority and the Trust to work together to (in relation to the vision and strategy):

- Set monitoring priorities, review results, and identify responses
- Prepare, review, change, or vary a RMA planning document
- Exercise duties, functions, or powers under Part 6 of the RMA in relation to applications for resource consents affecting the Waikato River.



Figure 7.15 Waikato River (Raewyn Peart) A co-governance arrangement has been put in place for the Waikato River under the Waikato-Tainui Raupatu Claims (Waikato River) Settlement Act

An *Integrated River Management Plan* will be developed between the Trust, iwi, local authorities, and relevant departments and agencies to manage aquatic

life, habitats and natural resources within the Waikato River. This plan will consist of at least three components, and each component will be required to be jointly approved by both the Trust and the relevant Minister in charge of each portfolio or Council. The components are conservation, fisheries, and regional council. Any other components of an Integrated River Management Plan must be jointly approved by the Trust and the agency that agreed on it.

Environment Canterbury (Temporary Commissioners and Improved Water Management) Act 2010

This legislation came into force on 13 April 2010 and has made some changes to the way the RMA is applied to freshwater management within the Canterbury region.

Part 2 of the legislation provides for the replacement of the elected regional councillors with commissioners appointed by the Minister for the Environment. The commissioners are to perform the role of elected councillors until the local body election due to be held in 2013. At that stage it is anticipated that a new council will be elected.

The legislation gives the Minister for the Environment broad regulatory-making powers which can be used to give the commissioners additional powers or to suspend the operation of parts of the RMA within the Canterbury region (section 31).

The legislation also enables the commissioners, with the prior approval of the Minister, to impose a moratorium on applications for water and discharge permits for any specified area. This power is designed to address the situation where freshwater resources are subject to high or increasing demand, to diminishing water quality or to full or over-allocation. In July 2010 the commissioners placed a moratorium on resource consent applications relating to water in the Hurunui River and its tributaries, operational from 2 August 2010 until 1 October 2011.¹² In November 2010 the commissioners also placed a moratorium on resource consent applications relating to water in the Waiau River, operational from 6 December 2010 until 1 October 2011.¹³

A significant part of the legislation deals with water conservation orders within the Canterbury region and substantially amends the provisions of Part 9 of the RMA in terms of the decision-making process. These provisions are described in more detail in chapter 11.

There is also special legislation which applies to the allocation of water within the Waitaki catchment in the Canterbury region (see Chapter 9).



Figure 7.16 Hurunui River (Chris Todd) *A moratorium has been put on resource consent applications relating to water in the Hurunui River and its tributaries*

Lake Wanaka Preservation Act 1973

The purposes of this Act are—

- To prevent the water in the body of the lake from being impounded or controlled by, or, as far as possible, obstructed by, any works except in an emergency
- To prevent the natural rate of flow of lake water between the outlet of the lake which forms the source of the Clutha River and the confluence of that river and the Cardrona River from being varied or controlled by any works except in an emergency
- To preserve, as far as possible, the water levels of the lake and its shoreline in their natural state
- To maintain and, as far as possible, to improve the quality of water in the lake.

The Otago Regional Council is required to have regard to the purposes of the legislation when exercising functions under the RMA, such as preparing a regional policy statement and regional plans, and determining resource consents

(Section 8). The Act places a prohibition on impounding or controlling the water in the lake, except in an emergency. It also establishes the Guardians of Lake Wanaka (section 5).

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Raewyn Peart

eight

Freshwater management bodies

Introduction

A large number of management authorities are involved in the management of freshwater in New Zealand. These include Ministers of the Crown, iwi and hapū, government departments, the Fish and Game Councils, regional councils, territorial authorities and guardian groups. Their respective roles are summarised in Figure 8.1 and described in more detail in the sections below.

Management body	Role in freshwater management	Key legislation
Minister for the Environment	Oversees freshwater management under the RMA Recommends the issue of national policy statements on freshwater Recommends the making of national environmental standards related to freshwater Recommends the approving of requiring authorities for water infrastructure Recommends the issue of water conservation orders Calls-in matters of national significance related to freshwater Oversees the Ministry for the Environment	Environment Act 1986 Resource Management Act 1991 Environment Canterbury (Temporary Commissioners and Improved Water Management) Act 2010
Ministry for the Environment	Provides policy advice to the Minister on freshwater management Disseminates information on freshwater management Engages in collaborative efforts to improve freshwater management	Environment Act 1986 Resource Management Act 1991
Environmental Protection Authority	Processes freshwater management matters of national significance which are 'called in'	Resource Management Act 1991
Minister of Conservation	Oversees coastal management under the RMA including freshwater within the coastal environment Recommends the issue of the New Zealand coastal policy statement Calls-in matters of national significance within the coastal environment Oversees the Department of Conservation and Fish and Game Councils	Conservation Act 1987 Resource Management Act 1991

Department of Conservation	Manages the conservation estate Manages wildlife Undertakes research into freshwater fisheries Advocates for the conservation of aquatic life and freshwater fisheries including involvement in RMA proceedings Promotes the benefits of conservation and prepares and disseminates conservation materials Manages the Taupo 'sports' fishery and whitebait fisheries Controls introduced species which cause damage to indigenous freshwater species or habitats	Conservation Act 1987 (especially Part 5B) Freshwater Fisheries Regulations 1983 Resource Management Act 1991 Wildlife Act 1953
Minister of Fisheries/ Ministry of Fisheries	Manages freshwater fisheries, excluding sports fish and whitebait	Fisheries Act 1996
Iwi and hapū	Exercise kaitiakitanga over freshwater bodies Co-governance and co-management arrangements for specific freshwater bodies	Deeds of Settlement Rotorua and Taupo fisheries regulations
Fish and Game Councils	Manages freshwater sports fish and gamebirds (mainly waterfowl) Advocates for the interests of sports and game including involvement in RMA proceedings Licences anglers and gamebird hunters Undertakes hatchery and breeding programmes for sports fish Undertakes research, information and education activities	Conservation Act 1987 (Part 5A) Wildlife Act 1953 (Part 2) Resource Management Act 1991
Regional councils	Controls discharges affecting freshwater bodies Controls the taking, use, damming and diverting of freshwater Allocates freshwater Controls the impact of land use on freshwater quality, quantity, ecosystems and natural hazards Controls the introduction of plants to the bed of freshwater bodies Maintains indigenous freshwater biological diversity	Local Government Act 2002 Resource Management Act 1991
Territorial authorities	Controls the impacts of land use on freshwater Controls activities on the surface of freshwater bodies Provides water and wastewater services May control drainage	Local Government Act 2002 Resource Management Act 1991 Land Drainage Act 1908
Guardians	Make recommendations to Ministers on management of hydro lakes	Conservation Act 1987 Lake Wanaka Preservation Act 1973

Figure 8.1 Summary of Freshwater management bodies

Minister for the Environment

The Minister for the Environment has the prime role for overseeing the management of freshwater in New Zealand as part of his or her environment portfolio. The Minister also has specific roles under the RMA with significance for freshwater management. The Minister (section 24):

- Recommends the issue of national policy statements including those on freshwater (but not those within the coastal environment)
- Recommends the making of national environmental standards including those related to freshwater
- Recommends the approval of requiring authorities, which can include authorities intending to construct water infrastructure such as dams, intakes, channeling, piping and wastewater treatment plants
- Recommends the issue of water conservation orders
- Can decide to call-in matters of national significance including those related to freshwater (but not those within the coastal environment)
- Monitors the effect and implementation of the RMA

Ministry for the Environment

The Ministry for the Environment was established under the Environment Act 1986. Its prime role is to provide policy advice to the Minister and other government authorities on environmental administration. Its functions, which are set out in section 31 of the Environment Act 1986, include providing advice on policies for influencing the management of freshwater amongst other things. The Ministry also supports the Minister's role in freshwater management under the RMA, develops and disseminates information on freshwater management, and engages in collaborative efforts to improve freshwater management.

The Ministry has an extensive website (www.mfe.govt.nz) which contains much useful information on freshwater management. This includes:

- *State of the environment* information on river, lake and groundwater water quality, water quality at freshwater swimming spots and demand for freshwater
- *Freshwater programmes* which the Ministry is currently engaged in such as the Lake Taupo Water Quality Protection Programme and the Rotorua Lakes Restoration Action Programme

- *Guidelines, tools and protocols* for monitoring and managing rivers, streams and lakes
- *General publications* on freshwater management topics

Environmental Protection Authority

A new Environmental Protection Agency (EPA) was established on 1 October 2009. The EPA is currently an office within the Ministry for the Environment, with its powers being exercised by the Secretary for the Environment. The EPA's role is currently focused on the 'call in' procedure which applies to proposals of national significance. These can include resource consent applications, private plan change requests, requests to prepare regional plans and notices of requirements for designations relating to freshwater.

Government has recently released proposals to expand the functions of the EPA and to establish it as a stand-alone agency from 1 July 2011. The expanded EPA will be a Crown agent responsible to the Minister for the Environment under a new Environmental Protection Authority Act. A key additional function of the expanded EPA will be close involvement in the development of national environmental standards through providing technical input during their preparation and monitoring their subsequent implementation by local authorities. These are likely to include new standards applying to freshwater management.

Minister of Conservation

The Minister of Conservation has a specific role under the RMA of overseeing the management of the coastal environment, which includes land and freshwater bodies which are influenced by the sea as well as the entire territorial sea which extends 12 nautical miles offshore. The coastal environment will usually include the downstream portion of rivers as they near the coastal edge, river mouths and estuaries, coastal lagoons and wetlands and dune lakes.

The Minister of Conservation undertakes the following functions relevant to freshwater management under the RMA:

- Recommends the issue of the New Zealand coastal policy statement which applies to the coastal environment

- Can decide to call-in matters of national significance within the coastal environment including those related to freshwater

The Minister also oversees the work of the Department of Conservation and undertakes oversight and some approvals of Fish and Game Councils, both of which undertake a range of freshwater management functions.

Department of Conservation

The Department of Conservation was established under the Conservation Act 1987 and has a wide range of functions related to conservation (see section 6). A major role of the Department is to manage the conservation estate which includes many lakes and rivers. The Department has a specific function under the Act ‘to preserve so far as is practicable all indigenous freshwater fisheries, and protect recreational freshwater fisheries and freshwater fish habitats.’

In addition, the Department has a broad role ‘to advocate the conservation of natural and historic resources generally’, as well as to promote the benefits of conservation to present and future generations, and to prepare and disseminate educational materials. The Department supports the Minister in exercising his or her functions, including providing advice on conservation matters and preparing draft New Zealand coastal policy statements. The Department, together with Fish and Game Councils, also administers the Wildlife Act 1953, which protects wildlife throughout New Zealand (section 3).

The Department is managed by the Director-General of Conservation who has specific powers under the Conservation Act. These include, of relevance to freshwater management (section 53(3)):

- Entering into agreements or arrangements with any person for the purposes of research and development work on freshwater fisheries
- Undertaking research, surveys and investigations into freshwater fisheries
- Issuing plans and publications relating to freshwater fish
- Advocating for the conservation of aquatic life and freshwater fisheries generally
- Managing the Taupo sports fishery
- Acquiring and protecting habitats
- Controlling any introduced species causing damage to any indigenous species or habitat

The Director-General can become a party in his or her own right in proceedings under the RMA through lodging submissions and appeals.

The Department of Conservation manages the whitebait fishery, which is mainly based on the young of īnanga, kōaro and banded kōkopu. Fishing is controlled under Whitebait Fishing Regulations which establish the season when whitebait fishing can take place and control what fishing gear can be used. The Department is also involved in seeking the protection of whitebait spawning habitat through encouraging landowners to fence off spawning areas of riparian vegetation from stock.¹

The Department of Conservation oversees the implementation of the Freshwater Fisheries Regulations 1983 which mainly apply to sports fish but also includes some special provisions for freshwater conservation. Regulation 68 provides for faunistic reserves, which can be established by the Minister of Conservation through a *Gazette* notice. These are areas where the taking of aquatic fauna, or introduction of any plants, are prohibited. The regulations in Part 6 also require fish passages or other mechanisms to enable fish to move along water bodies to be provided when culverts, fords, dams and diversions are constructed, unless written approval to do otherwise is provided by the Director-General of Conservation.

The Department has staff with considerable scientific expertise in freshwater species and habitats, and produces information and research reports on freshwater management issues, many which can be accessed from the Department's website (www.doc.govt.nz).

Minister of Fisheries

The Minister of Fisheries, with the assistance of the Ministry of Fisheries, manages freshwater fishing for species other than sports fish which are largely managed by the Fish and Game Councils and whitebait which is managed by the Department of Conservation. The main freshwater wild fishery managed by the Minister is shortfin and longfin eels. These are managed under the quota management system, within the framework set out in the Fisheries Act 1996, with limited recreational take of eels also provided for under the Amateur Fishing Regulations.

Iwi and hapū

Iwi and hapū were traditionally the managers of New Zealand's freshwater resources. More recently co-governance and co-management arrangements negotiated as part of Treaty of Waitangi settlements are reinstating this traditional role.

Māori management of freshwater

Māori have adopted a number of ways to deal with the management of water resources in the current environment such as kaitiakitanga, rāhui, wai tapu and wai māori. Kaitiakitanga gives the responsibility of specific iwi appointed representatives to carry out particular functions, to keep and guard iwi interests and taonga while rāhui is an act of prohibition used to conserve or replenish a resource and is often temporary.

Wai tapu and wai māori is where water may be considered tapu, or sacred because of its properties in relation to other water, tapu places or objects. Water bodies may be designated for common use as wai māori, and are not restricted by the prohibitions of tapu.

Within modern Māori resource management, restoring the ecological and spiritual integrity of degraded waterways is an important principle. This includes replenishing water quantity, improving water quality, and habitat restoration.

Quoted (with macrons inserted) from Wai Māori, 2008, Discussion on freshwater: A wai Māori perspective, <http://waimaori.maori.nz/innern/publications/resources/Discussion%20on%20Freshwater%20-%20A%20Wai%20Maori%20Perspective.pdf>

In the case of the Waikato River, for example, the *Waikato-Tainui Raupatu Claims (Waikato River) Settlement Act 2010* provides for a co-governance arrangement. Waikato-Tainui are an equal partner with the Crown on the Waikato River Authority which will review and oversee the implementation of the *Te Ture Whaimana* (Vision and Strategy for the River). There is also provision for Waikato-Tainui to participate directly in resource consent decision-making. The Authority will maintain a list of accredited iwi commissioners who can be appointed to hearings bodies considering river-related consents.² Each hearings committee must consist of an equal number of members appointed by the regional council and appointed by the Authority with an independent chairperson to be jointly appointed by the Authority and the council (section 28). The legislation has a range of other provisions including those enabling the development of a cross-agency integrated river management plan and joint management agreements (see chapter 7).

The *Te Arawa Lakes Settlement Act 2006* transfers the ownership of 13 lakebeds in the Rotorua area to the Te Arawa Lakes Trust trustees and provides for a role in ongoing management. This includes requiring any proposals for new commercial activities or structures within the lakes to obtain written consent of trustees. Regulations can be made to enable the trustees to manage

the non-commercial taking of a range of indigenous species from the lakes. There is also provision for the establishment of the Rotorua Lakes Strategy Group which is to include two Te Arawa Trust trustees as members.



Figure 8.2 Tikitapu (Blue Lake), Rotorua (Raewyn Peart) *The Te Arawa Lakes Trust has an ongoing role in the management of the Rotorua lakes under the Te Arawa Lakes Settlement Act 2006*

Under the *Fisheries (South Island Customary Fishing) Regulations 1999*, mātaimai reserves can be established in freshwater, but only in the South Island. This provides another legal mechanism for iwi or hapū to become directly involved in freshwater management. Mātaimai reserves are areas which are recognised traditional fishing grounds and with which tangata whenua have a special relationship. The reserve is managed by tangata tiaki/kaitiaki chosen by the tangata whenua. The tangata tiaki/kaitiaki can restrict or prohibit fishing in the mātaimai reserve by recommending bylaws. The first freshwater mātaimai was established in 2005. It is located in Southland on a 10 kilometre section of the Mataura River.³

New Zealand Fish and Game Council

Fish and Game Councils were established in 1990 to manage, maintain and enhance sports fish (under Parts 5A and 5B of the Conservation Act 1987) and game birds (under Part 2 of the Wildlife Act 1953) in 'the recreational interests of anglers and hunters' (Section 26Q). The New Zealand Council was established to represent nationally the interests of anglers and hunters and to coordinate the management, enhancement and maintenance of sports fish and game (section 26B).

The New Zealand Council has a broad function under section 26C(1)(g) 'to advocate generally and in any statutory planning process the interests of the New Zealand Fish and Game Council and, with its agreement, of any Fish and Game Council in the management of sports fish and game, and habitats'. The Fish and Game Council can therefore become a party to proceedings under the RMA through lodging submissions or appeals to protect freshwater game and freshwater habitats. The Council can also apply for water conservation orders and undertakes research into freshwater environments.



Figure 8.3 Freshwater anglers (Fish and Game New Zealand) *Regional Fish and Game Councils manage sports fish and gamebirds and issue licences to anglers which funds their operation*

The Conservation Act also provides for the establishment of 12 regional Fish and Game Councils (section 26P) which were elected from licence holders. These manage the sports fish and game resource at a regional level by setting licence fees and regulations. They are involved in issuing licences to take sports fish such as trout and salmon and gamebirds including waterfowl, advocacy to protect sports fish and game habitat, wetland management, hatchery and breeding programmes, compliance, enforcement, monitoring and education (section 26Q). The activities of the Fish and Game Councils are funded mainly through the sale of licences.

Regional councils

There are 12 regional councils within New Zealand. The boundaries of regional councils are broadly based on water catchments. There are also five unitary authorities which jointly exercise the functions of regional councils and territorial authorities under the RMA. Currently Gisborne District Council, Marlborough District Council, Nelson City Council, Tasman District Council and Chatham Islands Council are unitary authorities. A new Auckland Council will be established for the Auckland region on 1 November 2010 which will combine the functions of the Auckland Regional Council and seven territorial authorities.

The councils consist of councillors who are democratically elected every three years and officers who undertake the day-to-day work of the council and support and implement the decisions made by councillors. In the case of Environment Canterbury, the role of the councilors is currently being undertaken by commissioners.

Regional councils are the prime management authority for freshwater in New Zealand. As well as being tasked with the integrated management of natural and physical resources within their regions which includes freshwater bodies, regional councils have the following specific functions for freshwater management (section 30 RMA):

- The control of the use of land for the purpose of the maintenance of the quality of water in water bodies
- The control of the use of the land for the purpose of the maintenance of the quantity of water in water bodies
- The control of the use of land for the purpose of the maintenance and enhancement of ecosystems in water bodies

- The control of the taking, use, damming and diversion of water, and the control of the quantity, level and flow of water in any water body
- The control of discharges of contaminants into or onto water and discharges of water into water
- If appropriate, the establishment of rules in a regional plan to allocate the taking or use of water; the taking or use of heat or energy from water or material surrounding geothermal water; the capacity of water to assimilate a discharge of a contaminant
- The control of the introduction or planting or any plant on the bed of a freshwater body
- The establishment, implementation and review of objectives, policies and methods for maintaining indigenous biological diversity

Regional councils undertake their freshwater management role largely through preparing and administering regional policy statements and plans and determining resource consent applications in the first instance. Some regional councils supply bulk water and manage rivers for flooding. Many of the larger regional councils employ freshwater scientists and undertake research on freshwater management.



Figure 8.4 Northland Farmland (Raewyn Peart) *Regional councils are tasked with the integrated management of natural and physical resources with their regions which includes freshwater bodies*

Territorial authorities

There are currently 69 territorial authorities in New Zealand which are primarily charged under the RMA with controlling land use and subdivision. They control activities on the surface of water in rivers and lakes and may run drainage schemes. Territorial authorities are also often providers of water services within their districts such as stormwater systems and wastewater treatment works.



Figure 8.5 Lake Ōkareka settlement (Raewyn Peart) *Under the RMA Territorial authorities primarily control land use and subdivision*

Guardians

There is legislative provision for groups called Guardians, to provide advice to the Minister of Conservation on the management of specific lakes. The purpose of the Lake Wanaka Preservation Act 1973 is to ‘make provision for the preservation of the normal water levels and shoreline of Lake Wanaka, and the maintenance and improvement of its water quality’ (short title).

The legislation provides for the appointment of suitable people to the Guardians of Lake Wanaka (section 5). The persons appointed must include a person nominated by Te Runanga o Ngāi Tahu. The Guardians have the function of making recommendations to the Minister of Conservation on issues related to the purpose of the Act including the use of the lake for recreational purposes.

The Guardians of Lakes Manapouri, Monowai and Te Anau have been established under the Conservation Act 1987. The members of the Guardians are appointed by the Minister and also must include at least one person nominated by Te Runanga o Ngāi Tahu. The functions of the Guardians include making recommendations to the Minister of Conservation on:

‘any matters arising from the environmental, ecological, and social effects of the operation of the Manapouri-Te Anau hydro electric power scheme on the townships of Manapouri and Te Anau, Lakes Manapouri and Te Anau and their shorelines, and on the rivers flowing in and out of these lakes, having particular regard to the effects of the operations on social values, conservation, recreation, tourism, and related activities and amenities’ (section 6X(2)(a)).

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Billy Loban

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Allocation of freshwater

Rights to take water

Under common law, there was no ownership in water but there were rights to take and use water associated with land. Owners of land adjacent to a river or stream had riparian rights and could take water from the adjacent water body so long as it did not result in negative impacts on downstream water users. Land owners also had a right to abstract groundwater, but in this case, there was no restriction on negatively impacting on other users of the resource.

These common law rights have long been overridden by legislation in New Zealand - including by the RMA. Legislation has given the control of rights to use, divert and impound freshwater, and to discharge any material into freshwater, to the Crown.

Some uses of freshwater can still be undertaken as of right (section 14(3) of the RMA). These encompass the following non-commercial activities:

- Taking and using freshwater for an individual's reasonable domestic needs, so long as it does not or is not likely to have an adverse effect on the environment
- Taking and using freshwater for the reasonable needs of an individual's animals for drinking water, so long as it does not or is not likely to have an adverse effect on the environment
- Taking and using geothermal water in accordance with tikanga Māori for the communal benefit of the tangata whenua in the area, so long as it does not have an adverse effect on the environment
- Taking and using water for fire-fighting purposes

In most other cases, a water permit will be required before taking, diverting or impounding freshwater. This is because section 14(2) of the RMA makes it illegal to take, use, dam or divert freshwater unless expressly authorised by a national environmental standard, rule in a regional plan or a resource consent.

Restrictions on the use of freshwater under the RMA

Section 14 (2)

No person may take, use, dam, or divert any of the following, unless the taking, using, damming, or diverting is allowed by subsection (3):

- (a) Water other than open coastal water; or
- (b) Heat or energy from water other than open coastal water; or
- (c) Heat or energy from the material surrounding geothermal water.

Section 14(3)

A person is not prohibited by subsection (2) from taking, using, damming or diverting any water, heat, or energy if –

- (a) The taking, using, damming or diverting is expressly allowed by a national environmental standard, a rule in a regional plan as well as a rule in a proposed regional plan for the same region (if there is one), or a resource consent; or
- (b) to (e) which cover a range of non-commercial uses

Figure 9.1 Restrictions on the use of freshwater under the RMA

Provisions in regional plans

As outlined in chapter 8, regional councils are responsible for the control of the taking, use, damming, and diversion of water, and the control of the quantity, level, and flow of water in any water body (section 30(1)(e)). Their responsibilities include the setting of any maximum or minimum levels or flows of water, and the control of the range, or rate of change, of levels or flows of water. Councils make plans and policy statements which specify environmental baselines and how water will be shared between users.

If appropriate, regional councils can create rules in a regional plan to allocate the taking or use of water. This can override the default position that water will be allocated only on a ‘first come, first served’ basis. The RMA does not prescribe how or to whom water will be allocated. This means that regional councils have a broad discretion to devise allocation rules.

Rules in the RMA for water allocation
Section 30(4)

A rule to allocate a natural resource established by a regional council in a plan under subsection (1) (fa) or (fb) may allocate the resource in any way, subject to the following:

- (a) the rule may not, during the term of an existing resource consent, allocate the amount of a resource that has already been allocated to the consent; and
- (b) nothing in paragraph (a) affects section 68(7); and
- (c) the rule may allocate the resource in anticipation of the expiry of existing consents; and
- (d) in allocating the resource in anticipation of the expiry of existing consents, the rule may—
 - i) allocate all of the resource used for an activity to the same type of activity; or
 - (ii) allocate some of the resource used for an activity to the same type of activity and the rest of the resource to any other type of activity or no type of activity; and
- (e) the rule may allocate the resource among competing types of activities; and
- (f) the rule may allocate water, or heat or energy from water, as long as the allocation does not affect the activities authorised by section 14(3)(b) to (e) [small non-commercial uses].

Figure 9.2 Rules in the RMA for water allocation

Rules in a regional plan cannot allocate water that is already allocated under an existing resource consent. However, rules may allocate water in anticipation of the expiry of existing consents. They can allocate all or some of the water to the same type of activity, allocate some of the water to another type of activity, or allocate it to no type of activity (section 30(4)).

If the rule relates to maximum or minimum levels or flows, or rates of use, of water then the regional plan may state whether the rule will affect the exercise of existing resource consents which breach the rule. The plan may also prescribe whether the holders of water permits are able to comply with the terms of the rule in stages or over specified time periods (section 68(7)).

Environmental flows

When establishing how much water is available for allocation within a river, lake or groundwater body regional plans can determine the ‘ecological flow’ or ‘environmental flow’. The *ecological flow* refers to the amount of water and level of flow required in a water body to provide for the healthy ecological functioning of flora and fauna.

The *environmental flow* is a broader concept which includes ecological flows but also water flows and levels that are required to maintain other important

values such as tangata whenua, cultural, amenity, recreational, landscape or natural character.¹

The Ministry for the Environment has developed a proposed national environmental standard on ecological flows and water levels. The Ministry decided to focus on ecological flows rather than environmental flows which are harder to measure. The standard is intended to assist decision makers under the RMA to make better decisions on water allocation and to ensure that water is not over-allocated to the detriment of the species that live in the affected water body. It also aims to promote consistency in the way that decisions are made on how much water should be left in a freshwater body.²

The proposed standard includes measures to:

- Set interim limits on the alteration to flows and/or water levels for rivers, wetlands and groundwater systems that do not have limits imposed through regional plans
- Provide a process for selecting the appropriate technical methods for evaluating ecological flows and water levels in rivers, lakes and wetlands and groundwater systems

Water permits

Granting water permits

Water is mainly allocated by regional councils through the issuing of water permits, for which those wishing to take water apply.³ The RMA does not state on what basis or to whom water should be allocated, but when considering an application for a water permit the regional council must have regard to the following matters (section 104):

- Any actual and potential effects on the environment of allowing the activity (excluding adverse effects which are permitted in a national environmental standard or the relevant plan)
- Any relevant provisions of a national environmental standard, other regulations, a national policy statement, a New Zealand coastal policy statement, a regional policy statement or proposed regional policy statement, a plan or proposed plan
- Any other matter the consent authority considers relevant and reasonably necessary to determine the application.

When making decisions on water allocation the regional council must have regard to the purposes and principles of the RMA set out in Part 2 (see Chapter 7).⁴

Regional councils must not grant a water permit if it would be contrary to any provision of a water conservation order (section 217(2)). However, they can impose conditions on a water permit to ensure that the provisions of the water conservation order are maintained. The national environmental standard for sources of human drinking water is also relevant: a regional council must not grant a water permit if it means that the water no longer meets the health quality criteria.⁵

Councils have various means of dealing with the issues that arise when there is very high demand to take water from a particular source. They often identify the minimum flows that can be sustained by the water body, and specify the maximum amount of water that can be taken from it on that basis. Some councils operate a waiting list approach, so that where a water body is fully allocated no more consents to take water are granted until water becomes available. Alternatively, councils may use a rationing system so that when a water body becomes fully allocated, existing consents are reviewed and adjusted to reduce allocations.⁶

Nature of water permits

Permits for water takes can have a duration of up to 35 years, but they are often granted for shorter time periods. They can be cancelled by a regional council if not exercised for a continuous period of two or more years. Water permits can be transferred to other users in the same catchment if transfers are expressly authorised in the regional plan or have been approved by the consent authority (section 136). Transfers can be made on a temporary basis. In practice, there has been limited transfer of water permits⁷ although the number of transfers are increasing.

Water permits do not give rise to a property right to the resource, and the permit is not a guarantee that the water will be available. Section 122(1) of the RMA stipulates that a resource consent is neither real nor personal property. However, there is still some legal uncertainty over the extent to which regional councils have to take into account the potential for adverse effects on existing consent holders when new consent applications are received. Case law has generally supported the rights of existing consent holders in these circumstances.⁸

Regional councils may review conditions of water permits in a number of situations including where relevant national environmental standards have been made; when a regional plan has been made operative which sets rules relating to maximum or minimum levels or flows or rates of use of water; or

for any purpose specified in the consent (section 128), This is an increasingly important aspect of water management under the RMA given the greater role of adaptive management, particularly in relation to complex water permits. Further information about using the review process to address over-allocation is discussed below. In limited circumstances a regional council can cancel a water permit if there are significant adverse effects on the environment resulting from the exercise of the consent (section 132).

Priority between competing applications

The ‘first come, first served’ approach is the default position when there are competing applications to use a water resource.⁹ Case law has been evolving in this area, but the current position is that the first complete application lodged with the consent authority will have priority of hearing against all subsequent applications.¹⁰

The RMA contains provisions which give some priority to existing water permit holders at the end of the term of their permit. A regional council must not process an application for a water permit, if (section 124C)

- The applicant does not already hold a water permit, and
- The application is for a permit which cannot be fully exercised until an existing holder’s water permit expires in more than 3 months time.

When a competing application is made, the council must notify the holder of the existing permit and give that party the opportunity to apply to renew its consent. If the existing holder notifies the council that it does not propose to make such a renewal application, or if a renewal application has still not been made three months before the consent is due to expire, then the regional council must process the new competing application. A provision in a regional plan can overturn this priority afforded to existing water permit holders (section 124A).

Where a water resource is fully allocated, further consents for that water body cannot be granted. This is known as the principle of non-derogation and is designed to protect existing water permit holders.¹¹

Renewal of water permits

Section 124 of the RMA provides that when a holder of an existing water permit applies for a new consent for the same type of activity, at least 6 months before the expiry of the existing consent, then the holder of the permit may continue to operate until the application and all appeals are determined (section 124).

The regional council also has discretion to allow a consent holder to continue to operate if the application is made between 3 and 6 months before the expiry of the existing consent. When the consent authority considers the application, it must have regard to the value of the investment of the existing consent holder (section 104(2A)).

When determining whether to grant consent to the existing consent holder the consent authority must apply the relevant provisions of the RMA and consider the following matters (section 124B):

- The efficiency of the person's use of the resource
- The use of industry good practice by the person
- If the person has been served with an enforcement order or convicted of an offence under the Act

Measuring water takes

The Resource Management (Measurement and Reporting of Water Takes) Regulations 2010 have been made to improve the management of water resources by improving the way that water takes are measured. The Regulations have set minimum requirements for the installation and operation of new water measuring and recording devices and for the transfer of data to regional councils. There are obligations on consent holders to install and operate water measuring devices.

The new regulations apply to all water takes that require a resource consent enabling the taking of water at a rate of up to five litres per second or more. They are designed to help resource managers understand how the water resource is being used. This will enable resource managers to improve water use efficiency, and it is also critical to gaining an understanding of how the resource is responding to extractions that are actually occurring. The regulations came into force on 10 November 2010, but for existing water permits compliance with the requirements is introduced at differing times over the next six years dependent on differing rates of abstraction.

Prior to the regulations water measuring was not widespread in New Zealand. At the end of 2006 there were a total of 19,527 consents to take water, but of these, only 33 per cent of takes were measured. It is hoped that the regulations will improve this figure with it expected that 92 per cent will be measured within two years of the regulations taking effect and 98 per cent within six years.¹²

Addressing over-allocation

A water permit can be reviewed to deal with any adverse effect on the environment which may arise from the exercise of the consent and which it is appropriate to deal with at a later stage or for any other purpose specified in the consent (section 128(1)). Depending on the conditions of consent, this may enable a regional council to address over-allocation of a water body.

Regardless of the conditions of consent, regional councils can review a water permit if it is necessary to ensure compliance with a national environmental standard, or rules in an operative regional plan relating to (section 128):

- Maximum or minimum levels or flows
- Rates of use of water
- Minimum standards of water quality

As a result, in a situation of over-allocation, a regional council could create a rule in the regional plan relating to minimum levels and flows of water. This rule would have immediate legal effect (section 86B). The council could then review all the permits authorising extraction of water from the over-allocated resource and, if necessary, reduce the amount of water which is able to be taken under each water permit.

Regional councils also have the ability to issue directions and restrict the use of water when there are serious water shortages (section 329). In the event that this occurs, the council can suspend, restrict or apportion taking of water with immediate effect for 14 days. This can be renewed, and there is no provision for appeal. In some cases regional councils use this mechanism to manage the allocation of water during seasonal low flows.

Waitaki catchment legislation

The Waitaki River is one of New Zealand's largest braided rivers. Its catchment is predominantly located in Canterbury but covers a small part of Otago as well. The area suffers from frequent droughts due to low and sporadic rainfall, and high rates of evapotranspiration, caused by high winds and high temperatures.

There are many competing claims to the water in the Waitaki catchment, including hydroelectricity generation, agriculture (irrigation) industry, domestic water supply, recreation and tourism. The river is also very significant to Ngai Tahu, tangata whenua of the region.

The Resource Management (Waitaki Catchment) Amendment Act 2004 aimed to improve decision-making in respect of allocation in the Waitaki catchment. Although there was contention over whether it was necessary to make an amendment to the Resource Management Act (which is intended to create an overarching legislative system for all resource management issues) the significant competing proposals to use the limited water resources of the Waitaki Catchment convinced the Minister for the Environment that legislative action was necessary.¹³

Demand for water in the region was extremely high, and applicants for permits to take water were facing significant delays. In addition there was a proposal by Meridian Energy to create a new hydroelectric scheme on the river that would divert river water at a rate of up to 280 cubic metres per second into a separate canal for six power stations. The scheme would have generated enough power to supply a city the size of Christchurch. Although the project had much local support, there was also significant opposition from people concerned that the river would not be able to sustain this level of diversion. Ultimately, in March 2004, Meridian discontinued the scheme citing the uncertainties in gaining consent to access the water amongst their reasons.

The Amendment Act provided for the creation of a Waitaki Catchment Water Allocation Board which was an independent board charged with developing a regional plan to guide water allocation in the Waitaki catchment. It was given all the powers that a regional council would have to develop the plan. To create the plan, reports on catchment information were collated, including information on hydrology, recreational values, water demand and freshwater ecological values.

The Waitaki Catchment Water Allocation Regional Plan was approved in September 2005 and the Board was disbanded. The plan applies to 'the taking, use, damming or diverting of water from water bodies within the Waitaki catchment.' It was amended by the High Court following a joint appeal by Meridian Energy and an irrigation company.¹⁴ The plan is unusual in that it takes precedence over all other applicable policy statements and plans.

The plan sets out policies to guide decision making under the plan, identifying particular water bodies which have high natural character, providing that environmental flows should be established for all rivers in the catchment, and prohibiting the abstraction of water where flows have reached their minimum level. Extraction of any water is prohibited from specific water bodies. The plan did not, however, set rules for water quality.

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- 2 <http://www.mfe.govt.nz/laws/standards/ecological-flows-water-levels/index.html>
- 3 Water permits are a type of resource consent: section 87(d) of the RMA.

- 4 RMA, section 104(1)
- 5 Resource Management (National Environmental Standard for Sources of Human Drinking Water) Regulations 2007, Clause 7
- 6 Ministry for the Environment and Ministry of Agriculture and Forestry, 2004
- 7 Counsell K, 2003, 17
- 8 *Brooklands Properties 2000 Limited v Road Metals Company Limited* unreported, Environment Court Christchurch, C037/08, 3 April 2008, Jackson J, para 22
- 9 *Fleetwing Farms Limited v Marlborough District Council* (1997) 3 ELRNZ 249, 259
- 10 *Central Plains v Ngai Tahu Property Group* [2008] NZCA 71, para 67; *Central Plains v Synlait Limited* [2009] NZCA 609, para 94.
- 11 *Aoraki Water Trust v Meridian* unreported, HC Timaru, CIV-2003-476-733, 30 November 2004, Chisholm and Harrison JJ, paras 26-31, 36 and 46
- 12 Ministry for the Environment, 'Proposed national environmental standard for the measurement of water takes', <http://www.mfe.govt.nz/issues/water/prog-action/cabinet-paper-measurement-of-water-takes.html>
- 13 Hon Marian Hobbs, Minister for the Environment, Resource Management (Waitaki Catchment) Amendment Bill – First Reading, Hansard, Vol 614, page 10600; Jeanette Fitzsimmons, 2004, 'Resource Management (Waitaki Catchment) Amendment Bill', 25 March, <http://www.greens.org.nz/speeches/resource-management-waitaki-catchment-amendment-bill>
- 14 *Re McKenzie Irrigation Co Ltd v Meridian Energy* CIV 2005-485-2192, 3 July 2006



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ten

Managing freshwater pollution

Rights to pollute water

There are no general rights to pollute water. All discharges of contaminants into water, or of contaminants onto land which may result in contaminants entering water, are illegal unless the discharge is authorised by a national environmental standard, a rule in a regional plan or a resource consent. This applies equally to point and non-source pollutants. However, with some non-source pollutants it can be difficult to trace the responsible party. Therefore, activities such as fertilising pasture or grazing cattle, where they result in nutrients entering groundwater or a stream, are illegal under the RMA unless expressly authorised by a national environmental standard, a rule in a regional plan or by a resource consent.

Section 15 (1)

No person may discharge any –

- (a) Contaminant or water into water; or
- (b) Contaminant onto or into land in circumstances which may result in that contaminant (or any other contaminant emanating as a result of natural processes from that contaminant) entering water; or
- (c) N/A
- (d) Contaminant from any industrial or trade premises onto or into land –

unless the discharge is expressly allowed by a national environmental standard or other regulations, a rule in a regional plan as well as a rule in a proposed regional plan for the same region (if there is one), or a resource consent

Figure 10.1 Restrictions on discharges into freshwater under the RMA

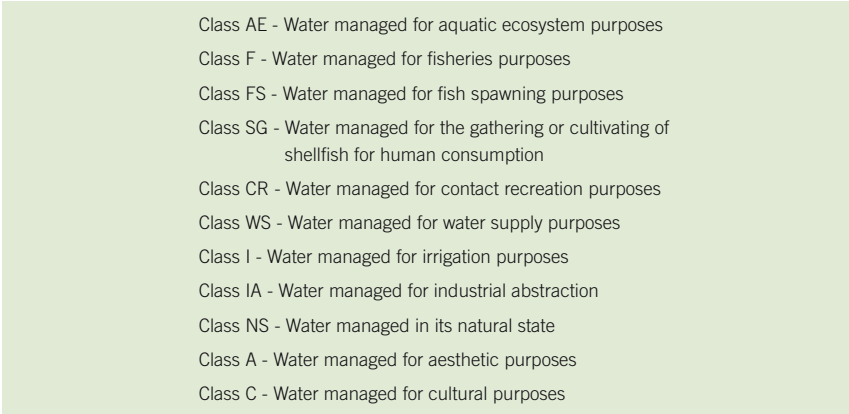
Regional plans

Regional plans provide the framework for the management of water pollution. Regional plans determine what activities will require consent, and what matters will be taken into account when consent is sought. Plans can classify different water bodies in terms of the water quality which needs to be maintained when water permits are sought.

Water standards

The RMA provides a framework for identifying water quality classes which can be adopted by councils in their regional plan. This is set out in Schedule

3 which describes particular standards that the water must achieve for each category, for example in relation to its visual clarity, pH level, levels of contamination and temperature. Where a water body is identified as belonging to one of these categories in the regional plan, these standards help to inform which activities are allowed to occur in the area. Many of the standards in the RMA are expressed in narrative form and need to be described in more detail and quantified in regional plans. The water quality classes in Schedule 3 have been applied to the Waimakariri River (see Figure 10.3).



- Class AE - Water managed for aquatic ecosystem purposes
- Class F - Water managed for fisheries purposes
- Class FS - Water managed for fish spawning purposes
- Class SG - Water managed for the gathering or cultivating of shellfish for human consumption
- Class CR - Water managed for contact recreation purposes
- Class WS - Water managed for water supply purposes
- Class I - Water managed for irrigation purposes
- Class IA - Water managed for industrial abstraction
- Class NS - Water managed in its natural state
- Class A - Water managed for aesthetic purposes
- Class C - Water managed for cultural purposes

Figure 10.2 Water classes under the RMA (Schedule 3)

Councils may choose to use other water quality standards. When establishing rules in their regional plans, regional councils must not set standards which result, or may result, in a reduction of the quality of the water in any freshwater body after reasonable mixing occurs unless it is ‘consistent with the purpose of’ the RMA to do so (section 69(3)).

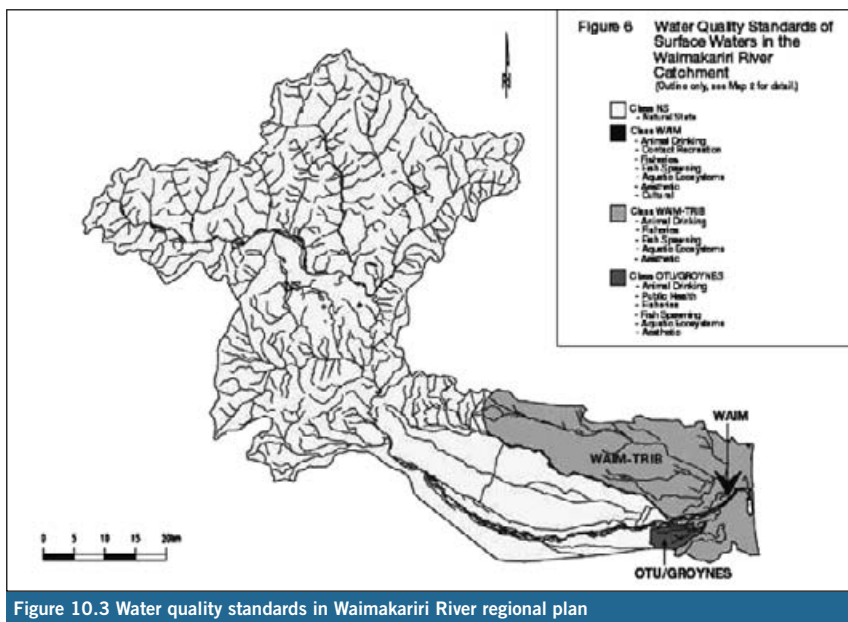


Figure 10.3 Water quality standards in Waimakariri River regional plan

Nutrient management

Councils have adopted a range of approaches in their regional plans for the management of nutrient run off. In the Bay of Plenty region, under rule 11 of the Bay of Plenty regional water and land plan, any proposed land use change which may increase nitrogen and phosphorus run-off above a benchmark requires a resource consent. Rule 11 applies only to a limited area in the Rotorua lakes catchments.¹

Under the RMA, it is also possible to provide for a cap of total nutrient discharges into a water body, and for trading of nutrient discharge rights. Such an approach has recently been applied to the Lake Taupo catchment in Variation 5 to the Waikato Regional Plan.

In Variation 5, nitrogen discharges have been capped at current levels and farmers are being ‘benchmarked’ to determine their nitrogen discharge allowance, which is the maximum amount of nitrogen that a farm is allowed to produce. The allowance is based on their highest year of leaching between July 2001 and June 2005. The allowance is established using a computer model called *Overseer*. Discharge rights will be tradeable, so that farmers can increase their discharge allowance, by buying someone else’s rights. This is intended to ensure that total nitrogen discharges from the catchment remain the same.

Variation 5 manages existing and new nitrogen leaching activities by providing that they are either permitted activities or controlled activities. Consent for controlled activities is conditional upon compliance with a nitrogen discharge allowance. Where a land use activity will result in an increase in the nitrogen leaching from the property this will be non-complying. Figure 10.4 indicates the rules for each category of activity in the catchment.

Rule	Application
Low nitrogen leaching farming activities (rule 3.10.5.1)	Permitted (e.g. rural residential lots) if low stocking rate (limits for various types of animals are set on a per hectare basis) and fertiliser standards (no more than 75kg per year) are met
Nitrogen leaching non farming activities (Rule 3.10.5.2)	Permitted (e.g. indigenous vegetation, forestry, existing golf courses)
Nitrogen leaching farming activities (rule 3.10.5.3)	Controlled (e.g. drystock, dairy farms and land with livestock that doesn't meet rule 3.10.5.1) to establish nitrogen allowances for properties and nitrogen management plans that ensure no increase in nitrogen leaching in the future unless corresponding decreases in nitrogen leaching are achieved elsewhere in the catchment as an offset
Nitrogen offsetting for new nitrogen leaching activities (rule 3.10.5.4)	Controlled (e.g. forestry changing to rural residential) to enable increased nitrogen leaching where there is a corresponding decrease in nitrogen leaching elsewhere in the catchment as an offset.
Land use activities that don't meet these rules	Non complying

Figure 10.4 Rules in Variation 5 for nitrogen leaching activities

Farms are required to establish a nitrogen management plan to show how they will comply with their discharge allowance. A nitrogen discharge allowance report and a nitrogen management plan must be provided with consent applications. Once consent has been granted on the basis of the documents, farmers can change their farm practices but the farm cannot exceed the discharge allowance, and a new nitrogen management plan must be completed setting out the changes. The new plan must be registered with Environment Waikato within 10 days of the farming activities changing.

Farmers can alter their nitrogen discharge allowance by applying via a consent application. If they intend to trade with another landowner both parties must apply to alter the consents and discharge allowances. New nitrogen

management plans to reflect the trade must be completed by both parties. Leases of nitrogen are also permitted, and follow the same process as a trade although the length of the lease will need to be specified. Farmers can also sell nitrogen to the Lake Taupo Protection Trust which has been set up as a charitable trust to implement nutrient-reducing programmes. This must also be done via the consent process, and farmers must submit a revised nitrogen management plan for their site based on the reduced allowance.

Discharge permits

In most cases, a discharge permit will be required before discharging water or pollutants into freshwater bodies. The RMA includes several extra matters which a consent authority (which is usually the regional council in the first instance) needs to consider when determining a discharge permit.

First, the consent authority must be sure that none of the following *adverse effects* are likely to occur after reasonable mixing of the contaminant with the water (section 70(7)):

- Production of conspicuous oil or grease films
- Conspicuous changes in colour or visual clarity
- Emissions of objectionable odours
- The rendering of freshwater unsuitable for consumption by farm animals
- Any significant adverse effects on aquatic life

The inclusion of the term ‘after reasonable mixing’ means that consent can still be granted to discharge a contaminant which causes any of the effects listed above when it enters the water, provided that the effects disappear once the contaminant has mixed with the water. What is ‘reasonable mixing’ is a question of fact to be assessed on a case-by-case basis.

This provision only sets out the minimum standard – the consent authority is still required to make a judgment that the proposed discharge is consistent with the purposes of the RMA in sections 5-8 (see chapter 7). In addition, discharges that do produce any of the effects listed above can be allowed if there are exceptional circumstances and if they are temporary, and if it is consistent with purposes of RMA to allow them.

Secondly, the consent authority must have regard to a range of extra matters including (section 105(1)):

- The nature of the discharge and the sensitivity of the receiving environment to adverse effects
- The application's reasons for the proposed choice
- Any possible alternative methods of discharge, including discharge into any other receiving environment

Thirdly, the consent authority may include a condition requiring the discharger to adopt the best practicable option to minimise adverse effects on the environment. The best practicable option is the best method for preventing or minimising the adverse effects on the environment, having regard to:

- The nature of the discharge and the receiving environment, and the sensitivity of the receiving environment to adverse effects
- The financial implications of the option
- The effects on the environment of the option
- The current state of technical knowledge and the likelihood that the option can be successfully applied

Where the best practical option is prohibitively expensive or onerous, it may not be appropriate. If a best practical option condition is imposed on a discharge permit, and the permit holder contravenes it, only the consent authority or the Minister for the Environment can apply to the Environment Court for an enforcement order enforcing the condition.

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Endnotes

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Opportunities for public involvement

Introduction

There are a number of opportunities for members of the public to participate in processes for making decisions on freshwater management. These are summarised in Figure 11.1 and described more fully in the following sections.

Decision	Opportunity for public involvement
<i>Resource Management Act 1991</i>	
National policy statements	Written submissions on proposed policy statement Presentation of submissions and evidence to Board of Inquiry (where the board of inquiry process is adopted)
National Environmental Standards	Must be adequate time and opportunity for public comment. Usually written submissions on the proposed standard will be sought
Regional policy statements	Written submissions on proposed policy or plan Written further submissions by some parties
Regional plans	Presentation of submissions and evidence to council hearing Appeal to Environment Court
District plans	Presentation of submissions and evidence to Environment Court hearing Appeal to High Court on points of law
Private plan changes	Anyone can lodge an application with council Council decides whether to accept the application Processed in the same way as a plan change initiated by council
Resource consents	Statutory right to participate only where application publicly notified or notice served if limited notification Written submissions on application Presentation of submissions and evidence to council hearing (where no direct referral or call-in) Appeal to Environment Court Presentation of submissions and evidence to Environment Court hearing where appeal, direct referral or some call-ins (or board or inquiry for some call-ins) Appeal to High Court on points of law
Declarations	Anyone can lodge proceedings in the Environment Court seeking to clarify matters under the RMA

Enforcement orders	Anyone can lodge proceedings in the Environment Court seeking to compel a person to comply with the provisions of the RMA
Fisheries Act 1996	
Fisheries plans	Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests. In practice, a draft plan is usually circulated for public written submissions
Sustainability measures	
Conservation Act 1987	
General policy	Written submissions on draft policy Presentation of submissions to the Director-General of Conservation
Conservation management strategies	Written submissions on draft strategy
Conservation management plans	Presentation of submissions at a meeting of representatives of the Director-General of Conservation and affected conservation boards
Sports fish and game management plans	Written submissions on draft plan Presentation of submissions to a meeting of representatives of the Fish and Game Council
Judicature Act 1908	
Judicial review	Any person can challenge the exercise of a statutory discretion by lodging proceedings in the High Court
Figure 11.1: Opportunities for public involvement in statutory decisions affecting freshwater	

Preparation of national RMA instruments

As indicated in chapter 7, there are two types of national instruments under the RMA, national policy statements and national environmental standards. Members of the public must be given the opportunity to make submissions when both these instruments are prepared.

National policy statements can be prepared through two processes. The first is the board of inquiry process which includes the following steps (sections 47 to 52):

- The Minister for the Environment prepares a proposed national policy statement
- The Minister appoints a board of inquiry which calls for submissions and holds a public hearing
- After considering the submissions and other relevant information, the board of inquiry prepares a written report and recommendations for the Minister
- The Minister considers the report and recommendations, may change the proposed national policy statement, and then may recommend approval to the Governor General in Council

In this case submitters will have the opportunity to present their case (including submissions and evidence) at a public hearing convened by an independent board of inquiry. The board of inquiry makes recommendations to the Minister who makes the final decision. There are no appeal rights against the Minister's decision and the only redress is through judicial review proceedings in the High Court (see below).

The second is an alternative process which the Minister establishes and which must give the public adequate time and opportunity to make a submission. This process requires a report and recommendations to be made to the Minister on the submissions and subject matter of the national policy statement (section 46A). This will almost certainly provide members of the public with an opportunity to make written submissions but may not involve a public hearing.

Section 46A(2) of the RMA sets out the matters which the Minister may consider when determining which process to adopt. These include the advantages and disadvantages of having the national policy statement made quickly, the extent to which the proposed policies differs from policies in other relevant documents such as regional policy statements and plans, district plans and national environmental standards, and the extent and timing of public debate and public consultation on the proposal.

No process for the preparation of *national environmental standards* is prescribed in the RMA and the Minister for the Environment must establish a process. The Act sets out the following steps which must be undertaken (section 44):

- The Minister must notify the public and iwi authorities of the proposed subject matter of the standard and the reasons why the Minister considers the standard to be consistent with the purposes of the RMA
- The Minister must establish a process that provides the public and iwi authorities with adequate time and opportunity to comment and that

requires a report and recommendations to be made on those comments and the proposed subject matter of the standard

- The report and recommendation must be publicly notified

In most cases members of the public will have the opportunity to make written submissions but there will be no public hearing.

Preparation of RMA policies and plans

At the regional level, regional policy statements and regional plans are prepared. At the district level, district plans are prepared. Plans typically contain objectives, policies, rules, other methods and anticipated environmental results which guide decision-making on resource consent applications affecting freshwater bodies.

There is a very open process for public participation in council policy and plan-making under the RMA. Councils are required to consult with a range of parties prior to proposed plans and policy statements being formally notified, including Ministers of the Crown, local authorities and iwi (clause 3, Schedule 1). Councils also commonly consult with a range of other stakeholders and this can be a useful opportunity to have some input outside the formal statutory process. Sometimes a draft plan is circulated for comment prior to the formal statutory process commencing.

Once the proposed policy statement or plan is formally notified, members of the public can make written submissions. The following persons are then able to lodge further submissions in support or opposition to the submissions already lodged:

- Any person representing a relevant aspect of the public interest. This is likely to include public interest environmental groups
- Any person that has an interest in the proposed policy statement or plan greater than the interest that the general public has. This is likely to include owners of land and users of resources directly affected by plan provisions. It is also likely to include iwi and hapū where their interests are directly affected
- The local authority itself

All submitters then have the opportunity to present their case before a council-initiated public hearing which the RMA requires to be held.

When preparing a plan or a plan change councils must take into account relevant iwi planning documents which have been lodged with them. This provides an opportunity for tangata whenua to have proactive input into planning initiatives.

Once the council has made decisions on submissions, any submitter has the right to appeal the decision to the Environment Court. The Court will consider the matter afresh at a public hearing.

Other parties can become party to an appeal (referred to as section 274 parties) only if they can show that they fall within one of the following categories:

- They have lodged a submission at the local authority stage; or
- They have an interest in the proceedings greater than the public generally (excluding trade competitors). This will normally include neighbours, downstream resource users and iwi authorities. It is uncertain whether public interest groups will be able to become parties under this category as well, due to recent changes to the RMA.

Decisions of the Environment Court can be appealed to the High Court on questions of law, but not on the merits.

In the Canterbury region there is currently no appeal rights to the Environment Court on RMA policy statements or plans. Decisions of the Council can only be appealed to the High Court on questions of law.¹

Private plan changes

Any person can seek a change to a regional or district plan and this is called a 'private plan change'. However, such a request can only be made when the plan is operative. The council can only reject the request if:

- It is frivolous or vexatious
- The local authority or the Environment Court has considered the substance of the request within the last two years
- The request is not in accordance with sound resource management practice
- The request would make the plan inconsistent with the provisions of the RMA
- The policy statement or plan has been operative for less than two years.

If none of these criteria apply the council must process the private plan change, and has the option of adopting the plan change as though it was initiated by the council.

Requesting a private plan change can be an expensive exercise as the application will need to include an assessment of environmental effects of the proposal and an assessment of its benefits and costs. Preparing these assessments is likely to require the assistance of environmental professionals. In addition, the council can charge the applicant the full costs of processing the proposed plan change including the costs of the submission process and hearing.

Resource consents

The opportunities to become involved in decisions about individual proposals impacting on freshwater bodies are much more limited than the opportunities provided for in the development of policies and rules.

Under the RMA there are broad rights of participation in resource consents, when they are publicly notified, and these usually include being heard at a council-initiated hearing, appealing the council's decision to the Environment Court and being heard again, and appealing the Environment Court's decision to the High Court on points of law only.

Notified resource consents can also be directly referred to the Environment Court for a decision at the request of the applicant. In this case there will be no hearing by the council in the first instance if the council agrees to directly refer the application. In order to participate in the Environment Court proceedings, submitters must give written notice to the Court (known as a section 274 notice). The decision of the Environment Court can be appealed to the High Court on points of law only.

Matters of national significance can be 'called-in' by the Minister of Environment (or Minister for Conservation for applications within the coastal marine area) or lodged directly with the EPA. In this case there will also be no council hearing in the first instance, but the matter will be processed by the EPA and determined either by the Environment Court or by a board of inquiry appointed by the Minister. The Court or board of inquiry will hold a public hearing. Decisions can be appealed to the High Court on points of law only.

There are no appeal rights against a council's decision not to notify a resource consent and the only redress is through initiating judicial review proceedings in the High Court (see below). In cases where the council decides not to publicly notify the application, it may still be given limited notification. This means that

parties who are directly affected by the proposal, and who have not given consent, will have the opportunity to lodge a submission and to subsequently appeal.

Water conservation orders

Anyone can apply for a water conservation order by writing to the Minister for the Environment specifying:

- The water body concerned
- The reasons for the application, with reference to sections 199, 200 and 2007 of the RMA
- The values for which protection is sought
- The provisions sought in the water conservation order
- The effect that the provisions would have on the water body

The cost of the application is \$1000. When the Minister has received an application, he or she can request further information. The Minister decides whether to reject the application, or to accept it, and appoint a special tribunal to hear and make recommendations. Once a tribunal is appointed, the application is publicly notified, with a call for written submissions. Anyone may make a submission within 20 days of the notification. The tribunal then holds a public hearing of the submissions.

When considering the application, the tribunal is required to have ‘particular regard’ to the purpose of water conservation orders and other matters set out in section 199 (see Figure 11.2). The tribunal is also required to have regard to:

- The application
- The submissions
- The needs of primary and secondary industry
- The needs of the community
- Relevant provisions of policy statements and plans

The tribunal prepares a report on the application with a recommendation, which will either include a draft water conservation order or a recommendation that the application be declined. This is sent to the applicant, the relevant local and iwi authorities, the Minister for the Environment, and to all parties who made a submission. Submitters then have a further right of submission to the Environment Court to respond to the tribunal’s report.

- (1) Notwithstanding anything to the contrary in Part 2, the purpose of a water conservation order is to recognise and sustain –
 - (a) outstanding amenity or intrinsic values which are afforded by waters in their natural state:
 - (b) where waters are no longer in their natural state, the amenity or intrinsic values of those waters which in themselves warrant protection because they are considered outstanding.
- (2) A water conservation order may provide for any of the following:
 - (a) the preservation as far as possible in its natural state of any water body that is considered to be outstanding:
 - (b) the protection of characteristics which any water body has or contributes to, and which are considered to be outstanding, –
 - (i) as a habitat for terrestrial or aquatic organisms:
 - (ii) as a fishery:
 - (iii) for its wild, scenic, or other natural characteristics:
 - (iv) for scientific and ecological values:
 - (v) for recreational, historical, spiritual, or cultural purposes:
 - (c) the protection of characteristics which any water body has or contributes to, and which are considered to be of outstanding significance in accordance with tikanga Māori.

Figure 11.2 Matters to which are to be paid 'special regard' when considering applications for water conservation orders (Section 199)

If the Environment Court receives one or more submissions it must hold an inquiry, and make recommendations on whether the tribunal's report should be accepted or rejected, with or without modifications. The Court can also include a draft water conservation order or recommend that the application for an order be declined. The Court must publicly notify its report (section 213).

The matter then goes to the Minister who makes a recommendation to the Governor General 'in accordance' with the Environment Court's report if there was an inquiry, or with the special tribunal report, if there were no submissions to the Environment Court. Finally, if the application is accepted, the Governor General makes the water conservation order by Order in Council.

If the Minister decides not to recommend the making of the order, he or she must present to the House of Representatives a written statement setting out the reasons for the decision and serve it on the applicant and submitters (section 215).

Within the first two years of a water conservation order being in place, no applications can be made to revoke it. Amendments can only be made if they are of a minor nature or of a technical nature that will enable the order to better achieve its purpose (section 216(1)).

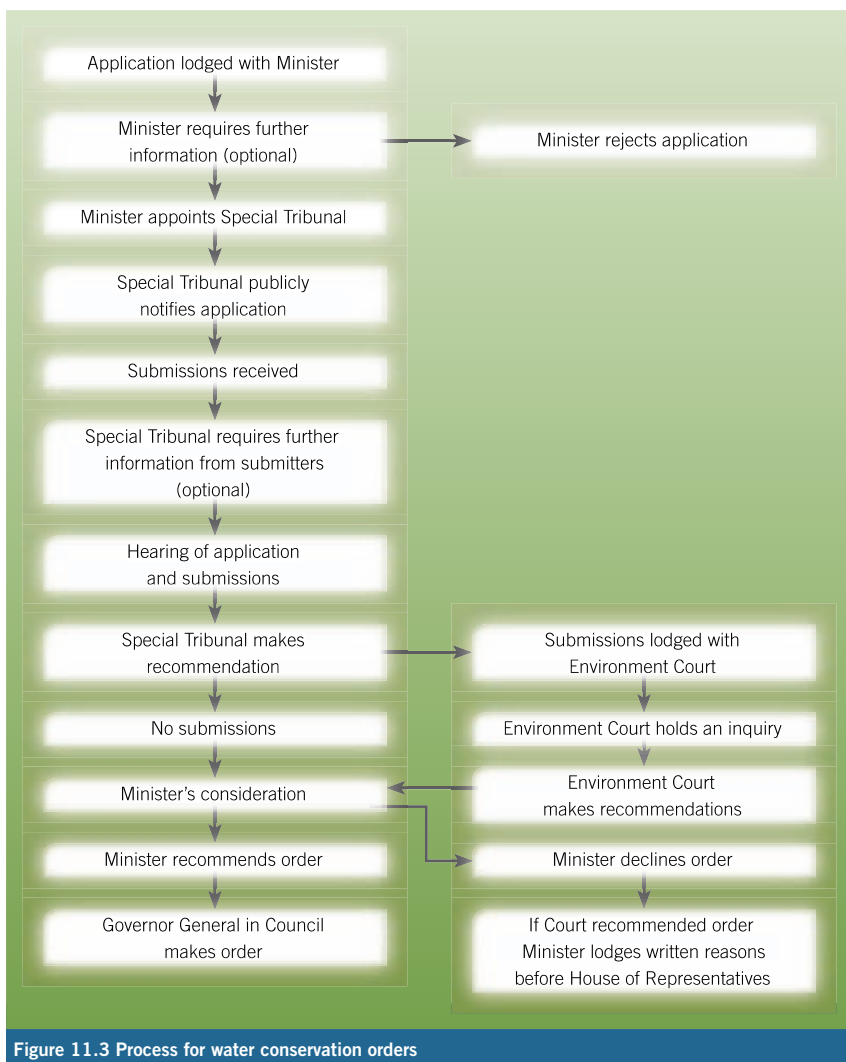


Figure 11.3 Process for water conservation orders

After the two year period has expired, any person may apply to revoke or amend the order. Unless the amendment applied for is minor, the application is required to go through the same process as the original water conservation order. Where the Minister for the Environment considers the effect of the amendment sought to be minor, and when the original applicant for the order

and the regional council agree to the amendment, it can be made directly through Order in Council without a public hearings process (section 216).

The legal provisions applying to the consideration of water conservation orders within the Canterbury region are different as a result of the Environment Canterbury (Temporary Commissioners and Improved Water Management) Act 2010.

Applications are still lodged with the Minister for the Environment but are considered by Environment Canterbury rather than by a special tribunal (section 50). There is also no right to make a submission to the Environment Court and the Court has no jurisdiction over water conservation orders in the Canterbury region. The decision of Environment Canterbury can only be appealed to the High Court on questions of law (sections 52 and 53).

When considering an application for a water conservation order, Environment Canterbury is required to have particular regard to the vision and principles of the Canterbury Water Management Strategy as well as the other matters set out in the legislation. The same process and considerations apply to applications to amend or revoke water conservation orders within the Canterbury region.

Conservation strategies and plans

Under the Conservation Act 1987, the Minister of Conservation can prepare general policy and conservation management strategies and plans. These guide the Department of Conservation's freshwater as well as other work. The public process for preparing each of these documents is similar. Members of the public must be given the opportunity to make written submissions on draft documents. Submitters must then also be given the opportunity to present their submissions to the Director-General of Conservation in person (for submissions on general policy) or to a meeting consisting of representatives of the Director-General and affected conservation boards (for strategies and plans).

Fish and Game Councils prepare sports fish and game management plans under sections 17L and 17M of the Conservation Act 1987. These establish management objectives. The Fish and Game Council prepares a draft plan which is made available for public written submissions. Submitters then have the opportunity to present their submissions in person at a meeting of representatives of the Fish and Game Council.

Fisheries decisions

Under fisheries management, there is statutory provision for the Minister of Fisheries to approve *fisheries plans*. Before approving a plan, the Minister must consult with representatives of stakeholder groups such as fishers and environmental organisations. The Minister must also provide for the ‘input and participation’ of tangata whenua (section 12, Fisheries Act). Although members of the public have no explicit consultation rights in relation to fisheries plans, in practice, draft plans are usually circulated for public comment before approval.

For the setting of *sustainability measures* under the Fisheries Act 1996 there are no statutory rights of public participation. Sustainability measures include setting the total allowable catch, regulatory measures such as restricting fishing methods or locations, and measures to manage by-catch. The Minister of Fisheries must consult with representatives of groups having an interest in the stock or the effects of fishing on the aquatic environment in the area concerned. This will normally include any commercial and recreational fishing interests and environmental organisations. The Minister must also provide for the ‘input and participation’ of tangata whenua (section 12, Fisheries Act).

When setting or varying the total allowable commercial catch, the Minister must consult with representatives of groups having an interest including Māori, environmental, commercial and recreational interests (section 21(2), Fisheries Act)

In practice, the Ministry of Fisheries makes available an initial advice paper and will accept written submissions from any party, before finalising its advice to the Minister. If you have an interest in particular fisheries sustainability matters, contact staff at the Ministry so they can add you to the relevant circulation lists.

Enforcement

Members of the public can initiate action to improve the health of the freshwater environment through reporting issues of concern to the relevant authorities, particularly where there is suspicion of an offence being committed. Before reporting the issue, it is useful to record as much relevant information as possible, including taking photographs or filming a video. Most authorities have hotlines and after hours numbers and these are often prominently displayed on their websites.

- Pollution incidents and suspected illegal extraction, damming or diversion of freshwater should be reported to the regional council
- Suspected illegal fishing (for fish other than sports fish and whitebait) should be reported to the Ministry of Fisheries
- Suspected illegal fishing for whitebait should be reported to the Department of Conservation
- Suspected illegal fishing for sports fish should be reported to the Fish and Game Council
- Suspected new invasive species should be reported to the Department of Conservation

In the case of offences under the RMA, it is possible for members of the public to seek enforcement in the event that the regional council does not take action. An *enforcement order* is an order made by the Environment Court compelling a person to comply with the provisions of the RMA, a rule in a regional or district plan, or the terms and conditions of a resource consent. An interim enforcement order can be made where there is imminent risk of irreparable environmental damage.

Anyone can apply for an enforcement order to require a person to comply with the provisions of the RMA (section 316). Any person may also apply for an enforcement order where the proper procedure was not carried out in preparing a policy statement or plan, but an application must be lodged within three months of the document becoming operative. Only a consent authority or the Minister for the Environment can apply for an enforcement order to change or cancel a resource consent or to enforce a condition requiring the adoption of the best practicable option. You should seek advice from a lawyer before initiating proceedings for an enforcement order.

Another useful procedure available to members of the public is to apply for a *declaration*. A declaration is a decision made by the Environment Court to clarify matters under the RMA.

A declaration can be sought for a range of reasons including to determine (section 310):

- The existence or extent of any function, power, right or duty under the Act
- Whether or not a provision of a regional policy statement, regional plan or district plan gives effect to provisions in a national policy statement or is likely to be inconsistent with a water conservation order
- Whether or not there is any inconsistency between the provision of a particular policy statement or plan and the provisions of other policy statements, plans or water conservation orders
- Whether an act or omission contravenes the Act, a rule in a plan, a requirement for a designation or heritage order, or a resource consent
- Whether an activity is a permitted, controlled discretionary, non-complying or a prohibited activity
- Any other matter relating to the interpretation, administration and enforcement of the Act except whether a resource consent application should be publicly notified

Any person may apply to the Environment Court for a declaration. However, only a consent authority or the Minister for the Environment can apply for a declaration to the effect that a consent holder or any other person is contravening a condition of a resource consent, or a rule in a proposed plan, which requires the adoption of the best practicable option (section 311). Again, you should seek advice from a lawyer before initiating proceedings for a declaration.

Judicial review

Judicial review involves challenging the exercise of a statutory discretion. The challenge cannot be based on the merits of the decision, but only on the basis that it was 'unlawful', that it was unreasonable or that the decision making process did not follow the proper procedure. Judicial review has been used to challenge decisions such as the setting of the total allowable commercial catch under the Fisheries Act and the failure to notify resource consents under the RMA.

In order to overturn a decision on the basis that it is unlawful, it will normally be necessary to establish one of the following grounds:²

- The decision was motivated by an improper purpose, which was not contemplated by the legislation
- The decision-maker took into account irrelevant matters
- The decision-maker failed to take into account relevant matters that were mandatory under the legislation
- The decision was made under an invalid delegation
- The decision-maker made an error of law
- The decision-maker had insufficient information on which to base the decision

It is also possible to challenge a decision on the basis that it was unreasonable. This is a hard test to meet and involves establishing that the decision was 'so unreasonable that no reasonable [decision-maker] could ever come to it' (known as the *Wednesbury* test).

Another ground on which a decision may be set aside is on the basis that the procedure which the decision-maker followed did not accord with the principles of natural justice, including acting in good faith and listening fairly to both sides.

References and further information

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Pearl R and K Mulcahy, 2010, *Managing the marine environment: An EDS guide*, Environmental Defence Society, Auckland – available at www.eds.org.nz as an e-book

Endnotes

- 1 Sections 64 and 66 of the Environment Canterbury (Temporary Commissioners and Improved Water Management) Act 2010
- 2 Kirman C and C Whata, 2005, 1024



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Best Practice Freshwater Management

As already described in previous chapters, freshwater management is undertaken primarily through developing policies and plans at the national, regional and district levels, granting resource consents and supporting voluntary actions. Elements of best practice which can be applied to these different functions are described in the sections below.

National instruments

National instruments under the RMA include national policy statements and national environmental standards. Experience with such national instruments to date (for example the New Zealand Coastal Policy Statement), has shown that high-level and general objectives and policies provide minimal direction, and result in poor implementation. An effective national policy statement for freshwater management would therefore need to include robust objectives and policies that set clear goals for achieving good freshwater quality, for allocating freshwater, and for avoiding over-allocation. It would also need to include specific timeframes for the attainment of objectives and a clear statement of implementation mechanisms.

The European Union *Water Framework Directive* is a good example of an instrument that provides a clear policy direction. The Directive sets a high standard for water quality and provides a relatively long time frame for the various milestones to be achieved. For example in Article 4(1)(a)(iii) the directive provides that, for artificial and heavily modified bodies of water, the aim is to achieve good surface water status at the latest 15 years after the date of entry into force of this Directive.¹ The requirements for 'good surface water quality' are described in detail in Annexure V of the Directive.

The Australian Intergovernmental agreement on a *National Water Initiative* is another example of a high level policy document which includes a clear timeline for addressing water over-allocation issues. Amongst many other things it includes a commitment to return all over-allocated or overused systems to environmentally-sustainable levels of extraction. In order to achieve this, state governments are to substantially complete plans to address any existing over-allocation for all river systems and groundwater resources by 2005. Substantial progress is then to be made by 2010 towards adjusting all over-allocated and/or overused systems in accordance with the timelines indicated in the plans.²

Section 55 of the RMA allows objectives and policies contained in a national policy statement to be inserted directly into district and regional plans, without the changes to the plans having to go through the Schedule 1 public submission

process. This mechanism can only be used, however, when the national policy statement has gone through a board of inquiry process before adoption.

Section 44A of the RMA provides that, where a rule in a plan conflicts with a provision of a national environmental standard, then the rule must be amended without using the process in Schedule 1. These provisions enable specific objectives, policies and rules to have immediate and direct effect throughout New Zealand and can help promote a consistent national approach to freshwater management.

Regional policy statements

As described in chapter 1, lakes, rivers, wetlands and groundwater are all part of inter-connected hydrological and ecological systems. As well as being connected to each other, freshwater bodies are directly affected by the use and management of land in the associated water catchment. Best practice freshwater management therefore encompasses an integrated catchment management approach which recognises that changes to land or water in one part of the catchment are likely to result in impacts in other parts. Such an approach includes management of the land, freshwater bodies and the receiving marine environment as an integrated ecological system. The NPS includes the objective to improve integrated management in catchments and requires every regional council to manage fresh water and land use and development in catchments in an integrated and sustainable way.

A key purpose of regional policy statements is to achieve integrated management of the natural and physical resources of the region so encompasses the concept of integrated catchment management. Regional policy statements are therefore key documents in providing a regional framework to drive effective freshwater management at the catchment level. This is recognised in the NPS which requires every regional council to make or change regional policy statements to provide for the integrated management of the effects of the use and development of land on freshwater. Specifically, the relevant policy in the NPS states that this should include encouraging the co-ordination and sequencing of regional and/or urban growth, land use and development and the provision of infrastructure.

In respect of freshwater management, regional policy statements could be expected to include (section 62):

- A statement of the significant freshwater management issues in the region

- A statement of the freshwater issues of significance to iwi
- Identification of the objectives sought to be achieved for freshwater management (quantified where possible)
- A description of policies to be applied to water management
- A description of policies to be applied to land use and development
- A description of the methods to be used to implement the policies
- The environmental results for freshwater bodies which are anticipated
- A freshwater monitoring programme

This would involve identifying the key natural and physical freshwater resources in the region. The regional policy statement should also identify the issues for freshwater management in some detail and describe in practical terms how the issues will be addressed. This could include information about what water bodies are under stress, what are the key causes of this stress, what will be done to reduce such pressures and what specific outcomes are to be achieved and by when. Identification of the key linkages between the land, freshwater bodies and the marine area in each catchment is also important, so that management responses in upper catchment areas are driven by the carrying capacity of fresh and marine water bodies downstream.

The policy framework provided by the regional policy statement needs to be underpinned by the identification of the values attached to fresh water bodies within the region. These may include a wide range of environmental, economic, social, cultural and spiritual matters which are specific for each water body.

Regional policy statements should also address section 6 matters of national significance which are of relevance to freshwater bodies within the region. This includes identifying freshwater bodies with high natural character, those which are outstanding natural features in their own right, those which are part of outstanding natural landscapes, those which provide significant habitats for indigenous fauna and those which have historic and cultural significance.

Water classification systems can be used to help identify appropriate management objectives for individual or groups of freshwater bodies. This approach recognises that water bodies may have different values and varying responses to impacts.

There are several systems for classifying water bodies. As described in chapter 10, the RMA includes a water classification system based on the purpose for which water is managed, such as for fisheries, water supply or contact recreation. The River Environment Classification system classifies rivers based on physical characteristics such as climate, source of flow, topography, geology, and catchment land use.³ The *Cultural Health Index for Streams and Waterways*

is a complementary classification tool that can be used by councils to recognise and provide for Māori values in objectives and policies.⁴

A regional policy statement can include different objectives and policies for each category or class of water body. Well drafted objectives will be ‘specific, quantitative and measurable descriptions of environmental state or condition.’⁵



Regional plans

A regional council was not required to prepare a regional plan to address freshwater management issues under the RMA. However, the NPS now requires every regional council to make or change regional plans to ensure the plans establish freshwater objectives, set freshwater quality limits and set environmental flows and/or levels for all bodies of freshwater in its region. ‘Freshwater objective’ describes the intended environmental outcome(s). The

interpretation section states that 'limit' is the maximum amount of resource use available, which allows a freshwater objective to be met. Preparing such a plan enables a strategic approach to be taken to freshwater management, rather than resource consents being considered solely on a case-by-case basis. It also enables the cumulative effects of numerous individual activities within a catchment to be addressed.

Plan-making involves the amalgamation of technical information with community values and aspirations. Relevant technical information includes data showing the quality of the existing environment (and its natural and physical resources) and trends in water quality, water allocation and land use. Other important information includes the identification of freshwater matters of national significance as described above, as well as iwi, hapū and community values.

A regional plan includes more detailed *objectives and policies* for freshwater management within the framework established by the regional policy statement. The plan can also include specific mechanisms to implement national and regional policy on the ground. The plan mainly has effect through *rules*. These determine what activities are permitted without the need for a resource consent, when a resource consent is required what category it will fall into (controlled, restricted discretionary or discretionary) and what matters will be considered when resource consent applications are determined.

Regional Councils should undertake extensive consultation with the community before notifying a plan establishing freshwater objectives, setting limits and environmental flows and/or levels. Best practice entails classifying each water body in terms of its physical type and its current state, for example a lowland river that is eutrophic. A range of options for freshwater objectives and associated limits, flows and levels should then be devised at a minimum for each class of water body. Ideally options for freshwater objectives and associated limits, flows and levels would be developed at the catchment level. Regional councils may decide to prioritise some areas and develop catchment level freshwater objectives and limits, flows and levels while developing options for other areas based on classes. The limits, flows and levels should be based on what would be required to implement the proposed freshwater objective. Best practice is to develop options for freshwater limits, flows and levels simultaneously as the quantity of water available in a water body affects its ability to assimilate contaminants. Therefore a change in the flow and/or level will have an effect on the corresponding freshwater quality in the particular water body.

The consequences of each option for economic, environmental, social and cultural values should then be assessed. It is important the options span the likely practical range of community aspirations. This allows the community

to fairly evaluate each option and means subsequent plan provisions can be justified. Based on community feedback regional councils should then notify a proposed plan or change including the desired options.

As noted in Chapter 7 on the Freshwater Management Framework, the freshwater objectives and limits set by regional councils must give effect to the objectives in the NPS. This includes Objective A1: to safeguard the life-supporting capacity, ecosystem processes and indigenous species including their associated ecosystems of freshwater. In order to give effect to this objective best practice is to recognise and respect fresh water's intrinsic values. Examples of these values are set out in the NPS and include: the interdependency of the elements of the freshwater cycle; and the natural form, character, functioning and natural processes of water bodies and margins, including natural flows, velocities, levels, variability and connections.

Plan may also identify voluntary measures to the extent that these require council resources. For example, a council may decide to provide funding support for riparian fencing, and if so will need to include such commitments in the annual budget and long term council community plan under the Local Government Act 2000.

The regional plan should define anticipated environmental results that are measurable and can be used to assess the extent to which the desired objectives are attained. Monitoring is essential to determine the existing environment at any point in time and to assess how effective the plan provisions are over time. Monitoring is often a condition of major consents. While this may assist with information collection, it is not a substitute for a region-wide programme of freshwater monitoring.

A regional plan should include a robust monitoring framework for the regular measurement of water quantity, water quality, and the ecological health of water bodies. It is important that the monitoring framework is designed as an integral part of the plan, so that there is a clear link between the plan's objectives and the council's ongoing monitoring programme. The cost of monitoring will need to be included in the long term council community plan and annual budgets.

Water quality can be measured using chemical and physical measures and/or biological measures. Interpreting water quality data can be a complex exercise and requires specialised skills.⁶ There are Australian and New Zealand Guidelines for Fresh and Marine Water Quality 2000 which indicate 'trigger values' for the protection of aquatic ecosystems.⁷ These are currently being revised to incorporate new data and are expected to be finalised in 2012.

Water allocation provisions

An effective regional plan could be expected to provide a robust framework to manage water allocation and address any cumulative impacts of water abstraction. As a first step, the NPS requires that regional councils set *environmental flows* and/or levels for all freshwater bodies in its region. These provisions may need to ensure a variation of flow when this is required to maintain a healthy ecological system or other important cultural or recreational values. The plan needs to include clear rules to ensure that these environmental flows are not breached.

The NPS requires that regional plans set *allocation limits*, which specify the maximum rate at which water can be taken from a river and the maximum volume that can be taken from a lake or aquifer over a specified period of time. Clear rules then need to be established to ensure that water cannot be over-allocated. Because estimating the yield of aquifers is a much less exact science than for surface water, there needs to be a margin between permitted takes from aquifers and estimated yield, to allow for the greater degree of uncertainty.

The NPS requires every regional council to make or change regional plans to provide for the efficient allocation of fresh water to activities, within the limits set in the plan.. The default first-in first-served system, which is described in Chapter 9, often fails to ensure that water is allocated to the uses with the highest value and will not necessarily promote sustainable management as is required under the RMA.⁸

Regional plans can also include a priority regime to establish how water takes are to be restricted when an environmental flow or minimum level has nearly been reached. The priority may be based on prior appropriation (last user to receive a consent is cut off first), pro-rata reduction (all users have take reduced by the same proportion) or users grouped into tiers according to priority.⁹ These measures could also be used to address over-allocation. The NPS requires every regional council to set a defined timeframe and methods in regional plans by which over-allocation must be phased out, including by reviewing water permits and consents to help ensure the total amount of water allocated in the water body is reduced to the level set in the plan.

Water permits should include conditions requiring water takes to *be measured and monitored*. There are now mandatory requirements for measuring water takes for some water permits. Many councils do not know how much water is currently being used which makes effective management difficult. Water metering provides councils with better information to assist with setting allocations based on reasonable use. It also enables councils to assess effects on water bodies and to ensure compliance with consent conditions. Water metering also supports demand management, as if people know the volume of

water they are using, they are more likely to conserve water.¹⁰ As described in Chapter 9, the regulations for the measurement and reporting of water takes establishes minimum requirements for measurement by consent holders of water permits that allow fresh water to be taken at a rate of 5 litres/second or more. The regulations do not apply immediately to existing water permits.

As discussed in Chapter 7, Objective B3 of the NPS is to improve and maximise the efficient allocation and efficient use of water. There are corresponding policies associated with this objective including Policy B4 which requires every regional council to identify methods in regional plans to encourage the efficient use of water. Measures to provide for technical efficiency in the use of water include repairing leaky infrastructure, replacing old inefficient technology, not watering soils over field capacity and re-designing infrastructure for example piping water instead of having open channels.

Water quality provisions

As well as addressing water allocation issues, a regional plan needs to provide a management framework to address water quality issues within the region. A core element of such a framework is the establishment of water quality limits which are implemented through clear rules. One example of a water quality limit is a *water quality standard* which may be a numerical threshold for a contaminant or a narrative description of an environmental state.¹¹ Standards are based on scientific information but also involve a value judgment as to what degree of protection is necessary.

The NPS requires that outstanding freshwater bodies and the significant values of wetlands are protected. For water bodies with degraded water quality, the NPS requires that these be improved by regional councils specifying targets and implementing methods (either or both regulatory and non-regulatory) to meet the targets within a defined time frame.

At the catchment level, regional plans need to include mechanisms to effectively manage the cumulative impacts of numerous activities on water quality to ensure that freshwater management objectives and limits are met. This is likely to require the setting of catchment contaminant load limits. It may also require rules controlling significant land use change, such as urban development or dairy intensification, where such activities may adversely impact on water quality in a sensitive or pressured catchment.

Where existing land uses are having a negative cumulative effect on water quality, restrictions on activities such as fertiliser application or stocking rates may be required. Another mechanism which can be used to control non-point source discharges at a catchment-level is the establishment of a 'cap and

trade' scheme which identifies the maximum allowable pollution loading for a catchment and allocates tradeable pollution allowances to individual land users.

At a more localised level, regional plans could be expected to include provisions which require the adoption of best practice management for individual activities, in order to reduce their impact on freshwater bodies. Examples of such best practice measures are described below in the sections focused on specific activities.

District plans

District plans focus on the management of the environmental effects of land use including the impacts of land subdivision, development and use on freshwater bodies. They could be expected to identify sensitive freshwater areas that are not suitable for urban development or subdivision such as wetlands, sites with significant indigenous vegetation and areas near important water bodies.

In order to protect waterways a district plan needs to contain provisions which effectively manage storm water, control the amount of impervious surfaces, regulate earthworks and manage traffic generation. There should also be limitations on the degree and extent to which water courses can be modified as this can result in the destruction of habitat and adversely affect flora and fauna. There is a discussion of the specific best practice measures for urban development later in this chapter.

Resource consents

When water and discharge permits are granted, it is important that the conditions are well drafted and provide for robust monitoring of the amount of water taken or the effects of the discharge. Resource consents should also include a condition which enables the council to regularly review the conditions of the permit. This is to provide for situations where water use exceeds water availability or where further information reveals that the allocation or discharge limits provide inadequate protection for the water body and need to be amended.¹²

Other matters which could be included as conditions on resource consents are described in the sections below focused on specific activities.

Voluntary action

There are many groups around New Zealand that undertake voluntary action to improve and restore the ecological health of water bodies. This includes fencing the margins of water bodies, undertaking planting on riparian margins and in wetlands, and monitoring water quality. Councils can support such activities by providing technical assistance and financial contributions as has occurred with the Twin Streams project in Waitakere City.

Project Twin Streams, Waitakere City

Project Twin Streams is an example of a community-based restoration project. The project is based in Waitakere City and has focused on the restoration of 56 kilometres of stream banks. Community groups contracted with the council to undertake the planting and maintenance work. The council purchased more than 80 properties which were subject to flooding freeing up additional land for planting. The project also included an extensive art component, where community members were involved in environmental art activities such as painting, sculpture, ceramics, music and drama.

Source: Waitakere City Council, 2010

An example of a voluntary action programme aimed at improving freshwater quality is the *Clean Streams Accord* spearheaded by Fonterra, a co-operative owned by the majority of New Zealand dairy farmers. Specific priorities for action identified as part of the accord include

- Excluding dairy cattle from streams, rivers and lakes and their banks
- Constructing bridges or culverts for farm races where stock cross a watercourse
- Appropriately treating and discharging effluent in accordance with regional plans and resource consents



Figure 12.2 Pamphlet promoting community planting day for Project Twin Streams (Waitakere City Council)

- Creating systems on all dairy farms to manage nutrient inputs and outputs.¹³

Another form of voluntary action is demand management. This entails providing information and establishing policies to assist people to conserve water and limit demand. The Ministry for the Environment has identified the main barriers that prevent people from conserving water as being:

- Having minimal information about personal water use
- The view that there is no need to conserve water because New Zealand has a plentiful supply
- The cost, time and effort involved in conserving water.¹⁴

To overcome these attitudes, people need greater information about the benefits of conserving water and the simple ways they can do so. If households are charged directly for their water use then this is likely to create an incentive to conserve water.¹⁵ Some councils have implemented water metering as a basis for charging.

New Zealand has recently adopted water efficiency standards for domestic water fittings. The water efficiency labelling scheme applies to six types of products: clothes washing machines, dishwashers, lavatories, showers, taps and urinals.¹⁶ Efficiency ratings will appear on these domestic fittings in retail outlets, similar to the familiar “star” ratings on some electric appliances.¹⁷ All labels will include a water consumption or flow figure.¹⁸

Environmental education plays an important role in increasing the willingness of people to engage in voluntary activities. This is through ensuring that people understand why it is important to protect the health of freshwater bodies and what actions they can take to help achieve this end.

Specific activities

Agriculture

Measures required to address non-point source discharges from agriculture comprise of three categories: reduction at source, interception of pollutants on land, and interception or processing of pollutants in water.¹⁹

Animal effluent can enter waterways either directly from animals which have access to water bodies, or indirectly as a result of effluent being transported from the land. Measures to reduce this problem include:²⁰

- Ensuring that stocking levels are within the carrying capacity of the catchment
- Preventing stock from accessing waterways and riparian areas through adequate and well-maintained fencing
- Keeping stock off water-logged soil and avoiding soil treading and pugging. This could include the use of feed pads during winter
- Restoring and maintaining well-vegetated wetlands, seepage areas, bogs and filter strips
- Use of adequate treatment systems for effluent from dairy sheds including ponds and wetland systems
- Ensuring that land irrigation of treated effluent does not exceed the capacity of the application area to absorb contaminants

Fertiliser and sediment can also enter waterways directly or through ground seepage and soil erosion. Measures to reduce this problem include:²¹

- Ensuring that fertiliser application levels are within the carrying capacity of the catchment
- Maintaining well vegetated riparian areas
- Matching the quantity of fertiliser applied to plant requirements
- Avoiding the application of fertiliser directly into streams or wetlands
- Applying fertiliser when grass growth is active
- Avoiding fertiliser applications when rain is likely or when the water table is high
- Keeping animals away from stream banks and wetlands
- Avoiding overgrazing, especially in winter, which can exacerbate soil erosion
- Constructing sediment ponds in areas where there is significant run-off
- Constructing farm roads and tracks on stable low-gradient sites and including frequent cut-off channels
- Constructing bridges and culverts at frequently used river crossings
- Minimising cultivation of soil especially in erosion prone areas



Figure 12.3 Hikui Stream, Coromandel Peninsula (Raewyn Peart) *The retention of riparian vegetation and exclusion of stock can help reduce the amount of contaminants entering waterways*



Figure 12.4 Mountain Landing, Bay of Islands (Raewyn Peart) *Planting of riparian areas and restoration of wetlands can help restore water quality and freshwater ecosystems*



Figure 12.5 Farm culvert, Waikino Gorge (Raewyn Peart) *The construction of culverts at farm crossing reduces erosion and damage to the stream bed*

The preparation and implementation of farm environmental management plans and nutrient budgets can enable the range of environmental impacts from a single farming operation to be identified and managed. However, the adoption of best practice farm management approaches at the individual farm level, will not necessarily be sufficient to reduce the impacts of farming activity at a catchment level, so that they are within the environmental carrying capacity of the catchment. This may require controls on the location of farming activity, on the use of fertilisers, and/or on stock numbers.

Urban development

The development of new urban areas can have a major negative impact on freshwater bodies through the release of large amounts of sediment during earthworks and construction. As an urban area becomes more established, sediment usually becomes less of a problem, but there can be increased discharges of pollutants including organic matter, nutrients and heavy metals.

It is therefore important that urbanisation does not take place in catchments where downstream freshwater bodies and estuaries are likely to be significantly affected by increased sediment flows. When urban development does take place, careful measures need to be taken to reduce and manage sediment run-off. Erosion and sediment control plans should be prepared and approved by the

council before any works are commenced. Measures to reduce sedimentation include:

- Use of sediment detention measures such as silt fences, earth bunds, sediment detention ponds and chemical sediment control (flocculation)
- Avoidance of major building works during times of the year when rainfall is likely to be heavy
- Retaining existing vegetation wherever possible and revegetating disturbed areas as soon as possible
- Limiting the size of the disturbed areas

Urbanised areas discharge large amounts of stormwater which collects on impervious surfaces such as roofs, roads and driveways. Stormwater is often directly piped into rivers or the marine area. It can transport toxic pollutants as well as sediment into freshwater bodies. The impacts of stormwater discharge can be reduced through measures such as:

- Reducing the area covered by impervious surfaces. District plans can include maximum site coverage provisions and encourage the use of permeable paving
- Using networks of swales and planted areas to reduce and filter stormwater runoff in urban areas
- Restoring wetlands
- Collecting and treating stormwater before discharge
- Using roof rainwater collection tanks²²

Wastewater treatment works should use the best available technology to ensure that the water discharged into the catchment contains minimal amounts of organic matter, micro-organisms and nutrients. The construction of wastewater treatment facilities needs to be adequately planned so that there is capacity to deal effectively with all waste generated by the community, and by future or seasonal growth in the area. Combined sewage and stormwater systems, which are common in older urban areas, need to be progressively replaced with separate networks, and the sewage pipes regularly monitored and maintained to prevent leakage.

Septic tanks should only be constructed in areas where the soil is suitable and where there is an adequate amount of land surrounding the property to cope with the amount of effluent likely to be generated. They also need to be regularly maintained.



Figure 12.6 Whisper Cove stormwater detention pond, Snells Beach (Raewyn Peart) *Urban development should where possible incorporate 'green infrastructure' such as planted riparian areas and stormwater detention ponds as shown here*



Figure 12.7 Whisper Cove permeable driveway, Snells Beach (Raewyn Peart) *Permeable paving enables rainwater to soak into the soil and reduces the runoff into stormwater networks*

Horticulture

Horticultural activities may result in nutrients from fertilisers, herbicides and pesticides leaching from the soil into freshwater. Such impacts can be reduced by carefully matching fertiliser use to plant requirements and maintaining riparian margins.

Forestry

To reduce the impact of forestry activity on freshwater bodies, riparian areas adjacent to streams and wetlands should not be planted in exotic forestry species, but instead be allowed to revert to indigenous vegetation. In areas where native seed sources no longer exist (such as along streams and rivers crossing the Canterbury Plains) planting or seeding of stream, river and lake margins in native species may be required when a commercial forest is first established or replanted.

The ideal width of the margin, and the most effective species to plant, depends on the size and nature of the water body and the impacts likely to arise from harvesting. The riparian vegetation should remain undisturbed by earthworks, harvesting machinery or the application of fertilisers²³ or herbicides.²⁴ If a riparian margin is successfully maintained, then this will act as a buffer between water bodies and the trees during harvest, and protect waterways from the majority of future runoff.

To reduce sediment run-off during the harvesting cycle, trees should not be extracted during periods when the ground is saturated in order to avoid soil compaction, or where surface runoff is entering the riparian area. Depending on the circumstances, low impact techniques (such as the use of Skyline cable harvesting or specialist low ground pressure tracked harvesting machines) and/or sediment control measures may be required when harvesting trees on steep and/or highly erodible soils.²⁵ Soil disturbance should be minimised as much as practicable.

Forestry roads, tracks and landings can be a significant source of sediment run-off. Flumed culverts and/or cut-off drains should be installed at regular intervals along the tracks so that water run-off is contained and directed away from waterways.

Dams and structures

There are few measures that can overcome the adverse effects of damming or diverting a river. A dam will change part of the river into a still-water habitat. As a consequence, water temperatures and nutrient levels will often increase. This can promote the growth of aquatic weeds which can threaten native

fish and invertebrates. Altering the natural flow and water level regime of the downstream river adversely affects aquatic ecosystems and riparian vegetation and habitat.

There are some measures which can lessen the adverse impacts of dams on aquatic habitat, native fish and recreational interests. The main measure is to release water on a regular basis to create variations in the flow of the water body. This is important for flushing the river and removing excess sediments and weeds. Mighty River Power has a regulatory requirement to release daily spills of water down the Aratiatia Rapids to meet recreational and tourism demands.²⁶ If the construction of a dam or reservoir is likely to disrupt the movement of sediment, then a sediment sluicing facility should be incorporated into the structure.²⁷

Some hydro lakes, such as Lake Benmore and Lake Hayes, are used for a diversity of activities including freshwater aquaculture, fishing, rowing and recreational boating, as well as for energy generation. Management therefore needs to balance the needs of these competing uses.

The life cycle of most native freshwater fish means that they must migrate to and from the sea. Any structure built within a freshwater environment should therefore enable fish to move upstream to allow such migration to occur. Fish passes are commonly used in association with dams, although their effectiveness is uncertain. The design of such a pass depends on the local circumstances and the type of fish affected area. Some species, such as inanga, are non-climbing.²⁸

Where culverts are used, their slope and alignment should match that of the stream, and allow sediment and debris to pass through during floods. The bottom of the culvert should be below the stream bed and the banks around the inlet and outlet protected from erosion.²⁹ You are not allowed to construct any culvert or ford in such a way that the passage of fish would be impeded, without the written approval of the Director-General of Conservation incorporating such conditions as they think appropriate.³⁰

Where water is abstracted from rivers and lakes, systems to prevent fish uptake should be used such as fish screens to obstruct passage and jets of water to create turbulence. Intakes should also be sited away from areas that fish commonly inhabit such as the edge of rivers.

Mining

Mining should only be undertaken in areas where acid mine drainage can be effectively contained in the long term or treated. Where mining is undertaken, a site specific management plan and rehabilitation plan should be prepared by a qualified expert.

There are two ways to deal with the problem of acid mine drainage. First, measures can be put in place to prevent the formation of contaminated drainage. For example management techniques can be applied during mining to limit one of the three components necessary for oxidation (water, oxygen and sulphide).³¹

The second way to deal with the problem is to ensure that waste material is effectively treated before it is discharged into a water body. Passive or active treatment systems can be used. Active treatment involves the use of chemicals and requires regular operation and maintenance. It is a reliable form of treatment in effectively removing contaminants. Passive treatment systems rely on natural physical, geochemical and biological processes to reduce the acidity in acid mine drainage. These systems are not as costly as active treatment but are not as reliable.³² It depends on each site and the status of the mine as to what will be the most acceptable method of treatment.

A site specific management plan should also minimise the potential for the generation of sediment laden runoff and ensure that any structures will not cause erosion. Resource consents should include a condition requiring a bond and other financial assurance to be provided to ensure that rehabilitation of a mining site and ongoing maintenance is provided for.³³

Extraction of river gravel can alter the natural character of the river and damage invertebrate communities by altering the composition of the riverbed both at the extraction site and further downstream. Gravel extraction should therefore be avoided in freshwater bodies with high biodiversity and natural character values.

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Conclusions

New Zealand has a very diverse range of freshwater bodies including fast flowing mountain streams, extensive braided river systems, slow meandering rivers and streams, numerous lakes, extensive aquifers and remnant wetlands. These aquatic systems support an abundance of freshwater plants and animals, many species which are endemic to New Zealand, and which are threatened.

Freshwater systems are extremely valuable to New Zealand, for their very high cultural and recreational values, for their crucial role in supporting the economy and for their provision of essential ecosystem services.

Freshwater is coming under increasing pressure, particularly from urban and agricultural activities. Many lowland rivers and lakes are polluted, numerous indigenous freshwater species are moving towards extinction and freshwater has been over-allocated in some regions.

It is essential that these growing pressures are managed more effectively. This Guide is designed to help achieve this. Its purpose is to promote greater awareness of the issues and the adoption of better management approaches to freshwater management.

New Zealand has an abundance of freshwater which flows through a multitude of natural systems. These include fast flowing mountain streams, slow moving meandering rivers, complex braided rivers, lakes, wetlands and extensive groundwater systems. Freshwater bodies support a diverse range of birds, fish and invertebrates which form complex communities. Many of these species are endemic and are now threatened.

Freshwater is enormously valuable to New Zealand. It is a taonga for Māori and culturally important for other New Zealanders. It supplies the water needs of households, industries and the rural economy. It also supports tourism, recreation and quality of life. But more importantly, freshwater systems often supply critical ecosystem services including biodiversity retention, waste assimilation and flood control.

Freshwater is coming under increasing pressure, particularly from urban and agricultural activities. Pollution has become evident in many lowland lakes and streams and in some catchments water has been over-allocated.

This Guide has been written for people who want to know more about how to promote effective management of New Zealand's freshwater systems and in order to obtain better environmental outcomes. It is also designed to assist people to reduce their impacts on freshwater. It explains what activities impact on freshwater systems and how these impacts can be better managed.

The Guide will be invaluable for anyone concerned with the use and management of New Zealand's lakes, rivers, streams, wetlands and groundwater systems.

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ISBN 0-9582641-8-X



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